
EECS 16B Designing Information Devices and Systems II

Gireeja Ranade, Vladimir Stojanovic Homework 1

1. Reading Lecture Notes

Staying up to date with lectures is an important part of the learning process in this course. This question is worth as much as the rest of the homework. Fortunately, it is also really easy. Here are links to the notes that you need to read for this week: [Note 0A](#) [Note 0B](#)

- (a) **Explain why the Thevenin and Norton equivalents of a circuit are useful.**
- (b) **What are the building blocks in 16B that allow us to build the robot car as the final project?**

2. Save Baby Yoda! (8 points)

Despite our best efforts, we have lost Baby Yoda to former agents of the Galactic Empire. Luckily we were able to conceal a receiver in his locket, so now it's time to save Baby Yoda using our 16A knowledge!

Baby Yoda has been delivered to an Imperial Star Destroyer. Rebel intel has provided us with access to their internal communication beacons. The ship's layout is 2-dimensional with 3 beacon locations specified in Table 1.

Beacon	Coordinates	Distance to Baby Yoda
A	(5,5)	$\sqrt{20}$
B	(2,3)	1
C	(1,1)	2

Table 1: Data from Destroyer Beacons and their coordinates.

Explicitly write out a *linear* system of equations (in matrix-vector form) using the data above for finding Baby Yoda's location $\vec{x} = \begin{bmatrix} x \\ y \end{bmatrix}$. Draw a box around your final linear system, then solve for Baby Yoda's location. Nonlinear terms are not permitted in your final system of equations. You must provide both the system and the location for full credit.

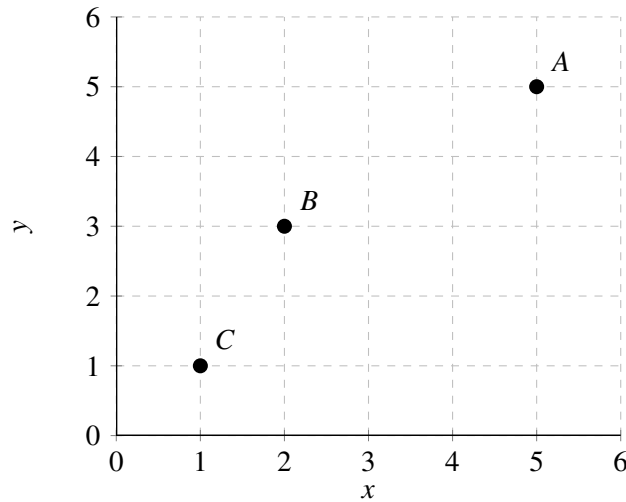


Figure 1: Diagram of the Destroyer's floor-plan with Beacon coordinates marked accordingly.

3. Ultrasound Sensing with Op-Amps (14 points)

The transresistance amplifier is often used to convert a current from a sensor to a voltage. In this problem we will use it to build an ultrasound sensor! When an ultrasonic wave hits our sensor, it generates a current, i_{ultra} . Whenever no ultrasonic wave hits our sensor zero current is generated, so $i_{\text{ultra}} = 0$.

*Note: An **ideal op-amp** is used in all subparts of this question. You can also assume that $V_{DD} = -V_{SS}$.*

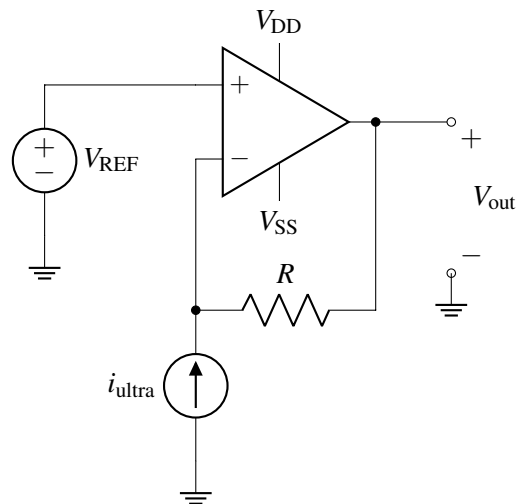


Figure 2: Transresistance sensor circuit

- (a) (4 points) Calculate the output voltage, V_{out} , of the transresistance sensor circuit shown in Fig. 2, as a function of the reference voltage, V_{REF} , the sensor input current, i_{ultra} , and the resistor, R , when an ultrasonic wave hits the sensor. Clearly show all your work and justify your answer. Writing only the final expression will not be given full credit.

(b) (5 points) Assume that the amplitude of the ultrasonic wave hitting the sensor is such that the current i_{ultra} fluctuates from a minimum value of $i_{\text{min}} = 1 \cdot 10^{-6} \text{A}$, to a maximum value of $i_{\text{max}} = 2 \cdot 10^{-6} \text{A}$. Also assume that the reference voltage is set to $V_{\text{REF}} = 1 \text{V}$. In this case, calculate the following:

- The maximum value of the resistor, R , so that the output voltage, V_{out} , does not drop below 0V.** Clearly show all your work.
- Assuming you picked $R = 250 \cdot 10^3 \Omega$ (which may or may not be the correct answer to part (i)), **calculate the maximum value of the output voltage, V_{out} .** Clearly show all your work.

(c) (5 points) Unfortunately, after a few hours of successful ultrasound sensing, our sensor got damaged. It now constantly generates a huge background current, I_{damage} . So when an ultrasonic wave hits it, the sensor produces $I_{\text{damage}} + i_{\text{ultra}}$, as shown in Fig 3(b). When no ultrasonic wave hits it, the sensor produces just I_{damage} . However, the huge background current causes our output to constantly be $V_{\text{out}} = V_{\text{SS}}$, so we are not able to tell whether an ultrasonic wave is present or not.

We would like to fix this in our circuit by canceling the background current and retaining only the useful signal. For this purpose we are going to use a current source, I_{fix} , shown in Fig. 3(a), whose value we can choose. **How would you connect this current source in your circuit and what value would you pick for it?** Redraw the entire circuit with the new current source, I_{fix} , added and give the value of I_{fix} in terms of I_{damage} , i_{ultra} , R , V_{REF} . **Explain how your design works.**

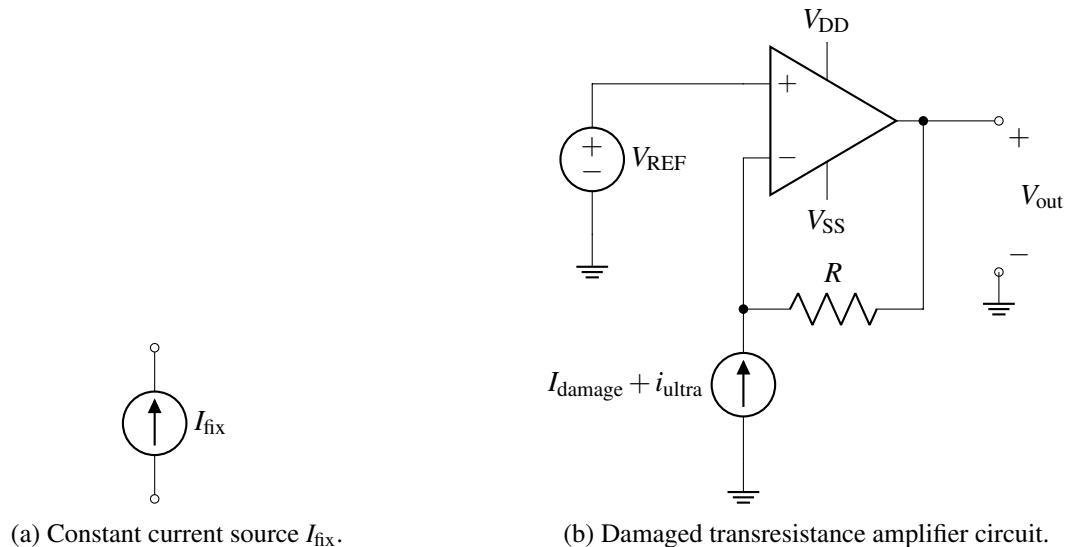


Figure 3: Circuits detailing the transresistance amplifier design, including the background signal I_{damage} .

4. Saving Lives with Op-Amps (19 points)

An electrocardiogram, or ECG, is a medical device used to detect electrical signals in your heart. Typically, the voltage signal from the human heart is only 1×10^{-3} V at maximum. However, in order for healthcare professionals to properly interpret ECGs, these signals need to be amplified so that abnormalities are more obvious. In this problem we will do so by using **ideal op-amps**.

Note: Assume that $V_{DD} = -V_{SS}$ in all subparts.

- (a) (3 points) We need to amplify the voltage signal recorded by the electrodes V_{in} by a factor of 1000. Using the op-amp in Figure 4 below and 2 resistors, **draw a circuit that achieves $V_{out} = 1000 \cdot V_{in}$. Write an equation for V_{out} in terms of V_{in} and the resistor(s), label the resistor(s) you use (i.e. R_1, R_2), and choose their values.** You should redraw the entire circuit in your answer sheet, but there is no need to draw the human as long as you label V_{in} . Clearly explain and show your work.

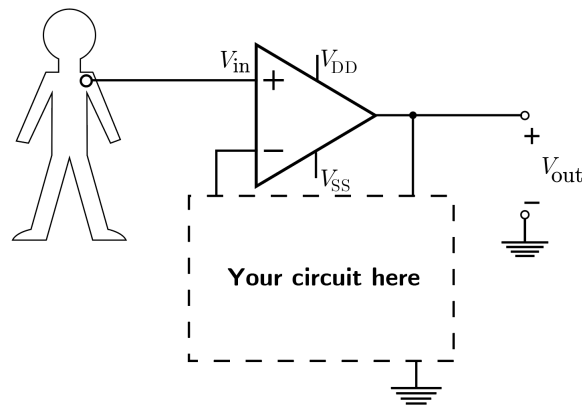


Figure 4: Unfinished ECG amplification circuit.

- (b) (4 points) A friend of yours is also working on an ECG amplification circuit, and shows you their design in Figure 5. Their design uses $R_{\text{electrode}} = 1 \cdot 10^3 \Omega$, $R_1 = 1 \cdot 10^3 \Omega$, and $R_2 = 1 \cdot 10^6 \Omega$. They claim their circuit gives, $V_{\text{out}} = -1000 \cdot V_{\text{in}}$. **Is their claim true?**

- If yes, justify why.
- If no, how would you choose the value of R_2 to achieve $V_{\text{out}} = -1000 \cdot V_{\text{in}}$, assuming that both $R_{\text{electrode}}$ and R_1 are fixed at $R_{\text{electrode}} = R_1 = 1 \times 10^3 \Omega$? Clearly show your work, and justify your answers.

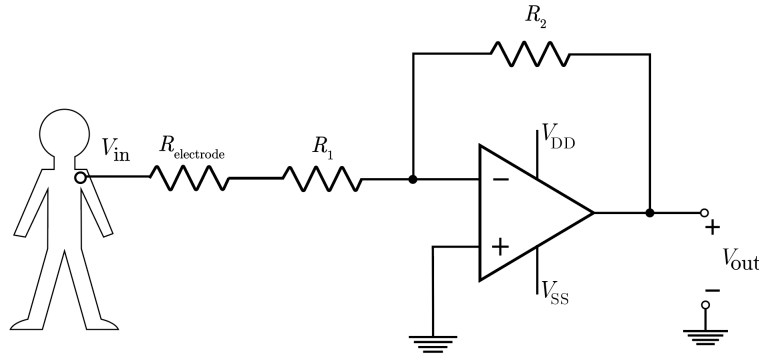


Figure 5: An alternative ECG op-amp circuit.

- (c) (4 points) Another configuration often used by healthcare professionals is to attach one electrode to the heart (recording its electrical signal, V_{in}) and another electrode to the right leg to serve as a reference voltage, as shown in Figure 6. **What is the output voltage, V_{out} , as a function of V_{in} , V_{RL} , R_{bottom} , and R_{top} ?** Clearly show your work.

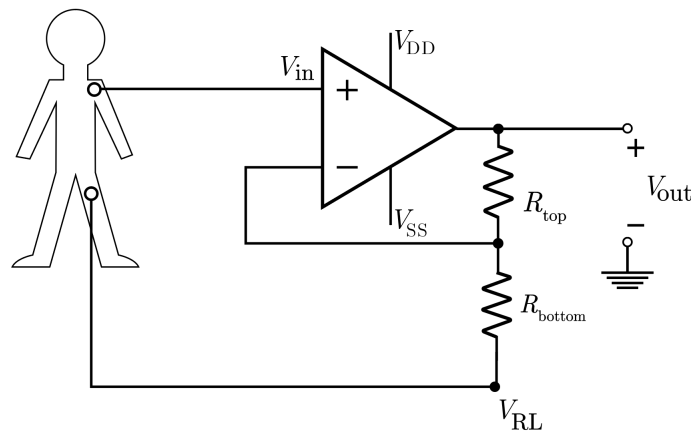


Figure 6: Alternative op-amp ECG topology.

- (d) (8 points) Even after amplification, certain peaks of your ECG signal are too low to be discerned. You want to sample them and amplify them a bit more. To this end, you use the circuit in Figure 7. The circuit cycles through two phases: in phase 1, switches labeled ϕ_1 are ON and ϕ_2 are OFF, while in phase 2, switches labeled ϕ_2 are ON and ϕ_1 are OFF. **Calculate the output voltage, V_{out} , during phase 2, after steady state has been reached, in terms of C_1 , C_2 and V_{in} .** Clearly show your work.

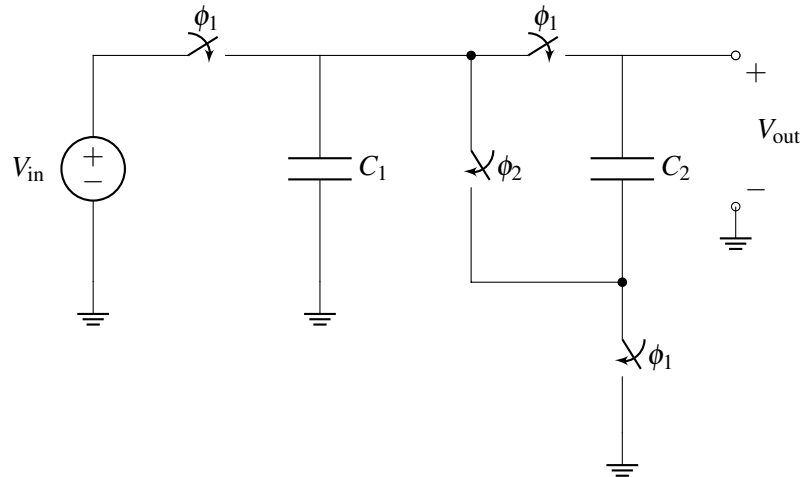


Figure 7: Switch capacitor voltage boosting circuit.

5. Hyperspectral Classification of Tomatoes (14 points)

You're a high-tech farmer who just bought a new hyperspectral sensor to monitor your crops.

NOTE: You do not need to understand how a hyperspectral sensor works to solve this problem.

You attach the sensor to a drone and fly it over your crops, taking measurements of the hyperspectral signature for different points along the field. You want to use these measurements to identify which crops are healthy and which crops are getting sick. Your sensor gives you a spectral signature for each plant as a length 5 vector, where each entry of the vector represents a different frequency. Scientists have determined that healthy versus sick tomato plants will have the following spectral signatures as shown in Figure 8.

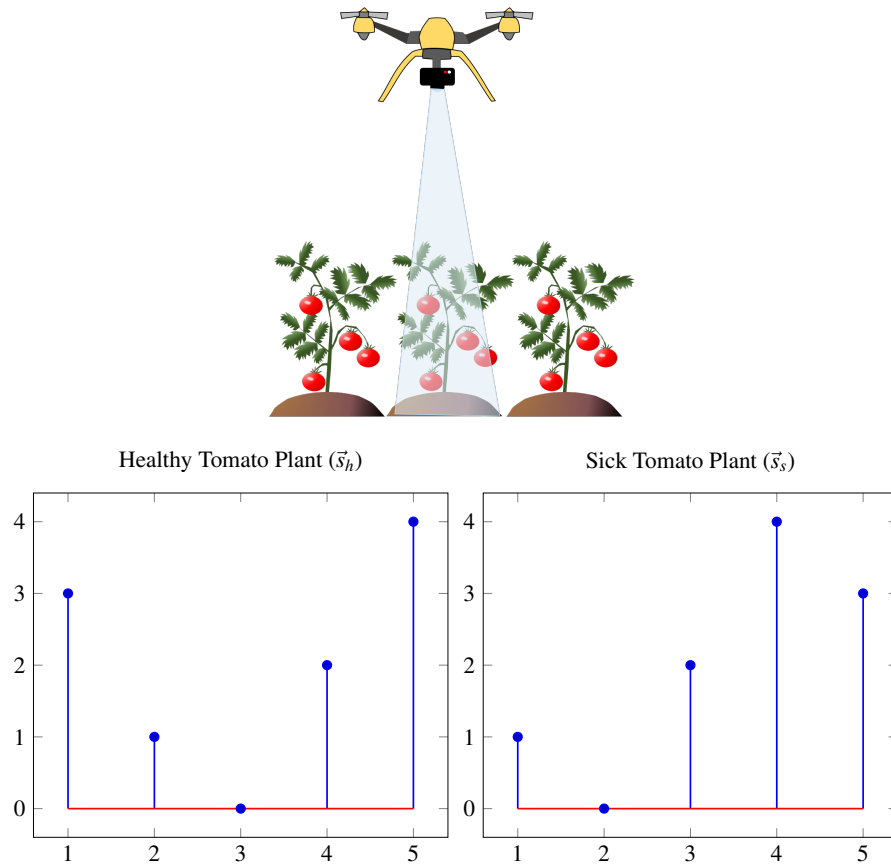


Figure 8: Spectral signature for healthy tomato plant ($\vec{s}_h$) and sick tomato plant ($\vec{s}_s$).

They can also be represented in vector form:

$$\vec{s}_h = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 2 \\ 4 \end{bmatrix}, \quad \vec{s}_s = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 4 \\ 3 \end{bmatrix}.$$

- (a) (6 points) Using your spectral sensor, you measure the following spectral signature for one of your tomato plants as shown in Figure 9. This measurement has some noise in it.

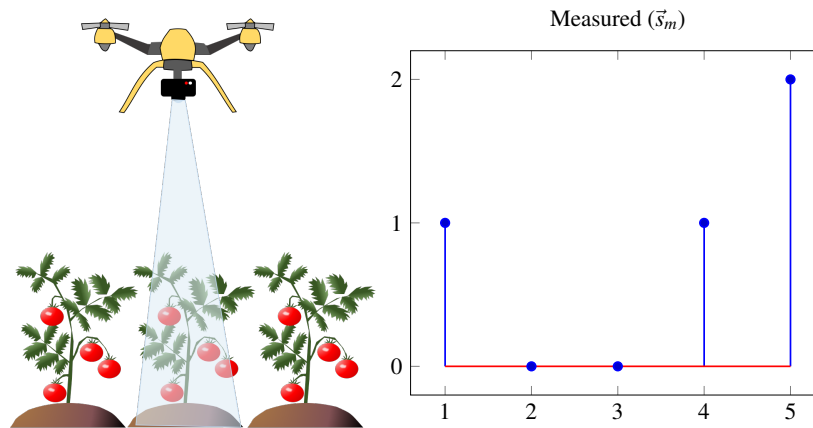


Figure 9: Spectral signature for the measurement, $\vec{s}_m$.

The spectral signatures for healthy, sick, and your measured tomato plants can also be represented in vector form as

$$\vec{s}_h = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 2 \\ 4 \end{bmatrix}, \quad \vec{s}_s = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 4 \\ 3 \end{bmatrix}, \quad \vec{s}_m = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 2 \end{bmatrix}.$$

Since spectral signatures never exactly match, the standard procedure is to calculate the angle between signature vectors to determine how close they are. **Compute the angle between $\vec{s}_m$ and $\vec{s}_h$ and the angle between $\vec{s}_m$ and $\vec{s}_s$. Is your measured vector closer to the sick plants or the healthy plants?**

Classify your plant's health based on the angle between your measured spectral signature ($\vec{s}_m$) and the known spectral signatures, ($\vec{s}_h, \vec{s}_s$). Show your work and justify your answer.

NOTE: Table 2 can be helpful for finding the angles.

Table 2: Cosine Table

$\cos(\theta)$	$\theta(^{\circ})$
$\frac{9}{\sqrt{180}}$	47.87
$\frac{10}{\sqrt{180}}$	41.81
$\frac{11}{\sqrt{180}}$	34.93
$\frac{12}{\sqrt{180}}$	26.57
$\frac{13}{\sqrt{180}}$	14.31

- (b) (4 points) It's a windy day and the drone got pushed as it was taking a measurement, so now the measurement has a linear combination of measurements for several different tomato plants (some of which are healthy and some of which are sick). So your measurement is

$$\vec{s}_m = \alpha \vec{s}_h + \beta \vec{s}_s + \vec{e}, \quad (1)$$

where $\vec{e}$ represents an error vector that is unknown.

The values you get for your measurement are:

$$\vec{s}_m = \begin{bmatrix} 5 \\ 1 \\ 4 \\ 10 \\ 10 \end{bmatrix}.$$

The measurement is also shown in Figure 10.

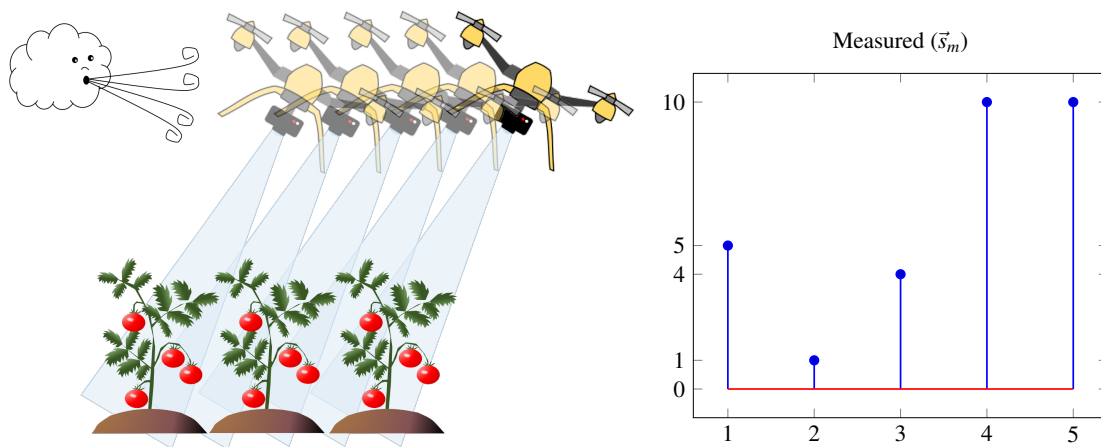


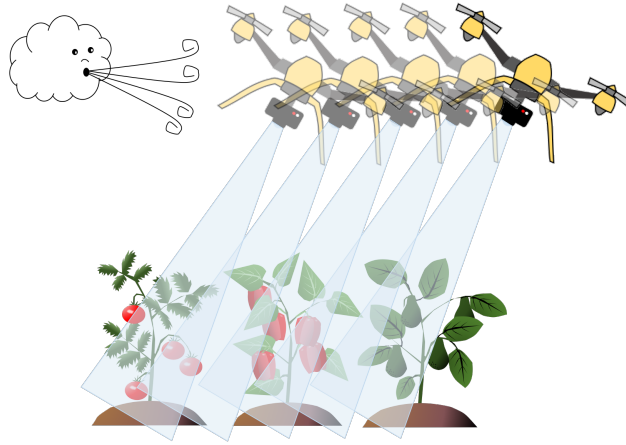
Figure 10: Spectral signature for your measurement, $\vec{s}_m$.

Recall that

$$\vec{s}_h = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 2 \\ 4 \end{bmatrix}, \quad \vec{s}_s = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 4 \\ 3 \end{bmatrix}.$$

You want to identify the unknowns α and β . **Write a least squares problem in the format $\mathbf{A}\vec{x} = \vec{b}$ to identify the unknowns α and β . Show your work.** You do not have to solve for α and β .

- (c) (4 points) Your drone got pushed by the wind again, but this time it was while it was taking a measurement on the border of three adjacent fields - your tomato, pepper, and avocado fields.



Tomato, pepper, and avocado plants have unique spectral signatures with a length of 5. The notations are described as the following:

- $\vec{s}_h$ and $\vec{s}_s$ represent the spectral signatures of healthy and sick tomato plants
- $\vec{s}_{ph}$ and $\vec{s}_{ps}$ represent the spectral signatures of healthy and sick pepper plants
- $\vec{s}_{ah}$ and $\vec{s}_{as}$ represent the spectral signatures of healthy and sick avocado plants

Your measurement is now a linear combination of 6 possible spectral signatures:

$$\vec{s}_m = \alpha_1 \vec{s}_h + \beta_1 \vec{s}_s + \alpha_2 \vec{s}_{ph} + \beta_2 \vec{s}_{ps} + \alpha_3 \vec{s}_{ah} + \beta_3 \vec{s}_{as}. \quad (2)$$

Here $\alpha_1, \alpha_2, \alpha_3$ are the unknown weights of healthy tomato, pepper, and avocado plants respectively. $\beta_1, \beta_2, \beta_3$ are the unknown weights of sick tomato, pepper, and avocado plants respectively. **Is it possible to uniquely determine the weights of healthy/sick tomatoes, peppers, and avocados from your measurement in equation 2? Why or why not? Show your work and justify your answer.**

6. Cross-correlation (28 points)

We are building our own Acoustic Positioning System.

NOTE: The signatures $\vec{s}_1$, $\vec{s}_2$ in each sub-part are different; each prompt is independent from the others.

- (a) (6 points) We have two signatures/gold codes of length-5, given by $\vec{s}_1$ and $\vec{s}_2$ as in Figure 11. So far we have numerically computed their linear cross-correlation $\text{Corr}_{\vec{s}_1}(\vec{s}_2)$, yet a few entries have been tragically lost! Fortunately we can compute these omitted terms by hand. **Please compute the missing cross-correlation values at shifts $k = -1$ and $k = +2$. Show your work and justify your answer.**

$$\vec{s}_1 = \begin{bmatrix} +1 \\ 0 \\ -1 \\ 0 \\ +1 \end{bmatrix} \quad \vec{s}_2 = \begin{bmatrix} +1 \\ +1 \\ 0 \\ -1 \\ +1 \end{bmatrix}.$$

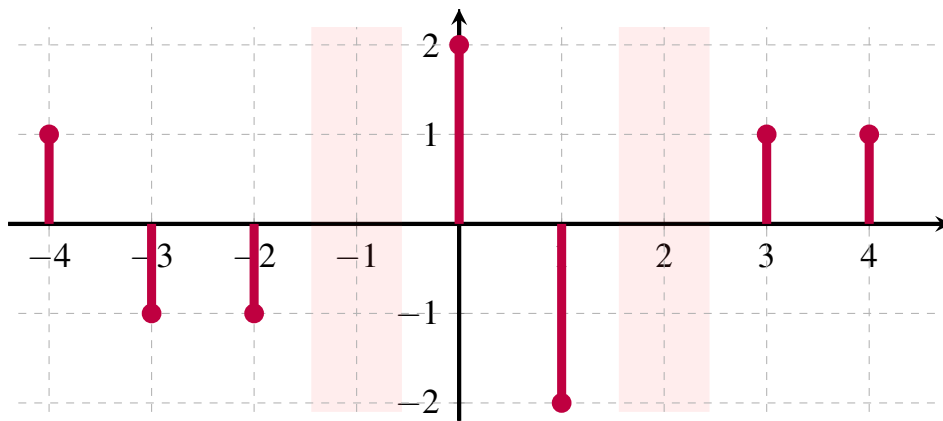


Figure 11: Linear cross-correlation plot of the two signals $\text{Corr}_{\vec{s}_1}(\vec{s}_2)$. The x-axis represents the shift.

- (b) (4 points) We are trying out some new codes $\vec{s}_1$ and $\vec{s}_2$. We only know that the codes are normalized ($\langle \vec{s}_1, \vec{s}_1 \rangle = 1$, $\langle \vec{s}_2, \vec{s}_2 \rangle = 1$) and their inner-product is $\langle \vec{s}_1, \vec{s}_2 \rangle = 0.3$. During our test we have received the signal $\vec{r} = \frac{1}{2}\vec{s}_1 + \frac{1}{3}\vec{s}_2$. Without knowing any more information about our codes, **compute $\text{Corr}_{\vec{r}}(\vec{s}_1)$ at the shift $k = 0$. Show your work and justify your answer.**

- (c) (4 points) We again have two new signals $\vec{s}_1$ and $\vec{s}_2$, and are now given the plot of $\text{Corr}_{\vec{s}_1}(\vec{s}_2)$ as shown in Figure 12. Our receiver identified a signal $\vec{r}$ which we know to be related to the code $\vec{s}_2$ by some scaling, shifting, and/or reflection. However, we only know the linear cross-correlation $\text{Corr}_{\vec{s}_1}(\vec{r})$ as shown in Figure 13. Can you express $\vec{r}$ in terms of $\vec{s}_2$? Show your work and justify your answer.

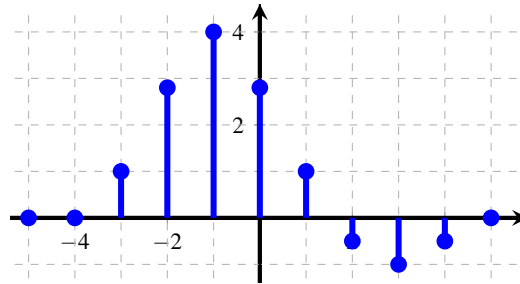


Figure 12: Linear cross-correlation plots for $\text{Corr}_{\vec{s}_1}(\vec{s}_2)$.

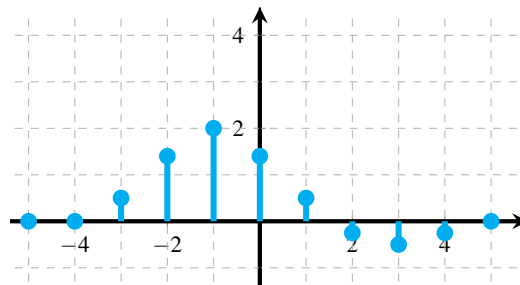


Figure 13: Linear cross-correlation plots for $\text{Corr}_{\vec{s}_1}(\vec{r})$.

- (d) (4 points) With a little effort we managed to create two *good* gold codes of length 100, $\vec{s}_1$ and $\vec{s}_2$. The linear cross-correlation of $\vec{s}_1$ and $\vec{s}_2$ is small at all shifts while the autocorrelation of each signal is also small, except at shift $k = 0$. We receive our first signal $\vec{r}$ which we know to be a combination of both codes

$$\vec{r}[n] = \vec{s}_1[n - k_1] + \vec{s}_2[n - k_2]. \quad (3)$$

The linear cross-correlation $\text{Corr}_{\vec{r}}(\vec{s}_1)$ has been computed and plotted in Figure 14, and similarly $\text{Corr}_{\vec{r}}(\vec{s}_2)$ is plotted in Figure 15. **Determine the shifts for $\vec{s}_1$ and $\vec{s}_2$ in the received signal $\vec{r}$, i.e. solve for k_1 and k_2 in equation (3). Explain your answer.**

Note: Don't worry too much about identifying the exact value for k_1 and k_2 . As long as your answer is reasonable, you will receive full credit.

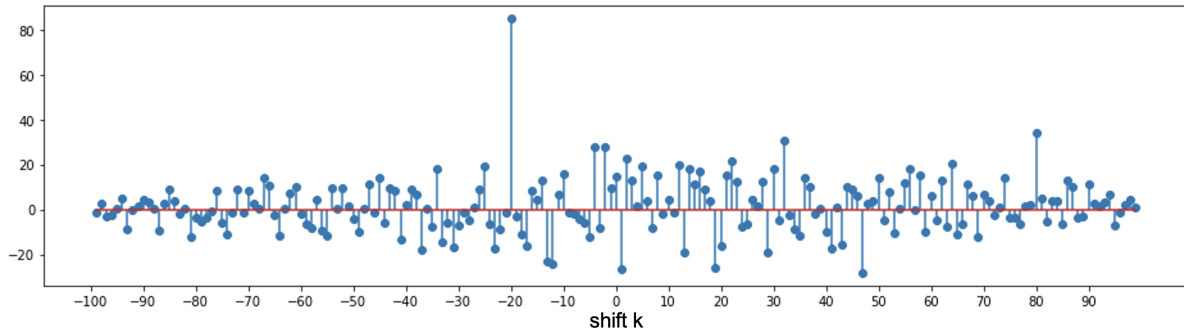


Figure 14: Linear cross-correlation plots for $\text{Corr}_{\vec{r}}(\vec{s}_1)$.

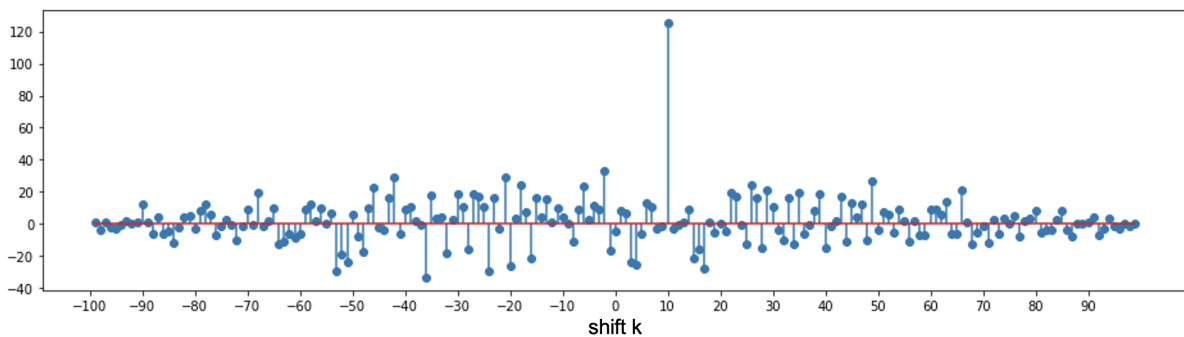


Figure 15: Linear cross-correlation plots for $\text{Corr}_{\vec{r}}(\vec{s}_2)$.

- (e) (4 points) It appears that making codes orthogonal to each other improves the robustness of our Acoustic Positioning System. Knowing this, we want to use our knowledge of projections to write our first code as $\vec{s}_1 = \vec{a} + \vec{b}$, where $\langle \vec{b}, \vec{s}_2 \rangle = 0$ and $\vec{a} = \alpha \vec{s}_2$ (for some constant α) as illustrated in Figure 16. **Compute α and $\vec{b}$ in terms of $\vec{s}_1$ and $\vec{s}_2$. Show your work and justify your answer.**

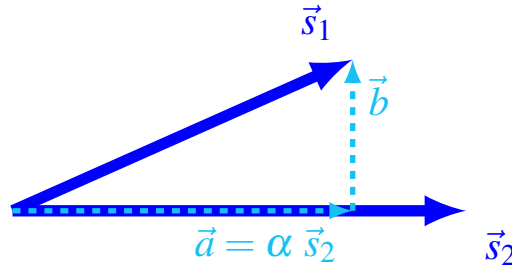


Figure 16: 2D figure of $\vec{s}_1 = \vec{a} + \vec{b}$.

- (f) (6 points) After optimizing two orthogonal codes $\vec{s}_1$ and $\vec{s}_2$ (i.e. $\langle \vec{s}_1, \vec{s}_2 \rangle = 0$), we would next like to include another code $\vec{s}_3$ and make it orthogonal to $\vec{s}_1$ and $\vec{s}_2$. We can start by writing $\vec{s}_3$ as $\vec{s}_3 = \vec{a} + \vec{b}$, such that $\vec{a}$ belongs to the $\text{span}\{\vec{s}_1, \vec{s}_2\}$ and $\vec{b}$ is orthogonal to $\text{span}\{\vec{s}_1, \vec{s}_2\}$, i.e. $\langle \vec{b}, \vec{s}_1 \rangle = 0$ and $\langle \vec{b}, \vec{s}_2 \rangle = 0$. **Use the idea of projections to write both $\vec{a}$ and $\vec{b}$ in terms of $\vec{s}_1$, $\vec{s}_2$, and $\vec{s}_3$, and inner-products thereof. (For full credit your final answer may not contain matrices nor matrix-vector products).** Show your work and justify your answer.

7. Warm for the Holidays (14 points)

Winter is coming, and both you and your roommate are in desperate need of electric heating eye pads to avoid overly dry eyes this holiday. Tragically the circuit for your eye pads broke, yet fortunately you've taken EECS16A and have come up with a clever fix by designing a voltage divider and a comparator circuit!

- (a) (4 points) First you build a circuit that converts temperature change to voltage change. Your design is shown in Fig. 17. In your design you use two temperature dependent resistors, whose values are given by $R_0 + \alpha T$, $R_0 - \alpha T$, where R_0 is the resistor value at 0 degrees centigrade, α is a thermal coefficient, and T is the temperature of your eye pads.

What is the temperature dependent output voltage, V_T , of this circuit, as a function of V_s , R_0 , α , and T ? Is V_T a linear function of T ? Clearly show all your work.

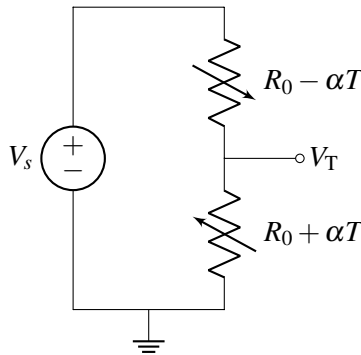


Figure 17: Temperature Sensing Circuit

- (b) (4 points) We want to use a comparator to turn the heat ON and OFF, and you set up the circuit in Fig. 18. You process the V_T to make $V_{in} = (1 - \frac{T}{T_0})[\text{Volts}]$, where $T_0 = 30^\circ\text{C}$. The heat will turn on when $V_{out} = V_{DD}$. **For what range of temperatures, T , is $V_{out} = V_{DD}$? Give your answer in terms of $^\circ\text{C}$.** Clearly show all your work.

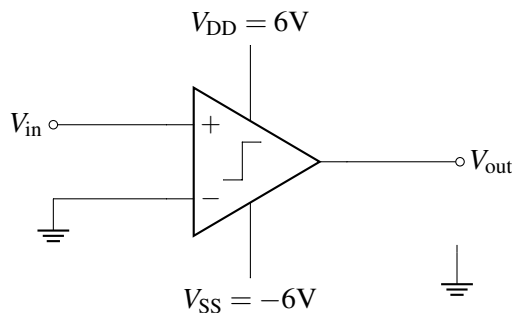


Figure 18: First attempt eye-pad control circuit.

- (c) (6 points) Your TA, Moses, points out that just using the circuit in Figure 18 will cause your heat to turn ON and OFF due to very small fluctuations. Instead, he suggests analyzing the following circuit in Figure 19. **Find the voltage u_+ at the positive terminal of the comparator, as a function of V_{out} , R_1 , R_2 , and V_{ref} .** Clearly show all your work.

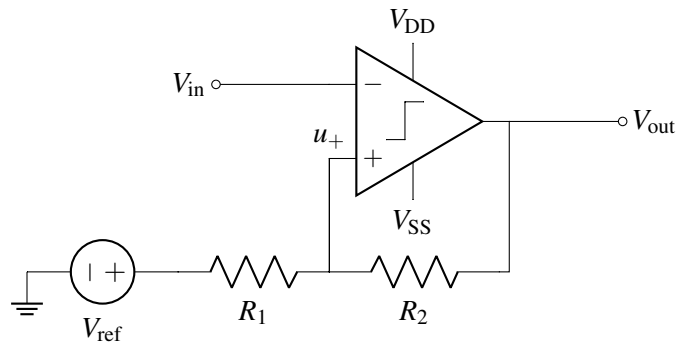


Figure 19: Proposed eye-pad control circuit.

8. Least Squares for Robotics (16 points)

Robots rely on sensors for understanding their environment and navigating in the real world. These sensors must be calibrated to ensure accurate measurements, which we explore in this problem.

(a) (3 points) Your robot is equipped with two forward-facing sensors – a radar and camera.

However, the sensors are placed with an offset (i.e. a gap) of ℓ in meters (m), as depicted in Fig. 20, and you want to find its value. The radar returns a range ρ in meters (m) and heading angle θ in radians (rad) with respect to the object. In contrast, the camera only returns an angle, ϕ in radians (rad), with respect to the object.

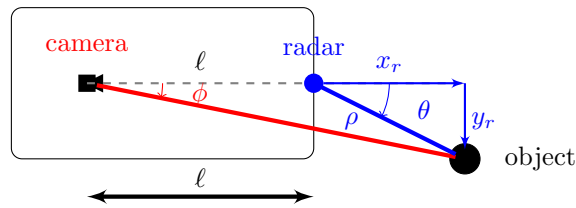


Figure 20: Sensor Placement and Offset ℓ .

These relationships are summarized by the following sensor model, where x_r and y_r are the Cartesian coordinates of the object with respect to the radar:

$$x_r = \rho \cos(\theta), \quad (4)$$

$$y_r = \rho \sin(\theta), \quad (5)$$

$$\tan(\phi) = \frac{y_r}{x_r + \ell}. \quad (6)$$

Assuming $\phi \neq 0$, use equations (4), (5), (6) to express ℓ in terms of ρ , θ , and ϕ .

- (b) (5 points) Often it is difficult to precisely identify the value of ℓ . To learn the value of ℓ you decide to take a series of measurements. In particular, you take N measurements and get the equations:

$$a\ell + e_i = b_i$$

for $1 \leq i \leq N$. Here $a \neq 0$ is a fixed and known constant. Each b_i represents your i^{th} measurement and e_i represents the error in your measurement. While you know all of the b_i values, you do not know the error values e_i .

We can write this equation in a vector format as:

$$\mathbf{A}\ell + \vec{e} = \vec{b},$$

where $\mathbf{A} = \begin{bmatrix} a \\ \vdots \\ a \end{bmatrix}$, $\vec{e} = \begin{bmatrix} e_1 \\ \vdots \\ e_N \end{bmatrix}$, $\vec{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_N \end{bmatrix}$.

In this simple 1-D case, the least squares solution is a scaled version of the average of $\{b_i\}_{i=1}^N$.

Find the best estimate for ℓ , denoted as $\hat{\ell}$, using least squares. Simplify your expression and express $\hat{\ell}$ in terms of a , b_i , and N . Your answer may not include any vector notation.

Note: A is a vector and not a matrix.

- (c) (8 points) Now we turn to the task of controlling the robot's velocity and acceleration, which is a key requirement for navigation.

We use the following model for the robot, which describes how the velocity and acceleration of the robot changes with timestep k :

$$\begin{bmatrix} v[k+1] \\ a[k+1] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v[k] \\ a[k] \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} j[k],$$

where

- k is the timestep;
- $v[k]$ is the velocity state at timestep k ;
- $a[k]$ is the acceleration state at timestep k ;
- $j[k]$ is the jerk (derivative of acceleration) control input at timestep k .

We start at a known initial state $\begin{bmatrix} v[0] \\ a[0] \end{bmatrix}$, and we want to find $j[0]$ to set $\begin{bmatrix} v[1] \\ a[1] \end{bmatrix}$ as close to $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ as possible. For this, we minimize:

$$E = \left\| \begin{bmatrix} v[1] \\ a[1] \end{bmatrix} \right\|^2.$$

Find the best estimate for the optimal choice of jerk, $\hat{j}[0]$, by using least squares method to minimize E . Express your solution in terms of $v[0]$ and $a[0]$. Show your work.

Hint: Rewrite E in terms of $j[0]$ and other relevant terms.

9. Proof (10 points)

Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$. The eigenvalues and eigenvectors of $\mathbf{A}$ are given by $(\alpha_1, \vec{v}_1), (\alpha_2, \vec{v}_2), \dots, (\alpha_n, \vec{v}_n)$, where all the α_i , $1 \leq i \leq n$, are distinct. Similarly the eigenvalues and eigenvectors of $\mathbf{B}$ are given by $(\beta_1, \vec{v}_1), (\beta_2, \vec{v}_2), \dots, (\beta_n, \vec{v}_n)$, where all the β_i , $1 \leq i \leq n$, are distinct.

NOTE: $\mathbf{A}, \mathbf{B}$ have identical eigenvectors.

Prove that:

$$\mathbf{A} \mathbf{B} \vec{x} = \mathbf{B} \mathbf{A} \vec{x},$$

for any vector $\vec{x} \in \mathbb{R}^n$.

10. Survey

Below we have attached a google form link for you to fill out. This is a demographic survey that will help us understand more about you and your background. Getting to know you will only help us improve your experience in EECS16B.

[Demographic Survey - Google Form](#)

11. Homework Process and Study Group

Citing sources and collaborators are an important part of life, including being a student!

We also want to understand what resources you find helpful and how much time homework is taking, so we can change things in the future if possible.

(a) **What sources (if any) did you use as you worked through the homework?**

(b) **If you worked with someone on this homework, who did you work with?**

List names and student ID's. (In case of homework party, you can also just describe the group.)

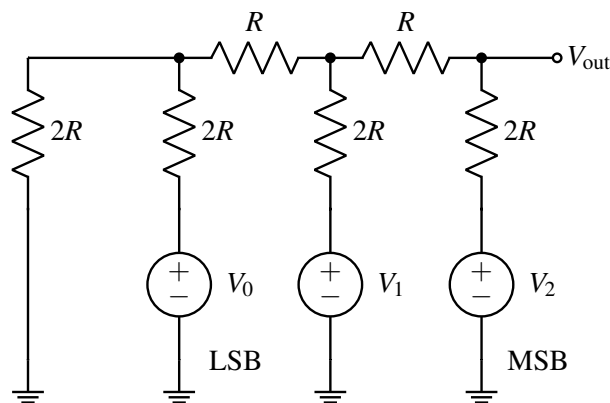
EECS 16B Designing Information Devices and Systems II

UC Berkeley

Homework 2

1. Digital-Analog Converter

A digital-analog converter (DAC) is a circuit for converting a digital representation of a number (binary) into a corresponding analog voltage. In this problem, we will consider a DAC made out of resistors only (resistive DAC) called the R - $2R$ ladder. Here is the circuit for a 3-bit resistive DAC.



Let $b_0, b_1, b_2 = \{0, 1\}$ (that is, either 1 or 0), and let the voltage sources $V_0 = b_0 V_{DD}$, $V_1 = b_1 V_{DD}$, $V_2 = b_2 V_{DD}$, where V_{DD} is the supply voltage.

As you may have noticed, (b_2, b_1, b_0) represents a 3-bit binary (unsigned) number where each of b_i is a binary bit. b_0 is the least significant bit (LSB) and b_2 is the most significant bit (MSB). We will now analyze how this converter functions.

- (a) If $b_2, b_1, b_0 = 1, 0, 0$, what is V_{out} ? Express your answer in terms of V_{DD} .
- (b) If $b_2, b_1, b_0 = 0, 1, 0$, what is V_{out} ? Express your answer in terms of V_{DD} .
- (c) If $b_2, b_1, b_0 = 0, 0, 1$, what is V_{out} ? Express your answer in terms of V_{DD} .
- (d) If $b_2, b_1, b_0 = 1, 1, 1$, what is V_{out} ? Express your answer in terms of V_{DD} .
- (e) Finally, solve for V_{out} in terms of V_{DD} and the binary bits b_2, b_1, b_0 .
- (f) Explain how your results above show that the resistive DAC converts the 3-bit binary number (b_2, b_1, b_0) to the output analog voltage V_{out} .

2. Complex Numbers

A common way to visualize complex numbers is to use the complex plane. Recall that a complex number z is often represented in Cartesian form.

$$z = x + jy \text{ with } \operatorname{Re}\{z\} = x \text{ and } \operatorname{Im}\{z\} = y$$

See Figure 1 for a visualization of z in the complex plane.

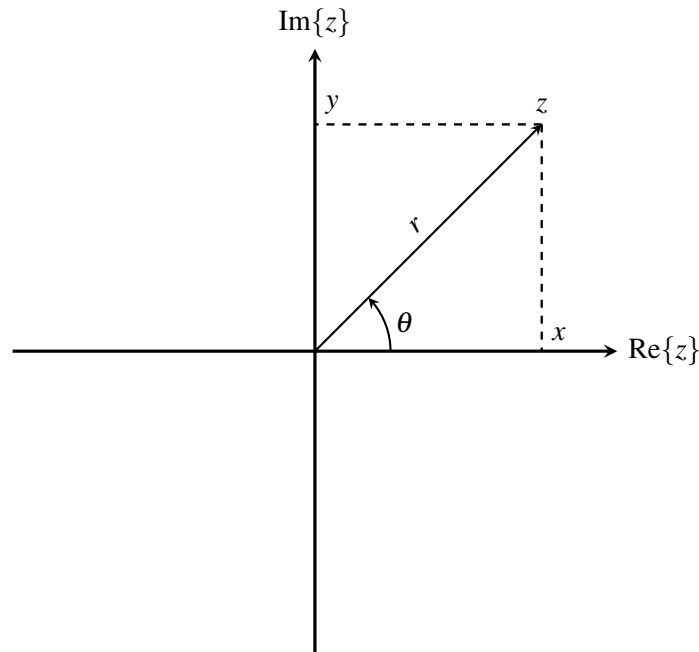


Figure 1: Complex Plane

In this question, we will derive the polar form of a complex number and use this form to make some interesting conclusions.

- (a) **Calculate the length of z in terms of x and y as shown in Figure 1.** This is the magnitude of a complex number and is denoted by $|z|$ or r .

(Hint: Use the Pythagorean theorem.)

- (b) **Represent x , the real part of z , and y , the imaginary part of z , in terms of r and θ .**

- (c) **Substitute for x and y in z .** Use Euler's identity¹ $e^{j\theta} = \cos \theta + j \sin \theta$ to **conclude that,**

$$z = re^{j\theta}.$$

- (d) In the complex plane, **sketch the set of all the complex numbers such that $|z| = 1$. What are the z values where the sketched figure intersects the real axis and the imaginary axis?**

- (e) If $z = re^{j\theta}$, **prove that $\bar{z} = re^{-j\theta}$.** Recall that the complex conjugate of a complex number $z = x + jy$ is $\bar{z} = x - jy$.

¹also known as de Moivre's Theorem.

(f) **Show (by direct calculation) that,**

$$r^2 = z\bar{z}.$$

3. Existence and uniqueness of solutions to differential equations

Let's show that if any function x satisfies

$$\frac{d}{dt}x(t) = \alpha x(t) \quad (1)$$

as well as

$$x(0) = x_0, \quad (2)$$

then it is unique: if y is *any function* that meets these two criteria then $x = y$.

In order to do this, we will first verify that a solution exists. Then we will compare it to a hypothetical alternative solution—and our goal will be to establish that these two solutions are equal.

- (a) **Verify that $x_d(t) = x_0 e^{\alpha t}$ satisfies (1) and (2).** (For this proof, x_d will be the “reference solution” against which alternates will be compared.)
- (b) To show that this solution is in fact unique, we need to consider a hypothetical $y(t)$ that also satisfies (1) and (2).

Our goal is to show that $y(t) = x(t)$ for all $t \geq 0$. (The domain $t \geq 0$ is where we have defined the conditions (1) and (2). Outside of that domain, we don't have any constraints.)

How can we show that two things are equal? In the past, you have probably shown that two quantities or functions are equal by starting with one of them, and then manipulating the expression for it using valid substitutions and simplifications until you get the expression for the other one. However, here, we don't have an expression for $y(t)$ so that style of approach won't work.

In such cases, we basically have a couple of basic ways of showing that two things are the same.

- Take the difference of them, and somehow argue that it is 0.
- Take the ratio of them, and somehow argue that it is 1.

We will follow the ratio approach in this problem. First assume that $x_0 \neq 0$. In this case, we are free to define $z(t) = \frac{y(t)}{x_d(t)}$ since we are dividing by something other than zero.

What is $z(0)$?

- (c) **Take the derivative $\frac{d}{dt}z(t)$ and simplify using (1) and what you know about the derivative of $x_d(t)$.** (HINT: The quotient rule for differentiation might be helpful since a ratio is involved.)

You should see that this derivative is always 0 and hence $z(t)$ does not change. **What does that imply for y and x_d ?**

- (d) At this point, we have shown uniqueness in most cases. Just one special case is left: $x_0 = 0$. The ratio technique omitted this case, because as $x_d(t) = 0$, x_d cannot be the denominator of a fraction.

To complete our proof we must show that if $x_0 = 0$, then $y(t) = 0$ for all t , and we will do so by assuming that $y(t)$ is not identically 0 for $t > 0$ —that is, at some $t_0 > 0$ $y(t_0) = k \neq 0$.

From (2), we know that $y(0) = 0$. In the subsequent sub parts, we will try to work backwards in time from the point $t = t_0$ to $t = 0$ and conclude that y violates (2) if $y(t_0) \neq 0$.

Apply the change of variables $t = t_0 - \tau$ to (1) to get a new differential equation for $\tilde{x}(\tau) = x(t_0 - \tau)$ that specifies how $\frac{d}{d\tau}\tilde{x}(\tau)$ must relate to $\tilde{x}(\tau)$. This should hold for $-\infty < \tau \leq t_0$.

- (e) Because the previous part resulted in a differential equation of a form for which we have already proved uniqueness for the case of nonzero initial condition, and since $\tilde{y}(0) = y(t_0) = k \neq 0$, we know what $\tilde{y}(\tau)$ must be. **Write the expressions for $\tilde{y}(\tau)$ for $\tau \in [0, t_0]$ and what that implies for $y(t)$ for $t \in [0, t_0]$.**

(f) **Evaluate $y(0)$ and argue that this is a contradiction for the specified initial condition (2).**

Consequently, such a $y(t)$ cannot exist and only the all zero solution is permitted — establishing uniqueness in this case of $x_0 = 0$ as well.

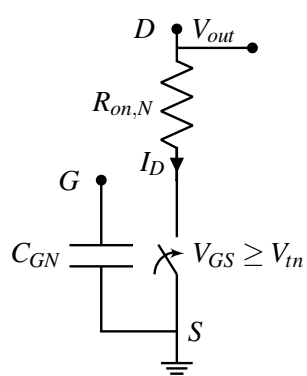
(g) Explain in your own words why it matters that solutions to these differential equations are unique.

Although we gave you lots of guidance in this problem, we hope that you can internalize this way of thinking.

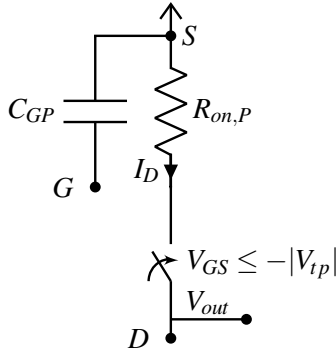
This elementary approach to proving the uniqueness of solutions to differential equations works for the kinds of linear differential equations that we will tend to encounter in EECS16B. For more complicated nonlinear differential equations, further conditions are required for uniqueness (appropriate continuity and differentiability) and proofs can be found in upper-division mathematics courses on differential equations when you study the Picard-Lindelöf theorem. (It involves looking at the magnitude of the difference of the two hypothetical solutions and showing this has to be arbitrarily small and hence zero. However, the basic elementary case we have established here can be viewed as a building block — the quotient rule gets invoked in the appropriate place, etc. The additional ingredients that are out-of-scope for lower-division courses are fixed-point theorems — which you can think of as more general siblings of the intermediate-value theorem you saw in basic calculus.)

4. Transistor Switch Model

We can improve our resistor-switch model of the transistor by adding in a gate capacitance. In this model, the gate capacitances C_{GN} and C_{GP} represent the lumped physical capacitance present on the gate node of all transistor devices. This capacitance is important as it determines the delay of a transistor logic chain.



a NMOS Transistor Resistor-switch-capacitor model



b PMOS Transistor Resistor-switch-capacitor model. Note we have drawn this so that it aligns with the inverter.

You have two CMOS inverters made from NMOS and PMOS devices. Both NMOS and PMOS devices have an “on resistance” of $R_{on,N} = R_{on,P} = 1 \text{ k}\Omega$, and each has a gate capacitance (input capacitance) of $C_{GN} = C_{GP} = 1 \text{ fF}$ (femto-Farads = 10^{-15}). We assume the “off resistance” (the resistance when the transistor is off) is infinite (*i.e.*, the transistor acts as an open circuit when off). The supply voltage V_{DD} is 1V. The two inverters are connected in series, with the output of the first inverter driving the input of the second inverter (Figure 3).

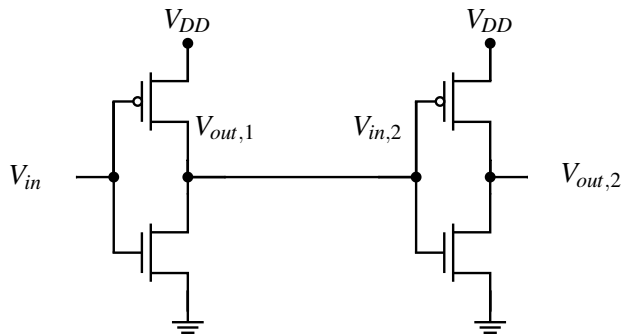


Figure 3: CMOS Inverter chain

- Assume the input to the first inverter has been low ($V_{in} = 0 \text{ V}$) for a long time, and then switches at time $t = 0$ to high ($V_{in} = V_{DD}$). **Draw a simple RC circuit and write a differential equation describing the output voltage of the first inverter ($V_{out,1}$) for time $t \geq 0$.** Don't forget that the second inverter is “loading” the output of the first inverter — you need to think about both of them.
- Given the initial conditions in part (a), **solve for $V_{out,1}(t)$.**
- Sketch the output voltage of the first inverter, showing clearly (1) the initial value, (2) the initial slope, (3) the asymptotic value, and (4) the time that it takes for the voltage to decay to roughly 1/3 of its initial value.**

- (d) A long time later, the input to the first inverter switches low again.

Solve for $V_{out,1}(t)$.

Sketch the output voltage of the first inverter ($V_{out,1}$), showing clearly (1) the initial value, (2) the initial slope, and (3) the asymptotic value.

- (e) For each complete input cycle described above ($V_{in} = 0V \rightarrow 1V \rightarrow 0V$), **how much charge is pulled out of the power supply?** Give both a symbolic and numerical answer. **Consider only the charge needed to charge up the $V_{out,1}$ node.**

5. Group formation survey

Participating in study groups is completely optional. In the following survey, please choose an option that is best for you. Note that submitting this form is required to get full credit for this HW question.

[Group Formation Survey - Google Form](#)

6. Homework Process and Study Group

Citing sources and collaborators are an important part of life, including being a student!

We also want to understand what resources you find helpful and how much time homework is taking, so we can change things in the future if possible.

(a) **What sources (if any) did you use as you worked through the homework?**

(b) **If you worked with someone on this homework, who did you work with?**

List names and student ID's. (In case of homework party, you can also just describe the group.)

Contributors:

- Edward Wang.
- Siddharth Iyer.
- Anant Sahai.
- Nikhil Shinde.
- Kris Pister.
- Regina Eckert.
- Sidney Buchbinder.
- Ayan Biswas.
- Gaoyue Zhou.

EECS 16B Designing Information Devices and Systems II

UC Berkeley

Homework 03

1. Reading Lecture Notes

Staying up to date with lectures is an important part of the learning process in this course. Here are links to the notes that you need to read for this week: [Note 2](#) and [Note 3A](#)

- (a) How do we deal with piecewise constant inputs as introduced in the notes?
- (b) What conclusions do we get after approximating any function $u(t)$ as being piecewise constant over fixed interval widths Δ ?

2. Simple scalar differential equations driven by an input

In class, you learned that the solution for $t \geq 0$ to the simple scalar first-order differential equation

$$\frac{d}{dt}x(t) = \lambda x(t) \tag{1}$$

with initial condition

$$x(t=0) = x_0 \tag{2}$$

is given for $t \geq 0$ by

$$x(t) = x_0 e^{\lambda t}. \tag{3}$$

In an earlier homework, you proved that these solutions are *unique*— that is, that $x(t)$ of the form in (3) are the only possible solutions to the equation (1) with the specified initial condition (2).

In this question, we will analyze differential equations with inputs and prove that their solutions are unique.

In particular, we consider the scalar differential equation

$$\frac{d}{dt}x(t) = \lambda x(t) + u(t) \tag{4}$$

where $u(t)$ is a known function of time from $t = 0$ onwards.

- (a) Suppose that you are given an $x_g(t)$ that satisfies both (2) and (4) for $t \geq 0$.

Show that if $y(t)$ also satisfies (2) and (4) for $t \geq 0$, then it must be that $y(t) = x_g(t)$ for all $t \geq 0$.

(HINT: You already used ratios in an earlier HW to prove that two things were necessarily equal. This time, you might want to use differences. Be sure to leverage what you already proved earlier instead of having to redo all that work.)

- (b) Suppose that the given $u(t)$ starts at $t = 0$ (it is zero before that) and is a nicely integrable function (feel free to assume bounded and continuously differentiable with bounded derivative — whatever conditions you assumed in your calculus course when considering integration and the fundamental theorem of calculus). Let

$$x_c(t) = x_0 e^{\lambda t} + \int_0^t u(\tau) e^{\lambda(t-\tau)} d\tau \tag{5}$$

for $t \geq 0$.

Show that the $x_c(t)$ defined in (5) indeed satisfies (4) and (2).

Note: the τ here in (5) is just a dummy variable of integration. We could have used any letter for that local variable. We just used τ because it visually reminds us of t while also looking different. If you think they look too similar in your handwriting, feel free to change the dummy variable of integration to another symbol of your choice.

(HINT: Remember the fundamental theorem of calculus that you proved in your calculus class and manipulate the expression in (5) to get it into a form where you can apply it along with other basic calculus rules.)

(c) **Use the previous part to get an explicit expression for $x_c(t)$ for $t \geq 0$ when $u(t) = e^{st}$ for some constant s , when $s \neq \lambda$ and $t \geq 0$.**

(d) **Similarly, what is $x_c(t)$ for $t \geq 0$ when $u(t) = e^{\lambda t}$ for $t \geq 0$.**

(HINT: Don't worry if this seems too easy.)

3. Uniqueness Counterexample

This problem explores an example of a differential equation that does not have a unique solution. The purpose is to show that uniqueness cannot always be assumed. There is a reason we are making you prove uniqueness to trust solutions.

Along the way, this problem will also show you a heuristic way to guess the solutions to differential equations that is often called “separation of variables.” The advantage of the separation of variables technique is that it can often be helpful in systematically coming up with guesses for nonlinear differential equations. However, as with any technique for guessing, it is not a proof and the guess definitely needs to be checked and uniqueness verified before proceeding.

The idea of separation-of-variables is to pretend that $\frac{d}{dt}x(t) = \frac{dx}{dt}$ is a ratio of quantities rather than what it is — a shorthand for taking the derivative of the function $x(\cdot)$ with respect to its single argument, and then writing the result in terms of the free variable “ t ” for that argument. This little bit of make-believe (sometimes euphemistically called “an abuse of notation”) allows one the freedom to do calculations.

To demonstrate, let’s do this for a case where we know the correct solution: $\frac{d}{dt}x(t) = \lambda x(t)$. This is how a separation-of-variables approach would try to get a guess:

$$\frac{d}{dt}x(t) = \lambda x(t) \tag{6}$$

$$\frac{dx}{dt} = \lambda x \tag{7}$$

$$\frac{dx}{x} = \lambda dt \text{ separating variables to sides} \tag{8}$$

$$\int \frac{dx}{x} = \int \lambda dt \text{ integrating both sides} \tag{9}$$

$$\ln x + C_1 = \lambda t + C_2 \tag{10}$$

$$x(t) = Ke^{\lambda t} \text{ exponentiating both sides and folding constants} \tag{11}$$

With the above guess obtained, $x(t) = Ke^{\lambda t}$ can be plugged in and seen to solve the original differential equation because the steps above are vaguely reversible, if a bit hallucinatory in nature. Then of course, a uniqueness proof is required, but you did that in the previous homework.

To see why this technique is a bit fraught, we will consider the following nonlinear differential equation involving a third root. (If we had more time in making this problem, we would have showed you how this sort of differential equation can arise from a toy physical setting of a inverted pyramidal container that had $x(t)$ liters of water in it, where the rate of water being poured in is proportional to the height of the water $x^{\frac{1}{3}}$. This fractional power arises since volume is a cubic quantity while the water is being poured in at a rate governed by a one-dimensional quantity of length. Similar equations can arise in microfluidic dynamics.)

Anyway, consider the differential equation

$$\frac{d}{dt}x(t) = \alpha x^{\frac{1}{3}}(t) \tag{12}$$

with the initial condition

$$x(0) = 0.$$

Let’s take the “separation-of-variables pill” and see what trip it takes us on:

$$\frac{d}{dt}x(t) = \alpha x^{\frac{1}{3}}(t) \quad (13)$$

$$\frac{dx}{dt} = \alpha x^{\frac{1}{3}} \quad (14)$$

Due to the “abuse of notation,” we can write it as follows

$$x^{-\frac{1}{3}}dx = \alpha dt \quad (15)$$

$$\int x^{-\frac{1}{3}}dx = \int \alpha dt \quad (16)$$

$$\frac{3}{2}x^{\frac{2}{3}} + C_1 = \alpha t + C_2 \quad (17)$$

$$x(t) = \left(\frac{2}{3}\alpha t + C_3\right)^{\frac{3}{2}} \quad (18)$$

That didn’t seem like too wild a ride. Let’s see what rabbit hole we have actually landed in.

- (a) Given our separation-of-variables based calculation, let us guess a solution of the form

$$x(t) = \left(\frac{2}{3}\alpha t + c\right)^{\frac{3}{2}}$$

Show that this is a solution to the differential equation (12), and find the c that satisfies the initial condition.

- (b) Let us guess a second solution:

$$x(t) = 0 \quad (19)$$

Show that this new guess also satisfies (12), and the initial condition ($x(0) = 0$).

- (c) **Show that any solution of the form**

$$x(t) = \begin{cases} 0, & \text{if } t < t_0 \\ \left(\frac{2}{3}\alpha(t - t_0)\right)^{\frac{3}{2}}, & t \geq t_0 \end{cases} \quad (20)$$

also satisfies (12) and the initial condition $x(0) = 0$, for any $t_0 > 0$.

So this actually has *infinitely* many solutions.

- (d) One of the actual sufficient conditions for the uniqueness of solutions to differential equations of the form $\frac{d}{dt}x(t) = f(x(t))$ is that the function $f(x)$ be continuously differentiable with a bounded derivative at the initial condition $x(0)$ and everywhere that the solution purports to go. (You will understand the importance of this condition and where it comes from better when we are in Module 2 of 16B.)

Does this differential equation problem satisfy this condition that would let us trust guessing and checking?

- (e) In this particular case, there is a little bit of a warning in the separation-of-variables trip (you can consider it a dream sequence if you’d like — it has a logic to it, just not a logic that you can rely upon in the real world). **Explain why (15) might be a bit problematic in this case.**
- (f) To see that separation of variables can actually give you reasonable guesses even for equations with interesting solutions, try to use it to solve the following differential equation $\frac{d}{dt}x(t) = \frac{1}{2}x^2$ for initial condition $x(0) = 1$. **Sketch your solution. Describe anything interesting that happens. Does this problem satisfy the uniqueness conditions for $t \in [0, 1]$?**

4. Op-Amp Stability

In this question we will revisit the basic op-amp model that was introduced in EECS 16A and we will add a capacitance C_{out} to make the model more realistic (refer to figure 1). Now that we have the tools to do so, we will study the behavior of the op-amp in positive and negative feedback (refer to figure 2). Furthermore, we will begin looking at the integrator circuit (refer to figure 3) to see how a capacitor in the negative feedback can behave. In the next homework, you will see why it ends up being close to an integrator.

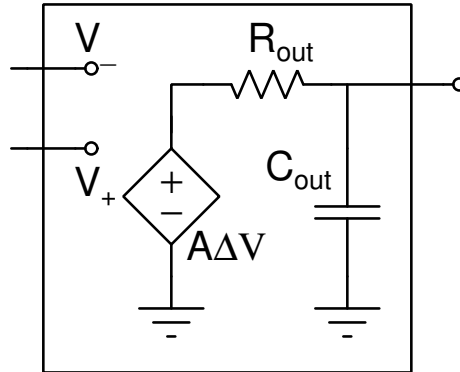


Figure 1: Op-amp model: $\Delta V = V_+ - V_-$

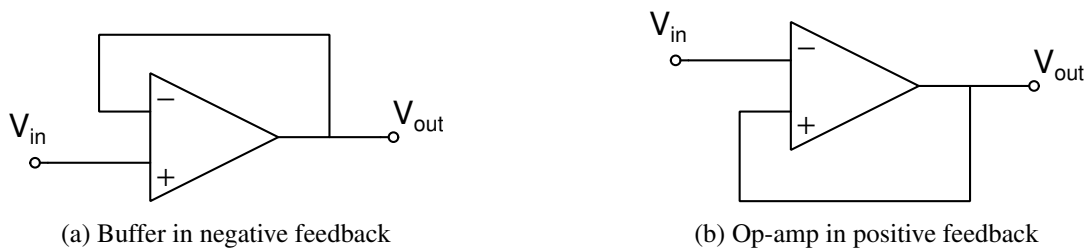


Figure 2: Op-amp in positive and negative feedback

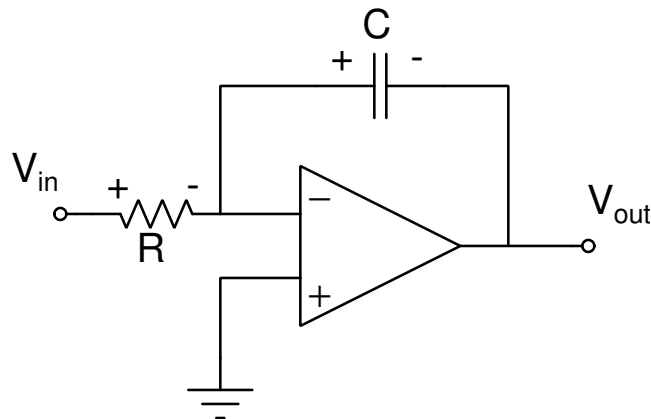


Figure 3: Integrator circuit

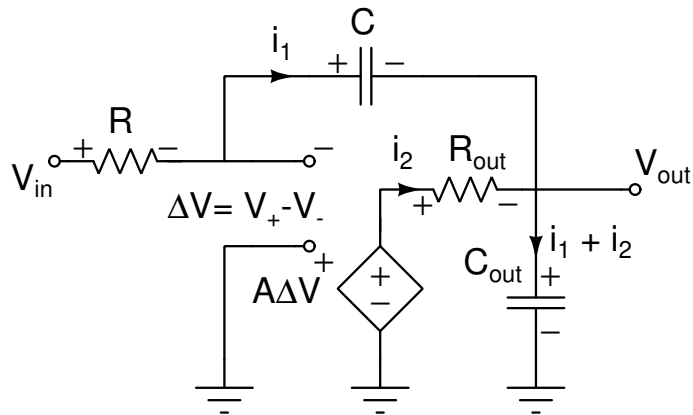


Figure 4: Integrator circuit with Op-amp model

- (a) Using the op-amp model in figure 1 and the buffer in negative-feedback configuration in figure 2(a), **draw a combined circuit**. Remember that $\Delta V = V_+ - V_-$, the voltage difference between the positive and negative labeled input terminals of the op-amp.
(HINT: Look at figure 4 to see how this was done for the integrator. That might help.)
Note: here, we have used the equivalent model for the op-amp gain. In more advanced analog circuits courses, it is traditional to use a controlled current source with a resistor in parallel instead.
- (b) Let's look at the op-amp in negative feedback. From our discussions in EECS 16A, we know that the buffer in figure 2(a) should work with $V_{out} \approx V_{in}$ by the golden rules. **Write a differential-equation for V_{out} by replacing the op-amp with the given model and show what the solution will be as a function of time for a static V_{in} . What does it converge to as $t \rightarrow \infty$?** Note: We assume the gain $A > 1$ for all parts of the question.
- (c) Next, let's look at the op-amp in positive feedback. We know that the configuration given in figure 2(b) is unstable and V_{out} will just rail. **Again, using the op-amp model in figure 1, show that V_{out} does not converge and hence the output will rail. For positive DC input $V_{in} > 0$, will V_{out} rail to the positive or negative side? Explain.**
- (d) For an ideal op-amp, we can assume that it has an infinite gain, i.e., $A \rightarrow \infty$. **Under these assumptions, show that the op-amp in negative feedback behaves as an ideal buffer, i.e., $V_{out} = V_{in}$.**
- (e) Let's extend our analysis to the integrator circuit shown in figure 3. Simplifying all the equations, we get a system of differential equations in two variables V_C and V_{out} , where V_C and V_{out} are the voltage drops across the capacitors C and C_{out} . **Fill in the missing term in the following matrix differential equation.**

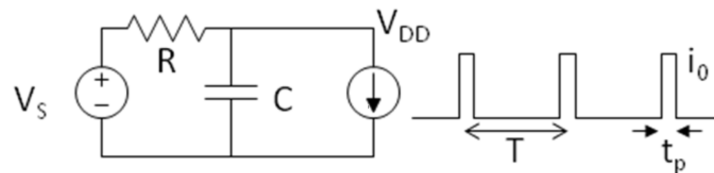
$$\frac{d}{dt} \begin{bmatrix} V_{out}(t) \\ V_C(t) \end{bmatrix} = \begin{bmatrix} -\left(\frac{A+1}{R_{out}C_{out}} + \frac{1}{RC_{out}}\right) & -\left(\frac{1}{RC_{out}} + \frac{A}{R_{out}C_{out}}\right) \\ -\frac{1}{RC} & ? \end{bmatrix} \begin{bmatrix} V_{out}(t) \\ V_C(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{RC_{out}} \\ \frac{1}{RC} \end{bmatrix} V_{in}(t) \quad (21)$$

(HINT: We picked an easier term to hide. You don't have to write out all the equations and do a lot of algebra to figure out what the missing term is.)

5. IC Power Supply

Digital integrated circuits (ICs) often have very non-uniform current requirements which can cause voltage noise on the supply lines. If one IC is adding a lot of noise to the supply line, it can affect the performance of other ICs that use the same power supply, which can hinder performance of the entire device. For this reason, it is important to take measures to mitigate, or “smooth out”, the power supply noise that each IC creates. A common way of doing this is to add a “supply capacitor” between each IC and the power supply. (If you look at a circuit board, and the supply capacitor is the small capacitor next to each IC.)

Here’s a simple model for a power supply and digital circuit:

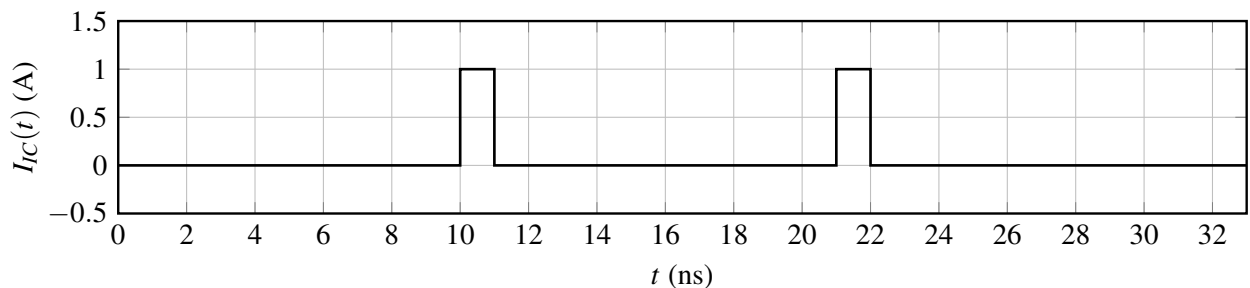


The current source is modeling the “spiky,” non-uniform nature of digital circuit current consumption. The resistor represents the sum of the source resistance of the supply and any wiring resistance between the supply and the load.

The capacitor is added to try to minimize the noise on V_{DD} . Assuming that $V_s = 3\text{V}$, $R = 1\Omega$, $i_0 = 1\text{A}$, $T = 10\text{ns}$, and $t_p = 1\text{ns}$.

- (a) Sketch the voltage V_{DD} vs. time for two periods assuming that $C = 0$. Assume the transient current drawn by IC behaves as follows.

Sketch of transient current drawn by IC



- (b) Give expressions for and sketch the voltage V_{DD} vs. time for two periods for each of three different capacitor values for C : 1pF , 1nF , $1\mu\text{F}$. ($1\text{pF} = 10^{-12}\text{F}$, $1\text{nF} = 10^{-9}\text{F}$, $1\mu\text{F} = 10^{-6}\text{F}$). You may assume $I_C = 0$ for a long time before $t = 0$. Feel free to use the script in the iPython notebook for plotting.
- (c) Launch the attached Jupyter notebook to interact with a simulated version of this IC power supply. Try to simulate the scenarios outlined in the previous parts. For one of these scenarios, keep the RC time constant fixed, but vary the relative value of R vs. C (e.g. compare $R = 1, C = 2 \times 10^{-9}$ to the case where $R = 2, C = 1 \times 10^{-9}$. You may use scientific notations $2\text{e-}9$ and $1\text{e-}9$ in the iPython notebook). **Is it better to have a lower R or lower C value for a fixed RC time constant when attempting to minimize supply noise? Give an intuitive explanation for why this might be the case.**

6. Homework Process and Study Group

Citing sources and collaborators are an important part of life, including being a student!

We also want to understand what resources you find helpful and how much time homework is taking, so we can change things in the future if possible.

(a) **What sources (if any) did you use as you worked through the homework?**

(b) **If you worked with someone on this homework, who did you work with?**

List names and student ID's. (In case of homework party, you can also just describe the group.)

Contributors:

- Anant Sahai.
- Nikhil Shinde.
- Sidney Buchbinder.
- Pavan Bhargava.
- Aditya Arun.
- Gaoyue Zhou.
- Ayan Biswas.
- Sanjit Batra.
- Kris Pister.
- Alex Devonport.

EECS 16B Designing Information Devices and Systems II

UC Berkeley

Homework 4

1. Reading Lecture Notes

Staying up to date with lectures is an important part of the learning process in this course. Here are links to the notes that you need to read for this week: Note 3A

- (a) Explain the process to solve a general vector differential equation $\frac{d}{dt}\vec{x} = A\vec{x}$ where $x \in \mathbb{R}^n$ and $\vec{x}(t_0) = \vec{x}_0$, including any necessary conditions.

2. Tracking Terry

Terry is a mischievous child, and his mother is interested in tracking him.

For this problem, the $\mathbb{R}^2$ standard basis vectors will be denoted by

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- (a) Terry texts his current location $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ with these coordinates in the basis $\begin{bmatrix} \vec{v}_1 & \vec{v}_2 \end{bmatrix}$. Write Terry's location in the standard basis in terms of $\vec{v}_1$ and $\vec{v}_2$.
- (b) Terry's friend tells you that Terry's location in the standard basis is $\begin{bmatrix} 8 \\ 9 \end{bmatrix}$. Determine the basis vectors he is using, or if it is impossible, explain why.
- (c) Terry's basis vectors get leaked to his mom on accident, so she knows they are

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and } \vec{v}_2 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

To hide his location, Terry wants to switch to a new coordinate system with the basis vectors

$$\vec{p}_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix} \text{ and } \vec{p}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

In order to do this, he needs a way to convert coordinates from the V basis to the P basis. Thus, find the matrix T such that if $\vec{x}_v$ is a location in V coordinates and $\vec{x}_p$ is the same location in P coordinates, then $\vec{x}_p = T\vec{x}_v$.

3. Eigenvectors and Diagonalization

- (a) Let A be an $n \times n$ matrix with n linearly independent eigenvectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$, and corresponding eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. Define V to be a matrix with $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ as its columns, $V = [\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n]$.
- Show that $AV = V\Lambda$, where $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$, a diagonal matrix with the eigenvalues of A as its diagonal entries.
 - Argue that V is invertible, and therefore, $A = V\Lambda V^{-1}$.
- (b) For a matrix A and a positive integer k , we define the exponent to be

$$A^k = \underbrace{A * A * \dots * A}_{k \text{ times}} \quad (1)$$

Let's assume that matrix A is diagonalizable with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, and corresponding eigenvectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ (i.e. the n eigenvectors are all linearly independent).

Show that A^k has eigenvalues $\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k$ and eigenvectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$. Conclude that A^k is diagonalizable.

4. Vector Differential Equations

In this problem, we consider ordinary differential equations which can be written in the following form

$$\frac{d}{dt}\vec{x}(t) = \begin{bmatrix} \frac{d}{dt}x_1(t) \\ \frac{d}{dt}x_2(t) \end{bmatrix} = A \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = A\vec{x} \quad (2)$$

where x_1, x_2 are variables depending on time t , and A is a 2×2 matrix with constant coefficients. We call (2) a vector differential equation.

- (a) Suppose we have a system of ordinary differential equations

$$\frac{dx_1}{dt} = 7x_1 - 8x_2 \quad (3)$$

$$\frac{dx_2}{dt} = 4x_1 - 5x_2 \quad (4)$$

Write this in the form of (2) and compute the eigenvalues of the A matrix.

- (b) Compute the eigenvectors of the matrix A . For consistency, assume that the smaller eigenvalue is λ_1 and the larger is λ_2 .
- (c) We now want to transform our current system to a new coordinate system in order to simplify our differential equation. What basis B should we use so that in the new coordinates $\vec{z} = B^{-1}\vec{x}$, the new A matrix is diagonal? Write out what this new system becomes in the z coordinates.
- (d) Solve the new system in the z coordinates, using the initial conditions that $x_1(0) = 1, x_2(0) = -1$.
- (e) Now convert your solution from the z coordinates back to the original x coordinates.
- (f) It turns out that all 2nd order linear differential equations with distinct eigenvalues will have this common form

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} c_0 e^{\lambda_1 t} + c_1 e^{\lambda_2 t} \\ c_2 e^{\lambda_1 t} + c_3 e^{\lambda_2 t} \end{bmatrix}$$

where c_0, c_1, c_2, c_3 are constants, and λ_1, λ_2 are the eigenvalues of A (this can be proven by just repeating the same steps in the previous parts and using the fact that distinct eigenvalues implies linearly independent eigenvectors). Thus, an alternate way of solving this type of differential equation in the future is to now use your knowledge that the solution is of this form and just solve for the constants c_i .

We will use this method to solve a different system. Consider another second-order ordinary differential equation

$$\frac{d^2 y(t)}{dt^2} - 5 \frac{dy(t)}{dt} + 6y(t) = 0, \quad (5)$$

First to make the problem familiar, write the system in the form of (2), by choosing appropriate variables $x_1(t)$ and $x_2(t)$.

- (g) Now solve the system in (5) with the initial conditions $y(0) = 1, \frac{dy}{dt}(0) = 1$, using the method from part (f).

5. Op-Amp Integrators: A continuation from the previous HW

In this question we will continue on from our analysis in Homework 3 and look at the eigenvalues of the integrator circuit (refer to Figure 3) in both non-ideal and ideal situations.

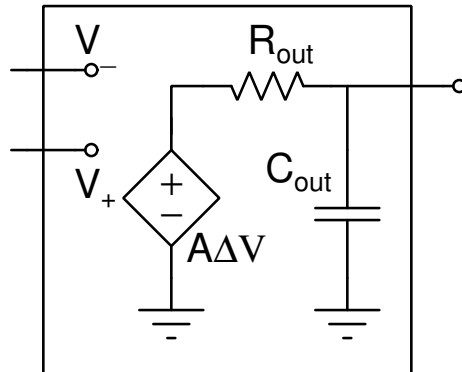


Figure 1: Op-amp model: $\Delta V = V_+ - V_-$

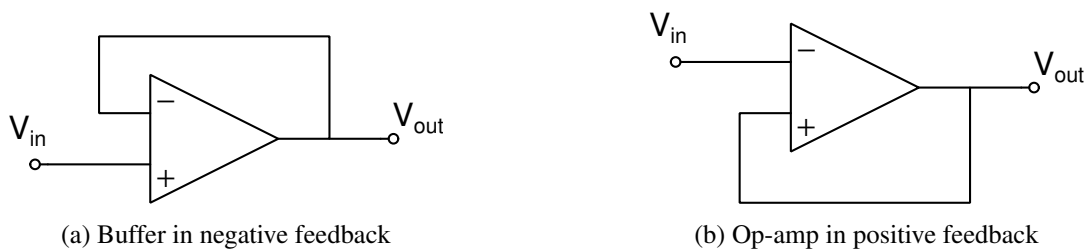


Figure 2: Op-amp in positive and negative feedback

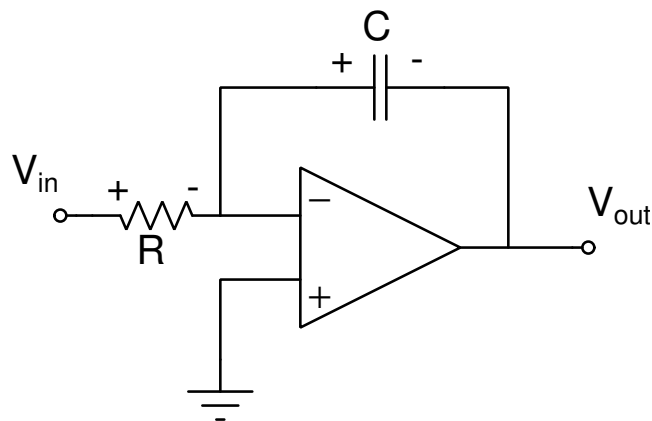


Figure 3: Integrator circuit

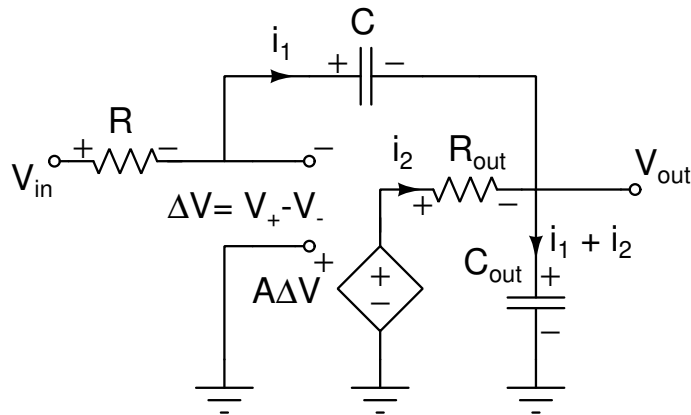


Figure 4: Integrator circuit with Op-amp model

- (a) Recall from Homework 3 we had the following analysis to the integrator circuit shown in Figure 4.

$$\frac{d}{dt} \begin{bmatrix} V_{out} \\ V_C \end{bmatrix} = \begin{bmatrix} -\left(\frac{A+1}{R_{out}C_{out}} + \frac{1}{RC_{out}}\right) & -\left(\frac{1}{RC_{out}} + \frac{A}{R_{out}C_{out}}\right) \\ -\frac{1}{RC} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} V_{out} \\ V_C \end{bmatrix} + \begin{bmatrix} \frac{1}{RC_{out}} \\ \frac{1}{RC} \end{bmatrix} V_{in} \quad (6)$$

Solve for the eigenvalues for the matrix/vector differential equation in Eq. (6).

For simplicity, assume $C_{out} = C = 0.01F$ and $R = 1\Omega$ and looking at the datasheet for the TI LMC6482 (the op-amps used in lab), we have $A = 10^6$ and $R_{out} = 100\Omega$.

Feel free to assume $A + 1 \approx 10^6$ when you finally need to plug in values, but do not make any other approximations. (Of course, such an approximation is not valid if you have a $A + 1 - A$ term showing up somewhere.) Feel free to use a scientific calculator or Jupyter to find the eigenvalues.

You should see that one eigenvalue corresponds to a slowly dying exponential and is close to 0. The other corresponds to a much faster dying exponential. The very slowly dying exponential is what corresponds to the desired integrator-like behavior. This is what lets it “remember.” (If you don’t understand why, think back to the HW problem you saw in a previous HW where you proved the uniqueness of the integral-based solution to a scalar differential equation with an input waveform.)

- (b) Again, assume we have an ideal op-amp, *i.e.*, $A \rightarrow \infty$. **Find the eigenvalues under this limit.** Feel free to make any reasonable approximations.

Here, you should see that the eigenvalue that used to be a slowly dying exponential stops dying out at all — corresponding to the ideal integrator’s behavior of remembering forever.

6. Multi-Capacitor Circuit

Consider the circuit below

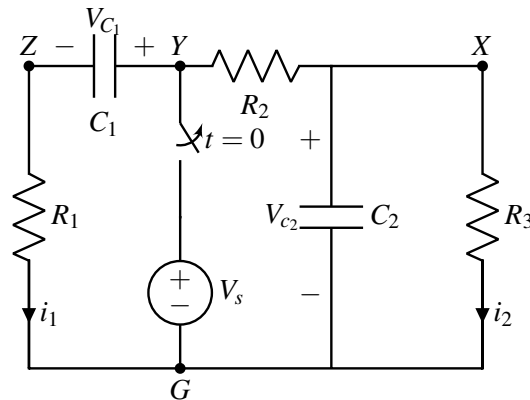


Figure 5: Circuit with multiple capacitors.

The resistors shown in the circuit have the same value $R_1 = R_2 = R_3 = R$. Capacitors C_1 and C_2 have the same capacitance $C_1 = C_2 = C$. Further, $RC = 1\text{s}$.

- Assume that the switch shown in Figure 5 was held in the closed position for a long time before $t = 0$. At $t = 0$, immediately after the switch is opened, what are the capacitor voltages $V_{C_1}(0)$ and $V_{C_2}(0)$?
- How are i_2 and V_{C_2} related? Using this, how are i_1 and V_{C_2} related?
- Using KVL on the loop comprising of both capacitors C_1 and C_2 , find a relationship between V_{C_1} , V_{C_2} and i_1 .
- Rewrite the equations derived above, eliminating the current i_1 to obtain a system of differential equations involving V_{C_1} and V_{C_2} . Write this system of equations in a matrix form

$$\frac{d}{dt} \begin{bmatrix} V_{C_1} \\ V_{C_2} \end{bmatrix} = A \begin{bmatrix} V_{C_1} \\ V_{C_2} \end{bmatrix}$$

What is the matrix A and what are its eigenvalues?

Hint: You can use the relation $i_1 = C_1 \frac{d}{dt} V_{C_1}$ in addition to the relations we have derived so far.

- In order to solve for the capacitor voltages, we need the initial values of the voltage derivatives. Immediately after the switch is opened, what are the voltage derivatives for the two capacitors, $\frac{dV_{C_1}}{dt}(0)$ and $\frac{dV_{C_2}}{dt}(0)$?

Hint: Calculate the currents i_1 and i_2 immediately after the switch is opened at $t = 0$. While the capacitor voltages do not change immediately, the current through them will change.

7. Homework Process and Study Group

Citing sources and collaborators are an important part of life, including being a student!

We also want to understand what resources you find helpful and how much time homework is taking, so we can change things in the future if possible.

(a) **What sources (if any) did you use as you worked through the homework?**

(b) **If you worked with someone on this homework, who did you work with?**

List names and student ID's. (In case of homework party, you can also just describe the group.)

Contributors:

- Ashwin Vangipuram.
- Archit Gupta.
- Lynn Chua.
- Anant Sahai.
- Aditya Arun.
- Kuan-Yun Lee.
- Gaoyue Zhou.

EECS 16B Designing Information Devices and Systems II

UC Berkeley

Homework 5

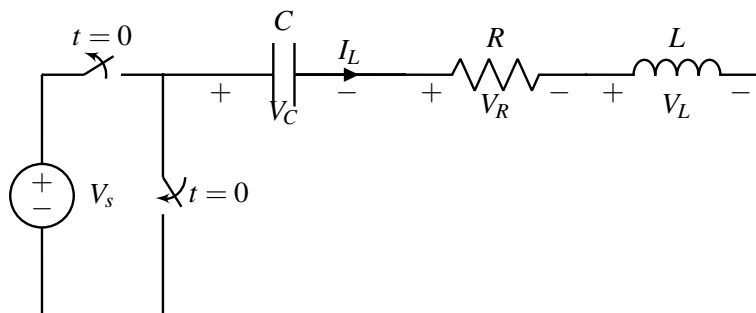
1. Reading Lecture Notes

Staying up to date with lectures is an important part of the learning process in this course. Here are links to the notes that you need to read for this week: [Note 3B](#).

- (a) What is the I-V relationship of an inductor? What is the behavior of the inductor under DC current (i.e. constant current)?

2. RLC Responses: Initial Part

Consider the following circuit like you saw in lecture:



Assume the circuit above has reached steady state for $t < 0$. At time $t = 0$, the switch changes state and disconnects the voltage source, replacing it with a short.

The sequence of problems 2 - 6 combined will try to show you the various RLC system responses and how they relate to how the eigenvalues of the A matrix changes. Note that the work you will do will also hold for any second-order system, like a mass-spring-damper, and is very common to study in controls as we'll see later on in Module 2 of the course.

- (a) We first need to construct our state space system. Our natural state variables are the current through the inductor $x_1(t) = I_L(t)$ and the voltage across the capacitor $x_2(t) = V_C(t)$ since these are the values that are changing in our circuit. Now, find the system of differential equations in terms of our state variables that describes this circuit for $t \geq 0$. Leave the system symbolic in terms of V_s , L , R , and C .

- (b) Write the system of equations in vector/matrix form with the vector state variable $\vec{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} I_L(t) \\ V_C(t) \end{bmatrix}$. This should be in the form $\frac{d}{dt}\vec{x}(t) = A\vec{x}(t)$ with a 2×2 matrix A .

- (c) Find the eigenvalues of the A matrix symbolically.

(Hint: the quadratic formula will be involved.)

- (d) Under what condition on the circuit parameters R, L, C are there going to be a pair of distinct purely real eigenvalues of A ?
- (e) Under what condition on the circuit parameters R, L, C are there going to be a pair of purely imaginary eigenvalues of A ? What will the eigenvalues be in this case?
- (f) Assuming that the circuit parameters are such that there are a pair of (potentially complex when conditions of part (d) fails) eigenvalues λ_1, λ_2 so that $\lambda_1 \neq \lambda_2$, find eigenvectors $\vec{v}_{\lambda_1}, \vec{v}_{\lambda_2}$ corresponding to them.

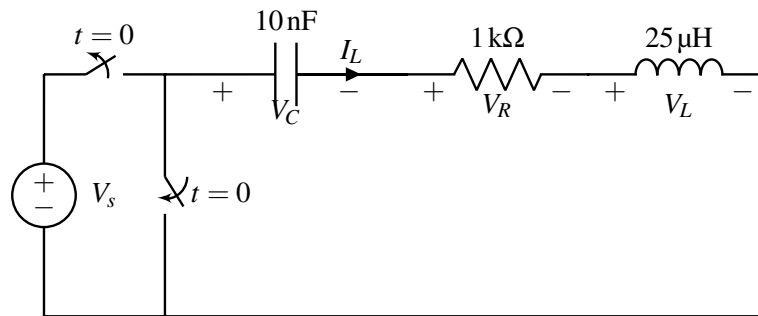
(HINT: Rather than trying to find the relevant nullspaces, etc., try to find eigenvectors of the form

$\begin{bmatrix} 1 \\ y \end{bmatrix}$ *where we just want to find the missing entry y . This works because we know the first entry of the eigenvector can not be 0 and we want to normalize it so that first entry is 1.)*

- (g) Assuming circuit parameters such that the two eigenvalues of A are distinct, let $V = [\vec{v}_{\lambda_1}, \vec{v}_{\lambda_2}]$ be a specific eigenbasis. Consider a coordinate system for which we can write $\vec{x}(t) = V\tilde{\vec{x}}(t)$. What is the $\tilde{A}$ so that $\frac{d}{dt}\tilde{\vec{x}}(t) = \tilde{A}\tilde{\vec{x}}(t)$? It is fine to have your answer expressed symbolically using λ_1, λ_2 .

3. RLC Responses: Overdamped Case

Building on the previous problem, consider the following circuit with specified component values:



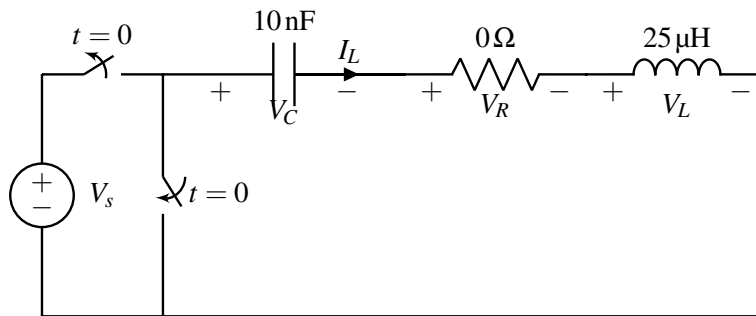
Assume the circuit above has reached steady state for $t < 0$. At time $t = 0$, the switch changes state and disconnects the voltage source, replacing it with a short.

For this problem, we use the same notations as in Problem 2. You may use a calculator or the attached 'RLC_Calc.ipynb' Jupyter Notebook for numerical calculations.

- Suppose $R = 1\text{ k}\Omega$ and the other component values are as specified in the circuit. Assume that $V_s = 1$ Volt. Find the initial conditions for $\tilde{x}(0)$. Recall that $\tilde{x}$ is in the changed “nice” eigenbasis coordinates from the first problem.
- Continuing the previous part, find $x_1(t) = I_L(t)$ and $x_2(t) = V_C(t)$ for $t \geq 0$.
- In the 'RLCSliders.ipynb' Jupyter notebook, move the sliders to approximately $R = 1\text{ k}\Omega$ and $C = 10\text{ nF}$. Comment on the graph of $V_C(t)$ and the location of the eigenvalues on the complex plane.

4. RLC Responses: Undamped Case

Building on the previous problem, consider the following circuit with specified component values:



Assume that the capacitor is charged to V_s and there is no current in the inductor for $t < 0$. At time $t = 0$, the switch changes state and disconnects the voltage source, replacing it with a short.

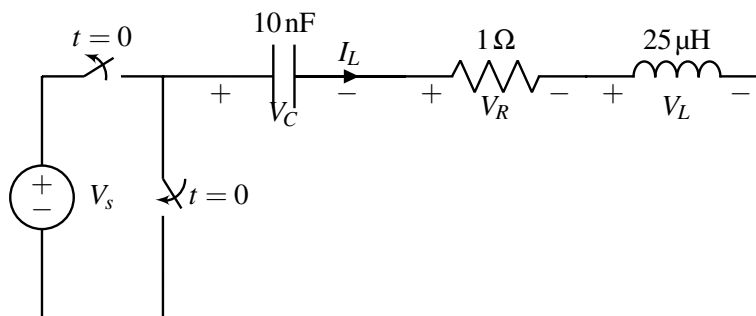
For this problem, we use the same notations as in Problem 2. You may use a calculator or the attached 'RLC_Calc.ipynb' Jupyter Notebook for numerical calculations.

- Suppose $R = 0\text{ k}\Omega$ and the other component values are as specified in the circuit. Assume that $V_s = 1$ Volt. Find the initial conditions for $\tilde{x}(0)$. Recall that $\tilde{x}$ is in the changed “nice” eigenbasis coordinates from the first problem.
- Continuing the previous part, find $x_1(t) = I_L(t)$ and $x_2(t) = V_C(t)$ for $t \geq 0$. Remember that your final expressions for $x_1(t)$ and $x_2(t)$ should be real functions (no imaginary terms).
- In the 'RLCSliders.ipynb' Jupyter notebook, move the sliders to approximately $R = 0\Omega$ and $C = 10\text{ nF}$. Comment on the graph of $V_C(t)$ and the location of the eigenvalues on the complex plane. Do the waveforms for $x_1(t)$ and $x_2(t)$ decay to 0?

Note: Because there is no resistance, this is called the “undamped” case.

5. RLC Responses: Underdamped Case

Building on the previous problem, consider the following circuit with specified component values:



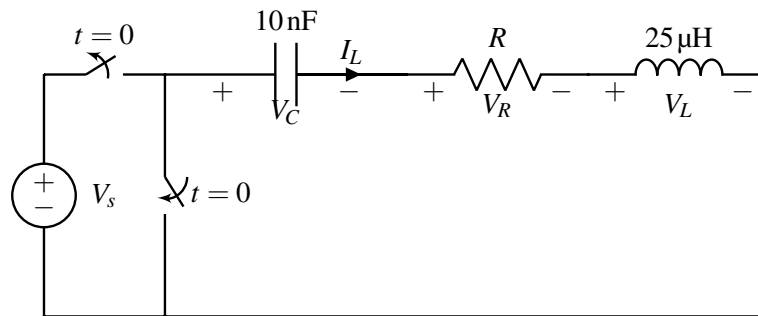
Assume the circuit above has reached steady state for $t < 0$. At time $t = 0$, the switch changes state and disconnects the voltage source, replacing it with a short.

For this problem, we use the same notations as in Problem 2. You may round numbers to make the algebra more simple. You may use a calculator or the attached 'RLC_Calc.ipynb' Jupyter Notebook for numerical calculations.

- Now suppose that $R = 1\ \Omega$ and the other component values are as specified in the circuit. Assume that $V_s = 1\ \text{V}$. Find the initial conditions for $\vec{x}(0)$. Recall that $\vec{x}$ is in the changed “nice” eigenbasis coordinates from the first problem.
- Continuing the previous part, find $x_1(t) = I_L(t)$ and $x_2(t) = V_C(t)$ for $t \geq 0$. Remember that your final expressions for $x_1(t)$ and $x_2(t)$ should be real functions (no imaginary terms).
(HINT: Remember that $e^{a+jb} = e^a e^{jb}$.)
- In the 'RLCSliders.ipynb' Jupyter notebook, move the sliders to approximately $R = 1\ \Omega$ and $C = 10\text{nF}$. Comment on the graph of $V_C(t)$ and the location of the eigenvalues on the complex plane. Do the waveforms for $x_1(t)$ and $x_2(t)$ decay to 0?
Note: Because the resistance is so small, this is called the “underdamped” case. It is good to reflect upon these waveforms to see why engineers consider such behavior to be reflective of systems that don't have enough damping.
- Notice that you got answers in terms of complex exponentials. Why did the final voltage and current waveforms end up being purely real?

6. RLC Responses: Critically Damped Case

Building on the previous problem, consider the following circuit with specified component values: (Notice R is not specified yet. You'll have to figure out what that is.)



Assume the circuit above has reached steady state for $t < 0$. At time $t = 0$, the switch changes state and disconnects the voltage source, replacing it with a short.

For this problem, we use the same notations as in Problem 2. You may use a calculator or the attached 'RLC_Calc.ipynb' Jupyter Notebook for numerical calculations.

- For what value of R is there going to be a single eigenvalue of A ?
- Find the eigenvalues and eigenspaces of A . What are the dimensions of the corresponding eigenspaces? (i.e. how many linearly independent eigenvectors can you find associated with this eigenvalue?)

For this part, assume the given values for the capacitor and the inductor, as well as the critical value for the resistance R that you found in the previous part. It is easier to do the algebra with a non-symbolic matrix to work with.

- We now create a new coordinate system V , with the first vector being $\vec{v}_\lambda$ — the eigenvector you found for the single eigenvalue λ above. For the second vector, just pick $\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. We will see later why we

chose such a vector. We then apply a change of basis to define $\tilde{\vec{x}}$ in the transformed coordinates such that $\vec{x}(t) = V\tilde{\vec{x}}(t)$. What is the resulting $\tilde{A}$ matrix defining the system of differential equations in the transformed coordinates?

- Notice that the second differential equation for $\frac{d}{dt}\tilde{x}_2(t)$ in the above coordinate system only depends on $\tilde{x}_2(t)$ itself. There is no cross-term dependence. This happened because we earlier chose $\vec{v}_2$ such that the transformation $\tilde{A}$ becomes upper triangular and results in the removal of the cross-dependency. We will later see that in fact, we had many other choices for $\vec{v}_2$. Now, compute the initial condition for $\tilde{x}_2(0)$ and write out the solution to this scalar differential equation for $\tilde{x}_2(t)$ for $t \geq 0$.

- With an explicit solution to $\tilde{x}_2(t)$ in hand, substitute this in and write out the resulting scalar differential equation for $\tilde{x}_1(t)$. This should effectively have a time-dependent input in it.

Note: this is just the differential-equations counterpart to the back-substitution step from Gaussian Elimination in 16A, once you had done one full downward pass of Gaussian Elimination. You went upwards and just substituted in the solution that you found to remove this dependence from the equations above. This is the exact same design pattern, except for a system of linear differential equations.

- Solve the above scalar differential equation with input and write out what $\tilde{x}_1(t)$ is for $t \geq 0$.

(HINT: You might want to look at a problem on an earlier homework for help with this.)

- (g) Find $x_1(t)$ and $x_2(t)$ for $t \geq 0$ based on the answers to the previous three parts.

This particular case is called the “critically damped case” for an RLC circuit. It is called this because the R value you found demarcates the boundary between solutions of the underdamped and overdamped variety.

- (h) In the ‘RLCSliders.ipynb’ Jupyter notebook, move the sliders to the resistance value you found in the first part and $C = 10nF$. Comment on the graph of $V_c(t)$ and the location of the eigenvalues on the complex plane. What happens to the voltage and eigenvalues as you slightly increase or decrease R ?
- (i) In part (c) we saw that A only had one eigenvalue, λ , and one eigenvector, $\vec{v}_\lambda$. This meant that we had a choice for $\vec{v}_2$ in the expression $V = \begin{bmatrix} \vec{v}_\lambda & \vec{v}_2 \end{bmatrix}$. We seemingly arbitrarily chose $\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. We

claim that there are many correct choices of $\vec{v}_2$ that will result in $\tilde{A} = V^{-1}AV = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$ where $a, c \neq 0$

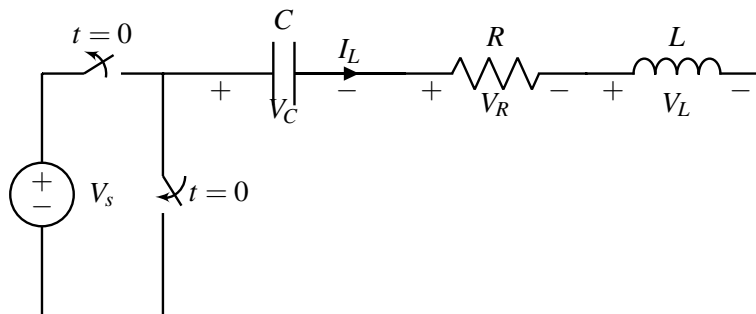
(i.e. it is upper triangular). Remember, we want $\tilde{A}$ to be upper triangular so that we have an uncoupled differential equation for $\vec{x}_2(t)$: $\frac{d}{dt}\vec{x}(t) = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \begin{bmatrix} \tilde{x}_1(t) \\ \tilde{x}_2(t) \end{bmatrix} \implies \frac{d}{dt}\tilde{x}_2(t) = c\tilde{x}_2(t)$

In fact, it turns out that we can pick any $\vec{v}_2$ as long as $\vec{v}_2 \neq k\vec{v}_\lambda$ for some $k \in \mathbb{R}$. We will try and prove this very claim. More concisely, prove the statement below:

if $V = \begin{bmatrix} \vec{v}_\lambda & \vec{v}_2 \end{bmatrix}$ and $\vec{v}_2 \neq k\vec{v}_\lambda$ for some $k \in \mathbb{R}$, then $\tilde{A} = V^{-1}AV = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$ where $a, c \neq 0$

7. Second order ODE perspective on solving the RLC circuit

Consider the following circuit like you saw in lecture, discussion, and the previous few problems:



Suppose now we insisted on expressing everything in terms of one waveform $V_C(t)$ instead of two of them (voltage across the capacitor and current through the inductor). This is called the “second-order” point of view, for reasons that will soon become clear.

For this problem, use R for the resistor, L for the inductor, and C for the capacitor in all the expressions until the last part.

- Write the current I_L through the inductor in terms of the voltage through the capacitor.**
- Now, notice that the voltage drop across the inductor involves $\frac{d}{dt}I_L$. **Write the voltage drop across the inductor in terms of the second derivative of V_C .**
- For this part, treat $V_s(t)$ as a generic input waveform — don’t necessarily view the switch as being thrown, etc.

Now write out a differential equation governing $V_C(t)$ in the form of

$$\frac{d^2}{dt^2}V_C(t) + a \cdot \frac{d}{dt}V_C(t) + b \cdot V_C(t) + c(t) = 0. \quad (1)$$

where a , b and $c(t)$ are terms you need to figure out by analyzing the circuit.

This is called a second-order system here as there is a 2nd derivative in the differential equation.

(*HINT: The $c(t)$ needs to involve $V_s(t)$ in some way.*)

- We don’t know how to solve equations like Eq. (1). To reduce this to something we know how to solve, we define the first derivative $\frac{d}{dt}V_C(t)$ as an additional state and label it as $X(t)$. Note that this directly gives us one equation in our matrix: $\frac{d}{dt}V_C(t) = X(t)$. This leaves us needing an equation for $\frac{d}{dt}X(t)$. **Express $\frac{d}{dt}X(t)$ in terms of $X(t)$, $V_C(t)$, and $V_s(t)$. Write a matrix differential equation in terms of $V_C(t)$ and $X(t)$.** Your answer should be in the form:

$$\frac{d}{dt} \begin{bmatrix} X(t) \\ V_C(t) \end{bmatrix} = \underbrace{\begin{bmatrix} ? & ? \\ ? & ? \end{bmatrix}}_A \begin{bmatrix} X(t) \\ V_C(t) \end{bmatrix} + \begin{bmatrix} ? \\ ? \end{bmatrix} \cdot V_s(t). \quad (2)$$

- Find the eigenvalues and eigenvectors of the matrix A from Eq. (2). Compare what you got to your answers for Problem 2 and explain why this is the case.**
(*Hint: use the same trick you did in problem 2. Don’t do this the hard way.*)
- Now use the same values of R, L, C and initial conditions from Problem 3 and 4 to solve for V_C in this system.** Did you get the same answer as in problems 3 and 4? Remember that you can use the attached ‘RLC_Calc.ipynb’ to help with computation.

8. Homework Process and Study Group

Citing sources and collaborators are an important part of life, including being a student!

We also want to understand what resources you find helpful and how much time homework is taking, so we can change things in the future if possible.

(a) **What sources (if any) did you use as you worked through the homework?**

(b) **If you worked with someone on this homework, who did you work with?**

List names and student ID's. (In case of homework party, you can also just describe the group.)

Contributors:

- Anant Sahai.
- Jaijeet Roychowdhury.
- Sanjeet Batra.
- Aditya Arun.
- Alex Devonport.
- Regina Eckert.
- Kuan-Yun Lee.
- Ashwin Vangipuram.

EECS 16B Designing Information Devices and Systems II

UC Berkeley

Homework 6

1. Exam Policy and Practice

Your midterm is coming up on **March 15th, 2021**. To prepare for midterm proctoring, and to receive credit on this question, we ask you to read and complete the **Proctoring Document**. Make sure to **complete all action items in the document in order to receive credit for this question**.

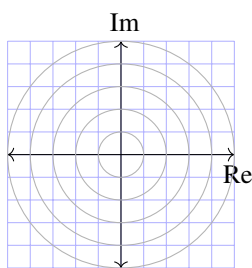
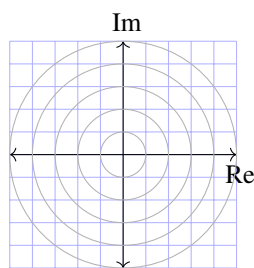
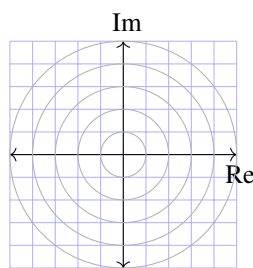
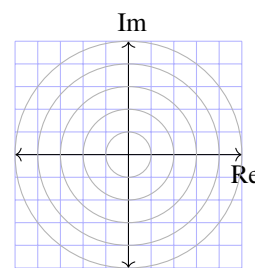
2. Reading Lecture Notes

Staying up to date with lectures is an important part of the learning process in this course. Here are links to the notes that you need to read for this week: [Note4](#), [Note5](#)

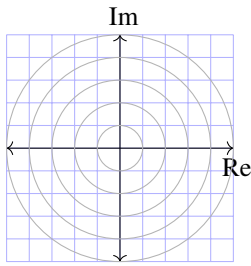
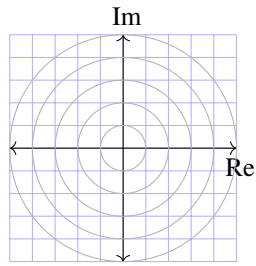
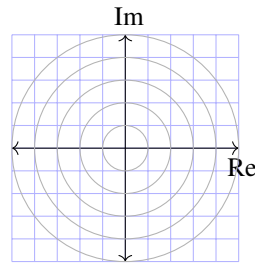
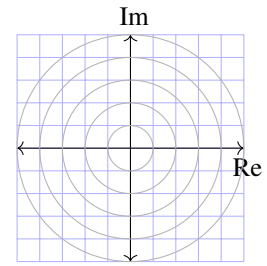
- Consider an RC circuit with a sinusoidal voltage input $V(t) = A \cos(\omega t)$. We are interested in finding the voltage on the capacitor in steady state (after a long time has passed). Can we solve this using our standard differential equation techniques? Can we solve this with phasors? Which one is more concise and why?
- There are two ways to make a low pass filter (discussed in the notes). What are they? Are there any major differences?

3. Phasors

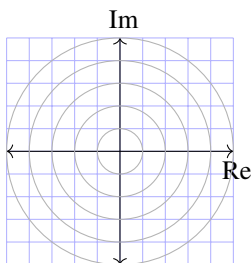
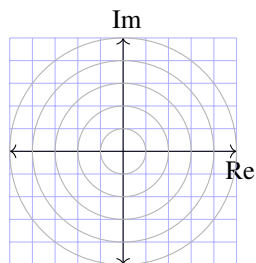
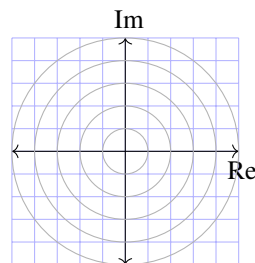
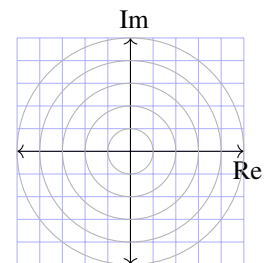
- Consider a resistor ($R = 1.5\Omega$), a capacitor ($C = 1\text{F}$), and an inductor ($L = 1\text{H}$) connected in series. **Give expressions for the impedances of Z_R, Z_C, Z_L for each of these elements as a function of the angular frequency ω .**
- Draw the individual impedances as “vectors” on the same complex plane for the case $\omega = \frac{1}{2}$ rad/sec. Also draw the combined impedance Z_{total} of their series combination. Then give the magnitude and phase of Z_{total} .**

a $Z_R(@\omega = 0.5)$ b $Z_C(@\omega = 0.5)$ c $Z_L(@\omega = 0.5)$ d $Z_{total}(@\omega = 0.5)$

- Draw the individual impedances as “vectors” on the same complex plane for the case $\omega = 1$ rad/sec. Also draw the combined impedance Z_{total} of their series combination. Then give the magnitude and phase of Z_{total} .**

a $Z_R(@\omega = 1)$ b $Z_C(@\omega = 1)$ c $Z_L(@\omega = 1)$ d $Z_{total}(@\omega = 1)$

- (d) Draw the individual impedances as “vectors” on the same complex plane for the case $\omega = 2$ rad/sec. Also draw the combined impedance Z_{total} of their series combination. Then give the magnitude and phase of Z_{total} .

a $Z_R(@\omega = 2)$ b $Z_C(@\omega = 2)$ c $Z_L(@\omega = 2)$ d $Z_{total}(@\omega = 2)$

- (e) For the previous series combination of RLC elements, what is the “natural frequency” ω_n (i.e. the frequency ω_n where the series impedance is purely real). Fact: We call this the natural frequency since it is the frequency that the circuit will oscillate at if it was underdamped (if R was small enough).

4. Color Organ Filter Design

In the fourth lab, we will design low-pass, band-pass, and high-pass filters for a color organ. There are red, green, and blue LEDs. Each color will correspond to a specified frequency range of the input audio signal. The intensity of the light emitted will correspond to the amplitude of the audio signal.

- (a) First, you remember that you saw in lecture that you can build simple filters using a resistor and a capacitor. **Design a simple first-order *passive* low-pass filter with the following frequency range using 1 μF capacitors.** (“Passive” means that the filter does not require any power supply to operate on the input signal. Passive components include resistors, capacitors, inductors, diodes, etc., while an example of an active component would be an op-amp).

- Low-pass filter: cut-off frequency $f_c = 2400\text{Hz}$, $\omega_c = 2\pi \cdot 2400 \frac{\text{rad}}{\text{sec}}$

Recall that the cutoff-frequency of such a filter is just where the magnitude of the filter is $\frac{1}{\sqrt{2}}$ of its peak value.

Show your work to find the resistor value that creates this low-pass filter. Draw the schematic-level representation of your design. Please mark V_{in} , V_{out} , and the ground node(s) in your schematic. Round your results to two significant figures.

- (b) **Now design a simple first-order *passive* high-pass filters with the following frequency range using 1 μF capacitors.**

- High-pass filter: cut-off frequency $f_c = 100\text{Hz}$, $\omega_c = 2\pi \cdot 100 \frac{\text{rad}}{\text{sec}}$

Show your work to find the resistor value that creates this high-pass filter. Draw the schematic-level representation of your design. Please mark V_{in} , V_{out} , and the ground node(s) in your schematic. Round your results to two significant figures.

- (c) You can try to build a bandpass filter by cascading the first-order low-pass and high-pass filters you designed in parts (a) and (b). To do this, you might be tempted to connect the V_{out} node of your low-pass filter directly to the V_{in} node of your high-pass filter. However, if you did this, just as you saw in 16A for voltage dividers, the purported high-pass filter would “load” the low-pass filter and you might get some potentially complicated mess instead of what you wanted.

Show how you can use an ideal op-amp configured as a unity gain buffer to eliminate this loading effect to cascade the low-pass and high-pass filters, and write the resulting transfer function of the combined circuit. Draw the magnitude and phase transfer functions of the combined circuit. What kind of filter is this? You can optionally use the included Jupyter notebook `tf_gen.ipynb`. (HINT: Read Section 2.1 in [Note6](#).)

(NOTE: In Python, use `1j` when your transfer function has a `j`.)

- (d) **Write down an expression for the time-domain output waveform $V_{\text{out}}(t)$ of this filter if the input voltage is $V_{\text{in}}(t) = 1 \sin(1000t)$ V.** Round your answer to 2 significant digits.

5. Phasors and Eigenvalues

Suppose that we have the two-dimensional system of differential equations expressed in matrix/vector form:

$$\frac{d}{dt}\vec{x}(t) = A\vec{x}(t) + \vec{b}u(t) \quad (1)$$

where for this problem, the matrix A and the vector $\vec{b}$ are both real.

- (a) **Give a necessary condition on the eigenvalues λ_k of A such that any impact of an initial condition will eventually completely die out.** (i.e. the system will reach steady-state.)

You don't have to prove this. The idea here is to make sure that you understand what kind of thing is required. (*HINT: Read Section 2 in [Note4](#).*)

- (b) Now assume that $u(t)$ has a phasor representation $\tilde{U}$. In other words, $u(t) = \tilde{U}e^{+j\omega t} + \overline{\tilde{U}}e^{-j\omega t}$.

Assume that the vector solution $\vec{x}(t)$ to the system of differential equations (1) can also be written in phasor form as

$$\vec{x}(t) = \tilde{X}e^{+j\omega t} + \overline{\tilde{X}}e^{-j\omega t}. \quad (2)$$

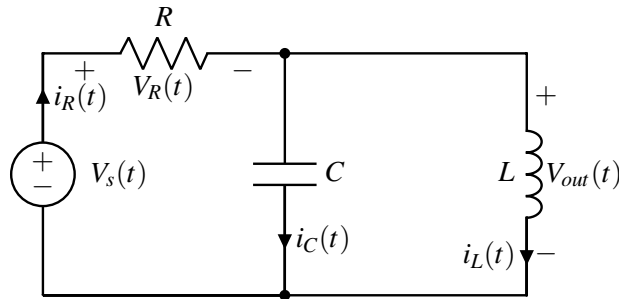
Derive an expression for $\tilde{X}$ involving $A, \vec{b}, j\omega, \tilde{U}$, and the identity matrix I .

(*HINT: Plug (2) into (1) and simplify, using the rules of differentiation and grouping terms by which exponential $e^{\pm j\omega t}$ they multiply.*)

6. Phasor-Domain Circuit Analysis

The analysis techniques you learned previously in 16A for resistive circuits are equally applicable for analyzing circuits driven by sinusoidal inputs in the phasor domain. In this problem, we will walk you through the steps with a concrete example.

Consider the following circuit where the input voltage is sinusoidal. The end goal of our analysis is to find an equation for $V_{out}(t)$.



The components in this circuit are given by:

$$V_s(t) = 10\sqrt{2}\cos\left(100t - \frac{\pi}{4}\right)$$

$$R = 5\,\Omega$$

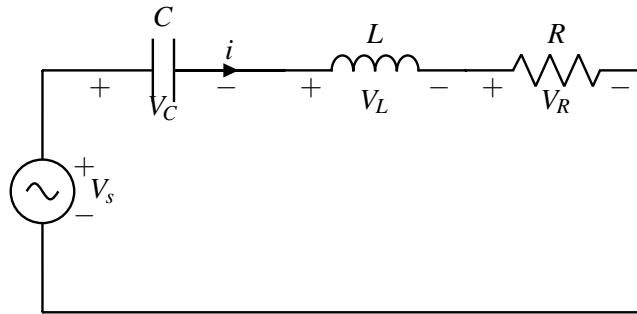
$$L = 50\text{mH}$$

$$C = 2\text{mF}$$

- Give the amplitude V_0 , oscillation frequency ω , and phase ϕ of the input voltage V_s .
- Transform the circuit into the phasor domain. **What are the impedances of the resistor, capacitor, and inductor? What is the phasor $\tilde{V}_s$ of the input voltage $V_s(t)$?**
- Use the circuit equations to **solve for $\tilde{V}_{out}$** , the phasor representing the output voltage.
- Convert the phasor $\tilde{V}_{out}$ back to get the time-domain signal $V_{out}(t)$.**

7. RLC filter

Consider the following RLC circuit:



- Write down the impedance of a series RLC circuit in the form $Z_{RLC}(\omega) = A(\omega) + jX(\omega)$, where $X(\omega)$ is a real valued function of ω .
- Write the transfer function from V_S to V_R — the voltage drop across the resistor.
- For the different specific values for R, L, C given by different cases (underdamped, overdamped, and critically damped) in the previous HW, use a computer to sketch plots of the magnitude and phase of the transfer function above. You can optionally use the included Jupyter notebook `tf_gen.ipynb`. (NOTE: In Python, use `1j` when your transfer function has a `j`.)
- To see how the values of R, L, C impact the impedance at different frequencies, run the included Jupyter notebook `hw6rlc_transfer.ipynb`. The script will generate two plots, the transfer function of the circuit as a function of frequency and the location of the eigenvalues in the imaginary, real plane. **Explain what happens at the following sets of values, and why it is interesting:**

	R	L	C
I	1	2.5E-5	1E-8
II	10	2.5E-5	1E-8
III	10	2.5E-5	2E-9
IV	500	.0001	2E-8

Table 1: Values for RLC Bandwidth problem, part d

8. Homework Process and Study Group

Citing sources and collaborators are an important part of life, including being a student!

We also want to understand what resources you find helpful and how much time homework is taking, so we can change things in the future if possible.

(a) **What sources (if any) did you use as you worked through the homework?**

(b) **If you worked with someone on this homework, who did you work with?**

List names and student ID's. (In case of homework party, you can also just describe the group.)

Contributors:

- Jaijeet Roychowdhury.
- Anant Sahai.
- Nathan Lambert.
- Kyoungtae Lee.
- Nikhil Shinde.
- Ayan Biswas.
- Geoffrey Négier.
- Regina Eckert.
- Sanjit Batra.
- Kris Pister.
- Daniel Abraham.

EECS 16B Designing Information Devices and Systems II
UC Berkeley

Homework 7

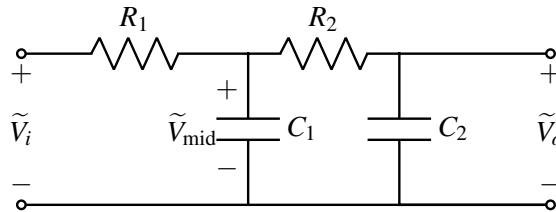
1. Reading Lecture Notes

Staying up to date with lectures is an important part of the learning process in this course. Here are links to the notes that you need to read for this week: [Note 5](#), [Note 6](#)

- (a) In the magnitude Bode plot for a first-order-low-pass filter drawn on a log-log graph (i.e. $|H(j\omega)|$ is plotted on a log scale), what is the slope of the straight line approximation at frequencies higher than the cut-off frequency? What is the slope for a third-order-low-pass filter (three identical first-order-low-pass filters cascaded with unity gain buffers)?
- (b) In the phase Bode plot for a first-order-low-pass filter, what is the approximation for the phase at frequencies higher than 10 times the cut-off frequency? What is the approximation for the phase for a third-order-low-pass filter?

2. Transfer functions and why loading is annoying

Consider the circuit below.



The circuit has an input phasor voltage $\tilde{V}_i$ at frequency ω rad/sec applied at the input terminals shown in the illustration above, causing an output phasor voltage $\tilde{V}_o$ at output terminals.

- (a) We are going to construct the transfer function $H(\omega) = \frac{\tilde{V}_o}{\tilde{V}_i}$ in two steps. We will compute two intermediate transfer functions, $H_1(\omega) = \frac{\tilde{V}_{\text{mid}}}{\tilde{V}_i}$ and $H_2(\omega) = \frac{\tilde{V}_o}{\tilde{V}_{\text{mid}}}$. Then, we will find the overall transfer function as the product of these two intermediate transfer functions, i.e. $H(\omega) = \frac{\tilde{V}_o}{\tilde{V}_i} = H_1(\omega)H_2(\omega)$.

This approach is valid since the $\tilde{V}_{\text{mid}}$ cancel.

For the first step, **find the intermediate transfer function** $H_2(\omega) = \frac{\tilde{V}_o}{\tilde{V}_{\text{mid}}}$. Have your expression be in terms of Z_{R2} and Z_{C2} , which are the impedances of R_2 and C_2 respectively.

- (b) Now, **compute the other intermediate transfer function** $H_1(\omega) = \frac{\tilde{V}_{\text{mid}}}{\tilde{V}_i}$. Have your expression be in terms of Z_{R1} , Z_{R2} , Z_{C1} , and Z_{C2} . (i.e. Don't forget to consider the impact of loading by R_2 and C_2 in this transfer function.) *Hint: You should try to find an expression for H_1 that has factors that H_2 can cancel out.*
- (c) Then, **use these two intermediate transfer functions to calculate the overall transfer function as** $H(\omega) = \frac{\tilde{V}_o}{\tilde{V}_i} = H_1(\omega)H_2(\omega)$.
- (d) Sometimes it is useful to collect all the frequency dependence into one place and to figure out how to think about what scale might be somewhat natural for the frequency. **Obtain an expression for** $H(\omega) = \frac{\tilde{V}_o}{\tilde{V}_i}$ **in the form**

$$H(\omega) = \frac{\tilde{V}_o}{\tilde{V}_i} = \frac{1}{1 + 2\xi \frac{j\omega}{\omega_c} + \frac{(j\omega)^2}{\omega_c^2}},$$

given that $R_1 = 100\Omega$, $R_2 = 1\Omega$, $C_1 = 2\text{F}$, and $C_2 = 2\text{F}$. What are the values of ξ and ω_c ?

- (e) **For the previous case, what is the magnitude of the transfer function at the $\omega = \omega_c$ you calculated?**

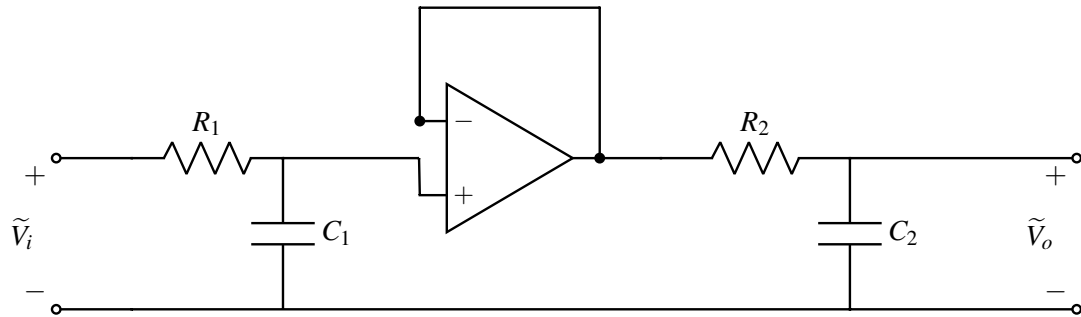
This is here so that you can see that just because we called it ω_c doesn't mean that the amplitude here is $\frac{1}{\sqrt{2}}$.

- (f) We can express the transfer function $H(\omega)$ in the polar form. That is,

$$H(\omega) = M(\omega)e^{j\phi(\omega)}$$

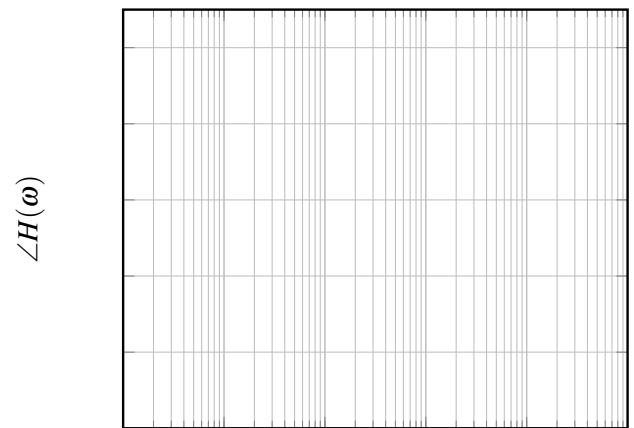
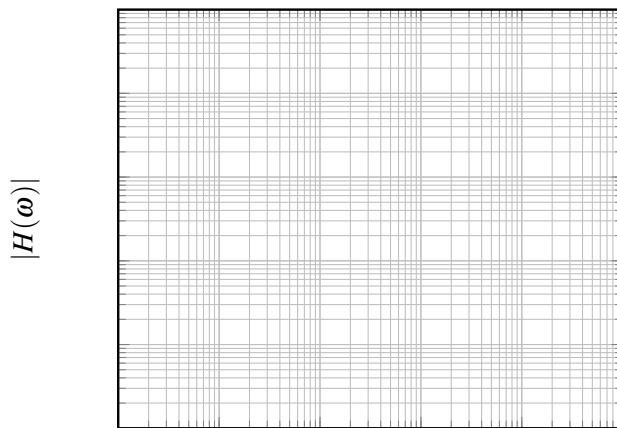
The functions $M(\omega)$ and $\phi(\omega)$ are the magnitude and the phase of $H(\omega)$, respectively. **Write down $M(\omega)$ and $\phi(\omega)$ using the transfer function you derived in part (d).**

(g) Now consider the same circuit but with a unity gain buffer to avoid the loading effect, as shown below.



Derive the new transfer function $H_{\text{wbuf}}(\omega) = \frac{\tilde{V}_o}{\tilde{V}_i}$ of the filter, using the same values of R_1 , R_2 , C_1 and C_2 from part (d).

(h) Use a computer and then **superpose the plots of $|H(\omega)|$ and $|H_{\text{wbuf}}(\omega)|$, and the plots of $\angle H(\omega)$ and $\angle H_{\text{wbuf}}(\omega)$** . Blank plots are provided here for you to use. Feel free to use the **tf_gen.ipynb** from HW6.



3. Bandpass Filter: Lowpass and Highpass Cascade

In lecture, you heard about how you can go through the design of a bandpass filter by cascading lowpass and highpass filters via buffers (Op-Amps in unity-gain negative feedback to prevent loading effects). In this problem, you will do this for yourself.

Consider an input signal that is composed of the superposition of:

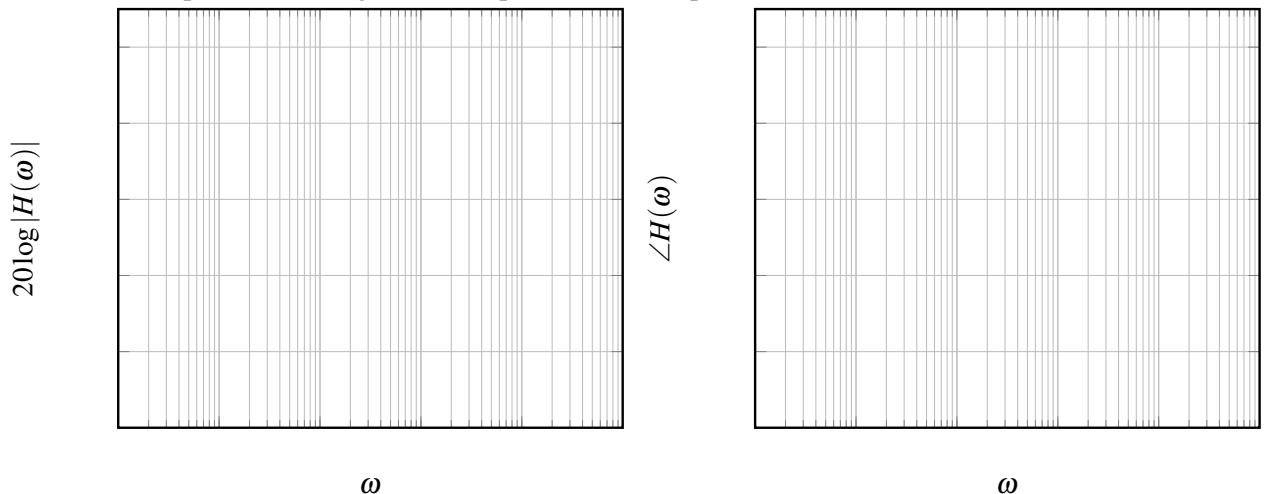
- 20mV level pure tone at 60Hz corresponding to power line noise.
- 1mV level pure tone at 600Hz corresponding to a voice signal.
- 10mV level pure tone at 60kHz corresponding to fluorescent light noise.

- (a) We would like to keep the 600Hz tone, which could correspond to a voice signal.

Ignoring any phase offset for each signal (i.e. set the phases to zero), **write the $V_{in}(t)$ that describes the above input in time domain, in the following format:**

$$V_{in}(t) = A_p \cos(2\pi \cdot f_p t) + A_v \cos(2\pi \cdot f_v t) + A_f \cos(2\pi \cdot f_f t)$$

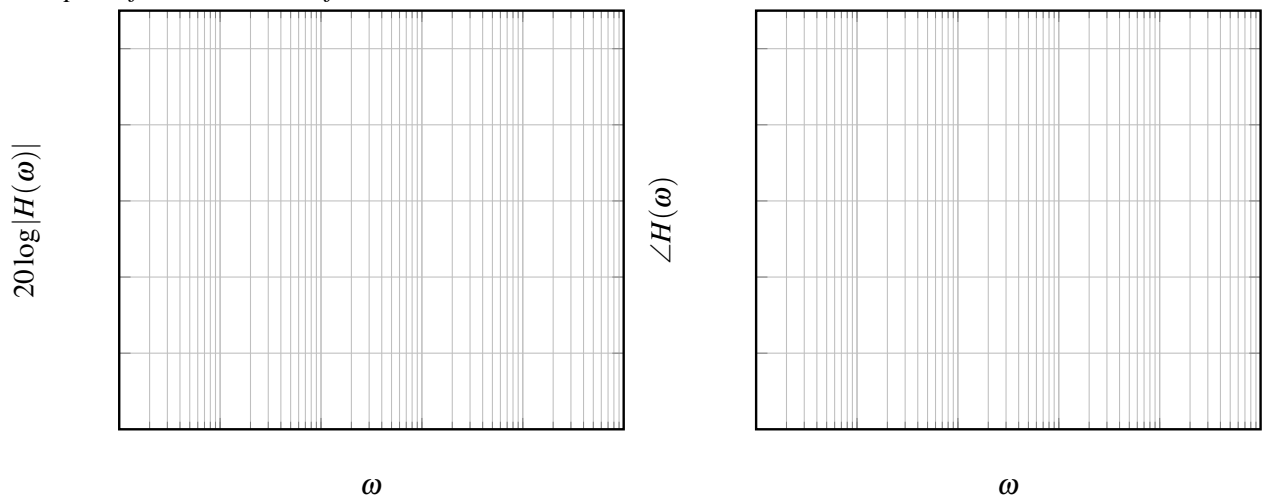
- (b) **What are the angular frequencies (i.e. $\omega_p, \omega_v, \omega_f$) involved and the phasors associated with each tone?** Remember that the frequencies of the tones were provided in Hz. To convert this to angular frequency we use $\omega = 2\pi f$
- (c) To achieve your goal of keeping the voice tone but rejecting the noise from the power-lines and fluorescent lights, **at what frequency do you want to have the knee or cutoff-frequency for the lowpass filters?** *Hint: To arrive at a unique solution consider computing the the geometric mean of the two frequencies of interest.*
- (d) Draw the Bode plot for the magnitude and phase of the lowpass filter.



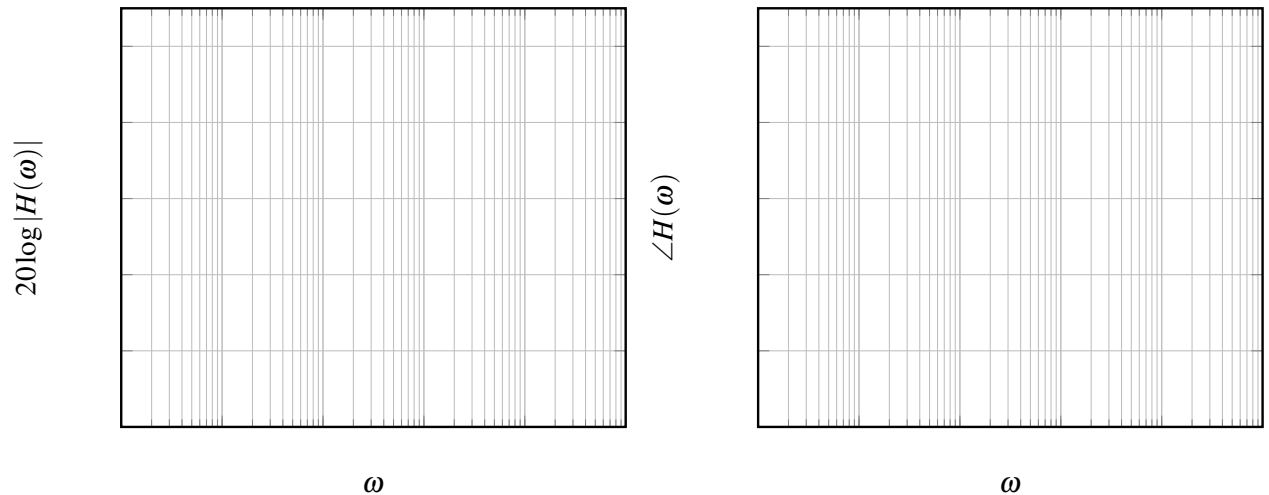
- (e) To achieve your goal of keeping the voice tone but rejecting the noise from the power-lines and fluorescent lights, **at what frequency do you want to have the knee or cutoff-frequency for the highpass filters?** *Hint: To arrive at a unique solution consider computing the the geometric mean of the two frequencies of interest.*
- (f) Draw the Bode plot for the magnitude and phase of the highpass filter.



- (g) For the following questions assume your cut-off frequency for lowpass and highpass are 6KHz and 189Hz , respectively. Suppose that you only had $1\mu\text{F}$ capacitors to use. **What resistance values would you choose for your highpass and lowpass filters so that they have the desired cutoff frequencies?**
- (h) The overall bandpass filter that is created by cascading the lowpass and highpass with ideal buffers in between. Draw the Bode plot for the magnitude and phase of the overall bandpass transfer function. *Hint: You should think about how the bode plot of cascade of two filters can be derived based on the bode plot of the lower-level filters.*



- (i) Suppose that the band-pass filter does not have enough suppression at 60Hz and 60kHz and you decide to use cascade of three band-pass filters (with unity-gain buffers in between) to mitigate the issue. **What are the phasors for each of the frequency tones after all three bandpass filters?** Feel free to use a computer to help you evaluate both the magnitudes and the phases here.
- (j) **Draw the Bode plot for the magnitude and phase of the 3rd order bandpass filter.** To highlight the difference between the 3rd and 1st order filters, please draw both Bode plots on a single figure.



(k) **Write the final time domain voltage waveform that would be present after the filter.**

- (l) The included jupyter notebook sets up the same problem described above. In the notebook, you can use the slider bars to play around with:
- highpass cutoff frequency (i.e. the knee frequency of the highpass filters)
 - lowpass cutoff frequency (i.e. the knee frequency of the lowpass filters)
 - Filter order N . Filter order means the number of lowpass filters and highpass filters that are used in a row. Here, N means that there are N lowpass filters and N highpass filters, so the overall order of the entire filter is actually $2N$.

The notebook will plot the magnitude and phase, the input voltage waveform, and the output waveform at the end of the filter.

Play around with the values for the highpass and lowpass cutoff frequencies, and N .

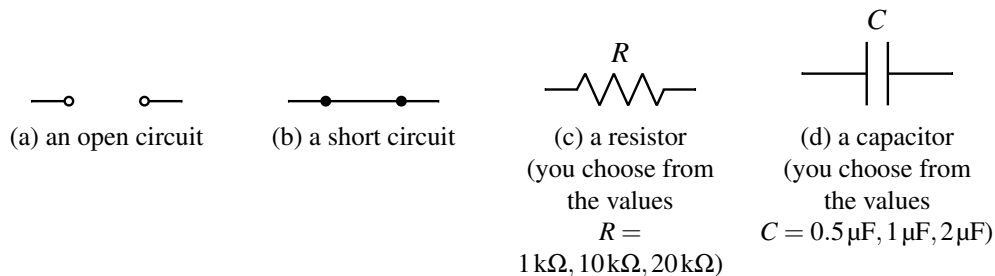
Observe the waveforms at the output of the filter. **Comment on the limits of f_{lowpass} , f_{highpass} , and N that you can use to successfully isolate the desired 600Hz tone. What happens if you keep f_{lowpass} and f_{highpass} constant, and just increase N ?**

4. Circuit Design

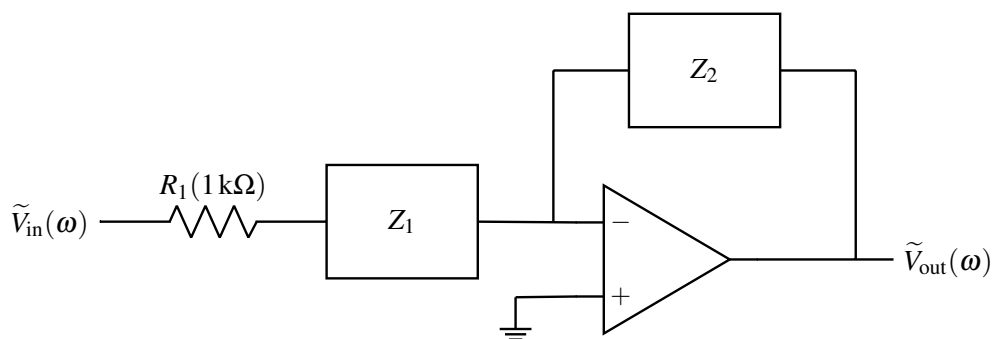
In this problem, you will find a circuit where several components have been left *blank* for you to fill in.

Assume that the op-amp is *ideal*. A *special note on op amps in frequency domain analysis*: The op-amps you learned about in 16A can be used in exactly the same way for setting up differential equations and even Phasor analysis in 16B. Treat them as ideal op-amps and invoke the Golden Rules.

You have at your disposal *only one of each* of the following components (not including R_1):



Consider the circuit below. The voltage source $V_{in}(t)$ has the form $V_{in}(t) = v_0 \cos(\omega t + \phi)$. The labeled voltages $\tilde{V}_{in}(\omega)$ and $\tilde{V}_{out}(\omega)$ are the phasor representations of $v_{in}(t)$ and $v_{out}(t)$. The transfer function $H(\omega)$ is defined as $H(\omega) = \frac{\tilde{V}_{out}(\omega)}{\tilde{V}_{in}(\omega)}$.



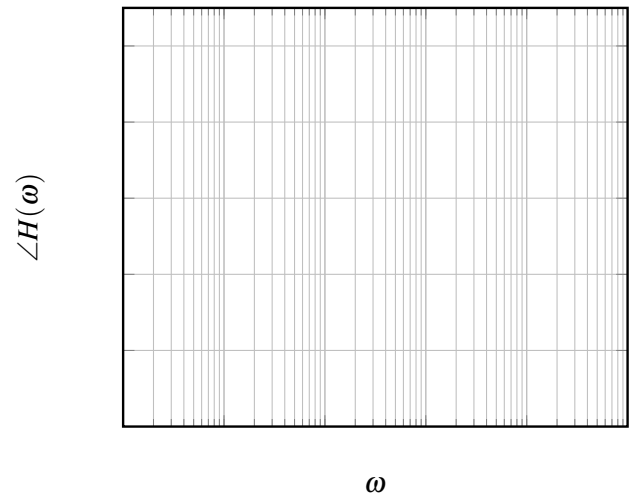
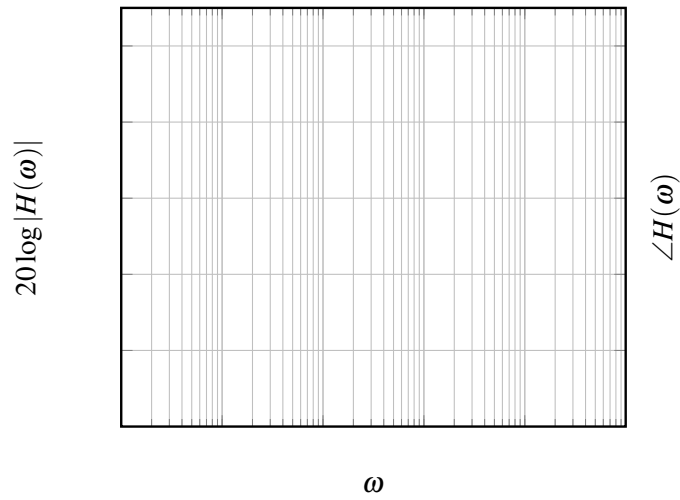
- Let the left box be Z_1 and the right box be Z_2 . **Write the expression of the transfer function $H(\omega)$.**
- Let R_1 be $1\text{ k}\Omega$. We have to find Z_1 and Z_2 , such that the circuit's transfer function $H(\omega)$ has the following properties:
 - It is a high-pass filter.
 - $|H(\infty)| = 10$.
 - $|H(10^3)| = \sqrt{50}$.

Using the fact that the circuit is a high pass filter, infer the components (we will find values later) of Z_1 and Z_2 . Write the transfer function $H(\omega)$ using these components.

Hint: Try method of elimination: figure out what Z_2 cannot be. Once you find what Z_2 is, what does Z_1 have to be for the circuit to be a filter?

- Now use the facts that $|H(\infty)| = 10$ and $R_1 = 1\text{ k}\Omega$ to find the component value of Z_2 .
- Finally use the fact that $|H(10^3)| = \sqrt{50}$ and the values of R_1 and Z_2 to find the component value of Z_1 .

- (e) **Draw the magnitude and phase Bode plots of this transfer function.** Blank plots are provided here for you to use.



5. Homework Process and Study Group

Citing sources and collaborators are an important part of life, including being a student!

We also want to understand what resources you find helpful and how much time homework is taking, so we can change things in the future if possible.

(a) **What sources (if any) did you use as you worked through the homework?**

(b) **If you worked with someone on this homework, who did you work with?**

List names and student ID's. (In case of homework party, you can also just describe the group.)

Contributors:

- Ayan Biswas.
- Yen-Sheng Ho.
- Alex Devonport.
- Sidney Buchbinder.
- Anant Sahai.
- Kris Pister.
- Pavan Bhargava.
- Kourosh Hakhamaneshi.

EECS 16B Designing Information Devices and Systems II
UC Berkeley Homework 8

on Monday, but we still will require a HW submission on Gradescope.

1. Reading Lecture Notes

Staying up to date with lectures is an important part of the learning process in this course. Here are links to the notes that you need to read for this week: [Note 7B](#), [Note 8](#)

- (a) For the overdetermined system $A\vec{x} = \vec{b}$, what condition is needed to use least squares to estimate $\hat{x}$?
- (b) What are the eigenvalue tests for stability for both discrete-time systems and continuous-time systems?
- (c) How do you use feedback control to change the eigenvalues of a closed-loop continuous-time system?

2. System Identification

You are given a discrete-time system as a black-box. You don't know the specifics of the system but you know that it takes one scalar input and has two states that you can observe. You assume that the system is linear and of the form

$$\vec{x}[t+1] = A\vec{x}[t] + Bu[t] + \vec{w}[t], \quad (1)$$

where $\vec{w}[t]$ is an external unseen disturbance that you hope is small, $u[t]$ is a scalar input, and

$$A = \begin{bmatrix} a_0 & a_1 \\ a_2 & a_3 \end{bmatrix}, \quad B = \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}, \quad x[t] = \begin{bmatrix} x_0[t] \\ x_1[t] \end{bmatrix}. \quad (2)$$

You want to identify the system parameters from measured data. You need to find the unknowns: a_0, a_1, a_2, a_3, b_0 and b_1 . However, you can only interact with the system via a blackbox model, i.e. you can see the states $\vec{x}[t]$ and set the inputs $u[t]$ that allow the system to move to the next state.

- (a) You observe that the system has state $\vec{x}[t] = [x_0[t], x_1[t]]^T$ at time t . You pass input $u[t]$ into the blackbox and observe the next state of the system: $\vec{x}[t+1] = [x_0[t+1], x_1[t+1]]^T$.

Write scalar equations for the new states, $x_0[t+1]$ and $x_1[t+1]$. Write these equations in terms of the a_i, b_i , the states $x_0[t], x_1[t]$ and the input $u[t]$. Here, assume that $\vec{w}[t] = \vec{0}$ (i.e. the model is perfect).

- (b) Now we want to identify the system parameters. We observe the system at the start state $\vec{x}[0] = \begin{bmatrix} x_0[0] \\ x_1[0] \end{bmatrix}$.

We can then input $u[0]$ and observe the next state $\vec{x}[1] = \begin{bmatrix} x_0[1] \\ x_1[1] \end{bmatrix}$. We can continue this for an m long sequence of inputs.

Let us define an m long trajectory to be $[x_0[0], x_1[0], u[0], x_0[1], x_1[1], u[1], x_0[2], x_1[2], u[2], \dots, x_0[m-1], x_1[m-1], u[m-1], x_0[m], x_1[m]]$. **Assuming there is no noise ($\vec{w}[t] = \vec{0}$), what is the minimum value of m you need to identify the system parameters?**

- (c) Now assume that there is a nonzero noise/disturbance $\vec{w}[t]$. **Would using more than equations than in part (b) help you in this case? If so, explain why.**
- (d) Say we feed in a total of 4 inputs $[u[0], u[1], u[2], u[3]]$ into our blackbox. This allows us to observe the following states $[x_0[0], x_0[1], x_0[2], x_0[3], x_0[4]]$ and $[x_1[0], x_1[1], x_1[2], x_1[3], x_1[4]]$, which we can use to identify the system.

To identify the system we need to set up an approximate (because of potential disturbances) matrix equation

$$D\vec{p} \approx \vec{y}$$

using the observed values above and the unknown parameters we want to find. We know our parameter vector should be $\vec{p} = [a_0 \ a_1 \ b_0 \ a_2 \ a_3 \ b_1]^T$. **Find the corresponding D and $\vec{y}$ to do system identification. Write out both explicitly.**

- (e) Now that we have set up $D\vec{p} \approx \vec{y}$, **explain how you would use this approximate equation to estimate the unknown values $a_0, a_1, a_2, a_3, b_0, b_1$.** In particular, give an expression for your estimate $\hat{\vec{p}}$ in terms of the D and $\vec{y}$. Assume that the columns of D are linearly independent.

(HINT: Don't forget that D is not a square matrix. It is taller than it is wide.)

3. Identifying systems from their responses to known inputs

In many problems, we have an unknown system, and would like to characterize it. One of the ways of doing so is to observe the system response with different initial conditions (or inputs). This problem is also called system identification. It is a prototypical example of a problem that today is called machine learning — inferring an underlying pattern from data, and doing so well enough to be able to exploit that pattern in some practical setting. Go through the attached Jupyter notebook “DemoSystemID_16b.ipynb” and answer the following questions.

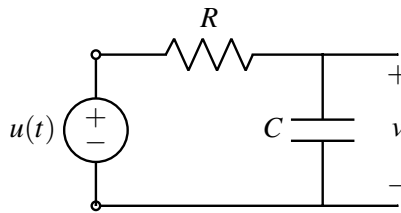
- (a) In Example 2, we assume that instead of measuring the state $\vec{x}$, we are instead measuring a transformation of the state $\vec{y} = T\vec{x}$ where T is a full rank matrix. Assume that we perform system ID on our observations $\vec{y}[t]$ to recover A_y, B_y such that $\vec{y}[t+1] = A_y\vec{y}[t] + B_y u[t]$. **How do the identified A_y and B_y matrices relate to the original A and B matrices in the dynamics of $\vec{x}$?** Remember that our original state dynamics are $\vec{x}[t+1] = A\vec{x} + B\vec{u}$.

HINT: The answer is given in the Jupyter notebook but remember to show your work.

- (b) **Please share your observations on Example 2.**
- (c) **Prove that for any full rank transformation matrix T , the eigenvalues of A_y and A from part (a) are the same.**
- (d) **Please share your observations on Example 3.**
- (e) **Please share your observations on Example 4.**
- (f) **Please share your observations on Example 5.**

4. BIBO Stability

- (a) Consider the circuit below with $R = 1\Omega$, $C = 0.5F$. Furthermore assume that $v(0) = 0$ (that the capacitor is initially discharged).



This circuit can be modeled by the differential equation

$$\frac{d}{dt}v(t) = -2v(t) + 2u(t) \quad (3)$$

Show that $v(t)$ remains bounded for all time if the input $u(t)$ is bounded, i.e. $|u(t)| < k, \forall t \geq 0$. *HINT: You may want to write the expression for $x(t)$ in terms of $u(t)$ and $x(0)$ and then use the triangle inequality to prove that $x(t)$ is bounded. Also use the fact that the norm of an integral is $\geq$ the integral of a norm. In other words $|\int_a^b f(\tau)d\tau| \leq \int_a^b |f(\tau)|d\tau$. This inequality is just the regular triangle inequality you have seen before, generalized to integrals. The proof strategy used in lecture for the discrete time case will extend here as well with the modified triangle inequality usage.*

Thinking about this helps you understand what bounded-input-bounded-output stability means in a physical circuit.

- (b) Consider a continuous-time scalar real differential equation with known solution

$$\frac{d}{dt}x(t) = ax(t) + bu(t) \quad x(t) = e^{at}x(0) + \int_0^t e^{a(t-\tau)}bu(\tau)d\tau.$$

Show that if the system has $\text{Re}\{a\} > 0$, then there exists a bounded input ($|u(t)| \leq \varepsilon$) that can result in an unbounded output, even with a zero initial condition. In other words, show the system is BIBO unstable.

- (c) **Repeat the previous part for the specific case of complex $a = r + j2\pi$ where $r > 0$ and zero initial condition $x(0) = 0$.**

All the other truly complex unstable cases are the same way for the same essential reason.

- (d) **[Optional] Repeat the previous part for the specific case of purely imaginary $a = j2\pi$ with zero initial condition $x(0) = 0$.**

All the other purely imaginary unstable cases are the same way for the same essential reason.

- (e) Consider the discrete-time system

$$x[t+1] = -2x[t] + 2u[t] \quad (4)$$

with $x[0] = 0$.

Is this system stable or unstable? If stable, prove it. If unstable, find a bounded input sequence $u[t]$ that causes the system to ‘blow up’.

- (f) For the example in the previous part, **give an explicit sequence of inputs that are not zero but for which the state $x[t]$ will always stay bounded.** (*HINT: building off of the previous part see if you can find any input pattern that result in an oscillatory behavior.*)

(g) Consider the discrete-time real system with known solution:

$$x[t+1] = ax[t] + bu[t] \quad x[t] = a^t x[0] + \sum_{\ell=0}^{t-1} a^{t-1-\ell} bu[\ell]$$

Show that if the system is unstable (has $|a| > 1$), then a bounded input can result in an unbounded output. Assume a zero initial condition here.

(h) **[Optional] Repeat the previous part for the specific case of $a = -1$.**

(i) [Optional: this part was derived in lecture] Now consider the discrete-time stable case where a is complex and has $|a| < 1$. **Show that as long as $|u[t]| < k$ for some k , that the solution $x[t]$ will be bounded for all time t .**

(HINT: There are a few helpful facts about absolute values and inequalities that are helpful in such proofs. First: $|\sum_j a_j| \leq \sum_j |a_j|$. Second $|ab| = |a| \cdot |b|$. Third: $|e^{j\theta}| = 1$ no matter what real number θ is. And fourth, if $a_i > 0$ and $b_i > 0$, and $b_i \leq B$, then $\sum_i a_i b_i \leq \sum_i a_i B = B \sum_i a_i$.)

5. Homework Process and Study Group

Citing sources and collaborators are an important part of life, including being a student!

We also want to understand what resources you find helpful and how much time homework is taking, so we can change things in the future if possible.

(a) **What sources (if any) did you use as you worked through the homework?**

(b) **If you worked with someone on this homework, who did you work with?**

List names and student ID's. (In case of homework party, you can also just describe the group.)

Contributors:

- Ashwin Vangipuram.
- Nikhil Shinde.
- Alex Devonport.
- Sidney Buchbinder.
- Nathan Lambert.
- Anant Sahai.

EECS 16B Designing Information Devices and Systems II
UC Berkeley Homework 9

1. Reading Lecture Notes

Staying up to date with lectures is an important part of the learning process in this course. Here are links to the notes that you need to read for this week: [Note 8](#) and [Note 9](#)

- (a) What is the matrix test for controllability of a general linear discrete time system $\vec{x}[t + 1] = A\vec{x}[t] + \vec{b}u[t]$?

2. Eigenvalue Placement through State Feedback

Consider the following discrete-time linear system:

$$\vec{x}[t+1] = \begin{bmatrix} -2 & 2 \\ -2 & 3 \end{bmatrix} \vec{x}[t] + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u[t].$$

In standard language, we have $A = \begin{bmatrix} -2 & 2 \\ -2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ in the form: $\vec{x}[t+1] = A\vec{x}[t] + Bu[t]$.

(a) **Is this system controllable?**

(b) **Is this discrete-time linear system stable in open loop (without feedback control)?**

(c) Suppose we use state feedback of the form $u[t] = \begin{bmatrix} f_1 & f_2 \end{bmatrix} \vec{x}[t]$

Find the appropriate state feedback constants, f_1, f_2 so that the state space representation of the resulting closed-loop system has eigenvalues at $\lambda_1 = -\frac{1}{2}, \lambda_2 = \frac{1}{2}$.

(d) We are now ready to go through some numerical examples to see how state feedback works. Consider the first discrete-time linear system. Enter the matrix A and B from (a) for the system

$$\vec{x}[t+1] = A\vec{x}[t] + Bu[t] + w[t]$$

into the Jupyter notebook “**eigenvalue_placement.ipynb**” and use the random input $w[t]$ as the disturbance introduced into the state equation. Observe how the norm of $\vec{x}[t]$ evolves over time for the given A. **What do you see happening to the norm of the state?**

(e) Add the feedback computed in part (c) to the system in the notebook and **explain how the norm of the state changes.**

(f) **[OPTIONAL]** Now suppose we’ve got a different system described by the controlled scalar difference equation $z[t+1] = z[t] + 2z[t-1] + u[t]$. To convert this second-order difference equation to a two-dimensional discrete time system, we will let $\vec{y}[t] = \begin{bmatrix} z[t-1] \\ z[t] \end{bmatrix}$ **Write down the system representation for $\vec{y}$ in the following matrix form:**

$$\vec{y}(t+1) = A_y \vec{y}(t) + B_y u(t). \quad (1)$$

Please specify what the matrix A_y and the vector B_y .

(g) **[OPTIONAL]** We will now show how the initial system for $\vec{x}[t]$ can be converted to the system for $\vec{y}[t]$ using a change of basis. Suppose we change coordinates with the transformation $\vec{y}[t] = P\vec{x}[t]$. **Write down the state-transition matrices of $\vec{y}[t]$ in terms of the state transition matrices of $\vec{x}[t]$, i.e., express A_y and B_y in terms of A, B , and P .** Additionally, confirm that for $P = \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix}$, the resulting state space representation of $\vec{y}[t]$ is the same as in the previous part (i.e. we get the same A_y, B_y).

(h) **[OPTIONAL]** For the $\vec{y}$ system from part (f), **design a feedback gain matrix $\begin{bmatrix} \bar{f}_1 & \bar{f}_2 \end{bmatrix}$ to place the closed-loop eigenvalues at $\lambda_1 = -\frac{1}{2}, \lambda_2 = \frac{1}{2}$. Confirm that $\begin{bmatrix} f_1 & f_2 \end{bmatrix} = \begin{bmatrix} \bar{f}_1 & \bar{f}_2 \end{bmatrix} P$.**

3. Tracking a Desired Trajectory in Continuous Time

The treatment in 16B so far has treated closed-loop control as being about holding a system steady at some desired operating point, by placing the eigenvalues of the state transition matrix. This control used the actual current state to apply a control signal designed to bring the eigenvalues in the region of stability. Meanwhile, the idea of controllability itself was more general and allowed us to make an open-loop trajectory that went pretty much anywhere. This problem is about combining these two ideas together to make feedback control more practical — how we can get a system to more-or-less closely follow a desired trajectory, even though it might not start exactly where we wanted to start and in principle could be affected by small disturbances throughout.

In this question, we will also see that everything that you have learned to do closed-loop control in discrete-time can also be used to do closed-loop control in continuous time.

Consider the specific 2-dimensional system

$$\frac{d}{dt}\vec{x}(t) = A\vec{x}(t) + \vec{b}u(t) + \vec{w}(t) = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \vec{x}(t) + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(t) + \vec{w}(t) \quad (2)$$

where $u(t)$ is a scalar valued continuous control input and $\vec{w}(t)$ is a bounded disturbance (noise).

- Would the given system be controllable if we viewed the parameters A and $\vec{b}$ as the parameters of a discrete-time system, i.e. $\vec{x}_d[t+1] = A\vec{x}_d[t] + \vec{b}u_d[t]$?**
- In an ideal noiseless scenario, the desired control signal $u^*(t)$ makes the system follow the desired trajectory $\vec{x}^*(t)$ that satisfies the following dynamics:

$$\frac{d}{dt}\vec{x}^*(t) = A\vec{x}^*(t) + \vec{b}u^*(t) \quad (3)$$

The presence of the bounded noise term $\vec{w}(t)$ makes the actual state $\vec{x}(t)$ deviate from the desired $\vec{x}^*(t)$ and follow (2) instead. In the following subparts, we will analyze how we can adjust the desired control signal $u^*(t)$ in (3) to the control input $u(t)$ in (2) so that the deviation in the state caused by $\vec{w}(t)$ remains bounded.

Represent the state as $\vec{x}(t) = \vec{x}^*(t) + \vec{v}(t)$ and $u(t) = u^*(t) + u_v(t)$. Using (2) and (3), we can represent the evolution of the trajectory deviation $\vec{v}(t)$ as a function of the control deviation $u_v(t)$ and the bounded disturbance $\vec{w}(t)$ as:

$$\frac{d}{dt}\vec{v}(t) = A_v\vec{v}(t) + \vec{b}_v u_v(t) + \vec{w}(t) \quad (4)$$

What are A_v and $\vec{b}_v$ in terms of the original system parameters A and $\vec{b}$? (HINT: Write out equation (2) in terms of $\vec{x}^*(t)$, $\vec{v}(t)$, $u^*(t)$ and $u_v(t)$.)

- Are the dynamics that you found for $\vec{v}(t)$ in part (b) stable? Based on this, in the presence of bounded disturbance $\vec{w}(t)$, will $\vec{x}(t)$ in (2) follow the desired trajectory $\vec{x}^*(t)$ closely if we just apply the control $u(t) = u^*(t)$ to the original system in (2), i.e. $u_v(t) = 0$?**
(HINT: Use the numerical values of A and $\vec{b}$ from (2) in the solution from part (b) to determine stability of $\vec{v}(t)$.)

Now, we want to apply state feedback control to the system using $u_v(t)$ to get it to more or less follow the desired trajectory $\vec{x}^*(t)$.

- (d) **[OPTIONAL]** For the $\vec{v}(t), u_v(t)$ system, **apply feedback control by choosing $u_v(t)$ as a function of $\vec{v}(t)$ that would place both the eigenvalues of the closed-loop $\vec{v}(t)$ system at -10 .** (*HINT: $u_v(t) = \begin{bmatrix} f_0 & f_1 \end{bmatrix} \vec{v}(t)$. Find f_0 and f_1 .*)
- (e) **[OPTIONAL]** Based on what you did in the previous parts, and given access to the desired trajectory $\vec{x}^*(t)$, the desired control $u^*(t)$, and the actual measurement of the state $\vec{x}(t)$, **come up with a way to do feedback control that will keep the trajectory staying close to the desired trajectory no matter what the small bounded disturbance $\vec{w}(t)$ does.** (*HINT: Express the control input $u(t)$ in terms of $u^*(t)$, $\vec{x}^*(t)$, and $\vec{x}(t)$.*)

4. [OPTIONAL] Group Re-assignment Survey

How are your study groups working out? We hope they have been helpful so far. If you feel things are not going as well as you hoped and you would prefer to be assigned to a new group, please fill out the following form:

[Group Re-assignment Survey - Google Form](#)

5. Homework Process and Study Group

Citing sources and collaborators are an important part of life, including being a student!

We also want to understand what resources you find helpful and how much time homework is taking, so we can change things in the future if possible.

(a) **What sources (if any) did you use as you worked through the homework?**

(b) **If you worked with someone on this homework, who did you work with?**

List names and student ID's. (In case of homework party, you can also just describe the group.)

Contributors:

- Ashwin Vangipuram.
- Varun Mishra.
- Anant Sahai.
- Ming Jin.
- Ayan Biswas.

EECS 16B Designing Information Devices and Systems II

UC Berkeley

Homework 10

1. Reading Lecture Notes

Staying up to date with lectures is an important part of the learning process in this course. Here are links to the notes that you need to read for this week: [Note 10A](#)

- (a) Consider $A \in \mathbb{R}^{n \times n}$ where the columns of A (denoted by $\vec{a}_k, 1 \leq k \leq n$) are orthonormal. What does the least squares solution $(A^T A)^{-1} A^T \vec{y}$ simplify to?
- (b) Suppose we have two vectors $\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \in \mathbb{R}^2$. Then, we use Gram-Schmidt Orthonormalization to construct $\vec{q}_1$ and $\vec{q}_2$. Are $\vec{q}_1$ and $\vec{q}_2$ the only vectors that form an orthogonal basis for the $\text{span}\{\vec{v}_1, \vec{v}_2\}$?

2. Gram-Schmidt Basic

- (a) **Use Gram-Schmidt to find a matrix U whose columns form an orthonormal basis for the column space of V .**

$$V = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

- (b) Show that you get the same resulting vector when you project $\vec{w} = \begin{bmatrix} 1 \\ -1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$ onto the columns of V as you do when you project onto the columns of U , i.e. **show that**

$$V(V^T V)^{-1} V^T \vec{w} = U(U^T U)^{-1} U^T \vec{w}.$$

Feel free to use numpy. No need to grind this out by hand.

3. Symmetric Matrices

We want to show that every real symmetric matrix can be diagonalized by a matrix of its orthonormal eigenvectors. In other words, a symmetric matrix has an orthonormal eigenbasis. This is called the spectral theorem for real symmetric matrices.

In discussion section, you have seen a recursive derivation of a related fact. Formally however, such recursive derivations are usually turned into proofs by using induction. This problem serves to both freshen your mind regarding induction as well as to give you a chance to prove for yourself this very important theorem. (This is the same essential proof as that of Schur upper-triangularization. So understanding this problem will help solidify your understanding of that proof as well.)

- (a) You will start by proving a basic lemma about real symmetric matrices under an orthonormal change of basis. **Prove that if S is an $m \times m$ symmetric matrix ($S = S^T$) and U is any $m \times n$ matrix, then $U^T S U$ is also symmetric.**

- (b) Another useful lemma is one about finding orthonormal bases. **Show that given a single nonzero vector $\vec{u}_0$ of dimension n , it is possible to find an orthonormal set of n vectors, $\vec{v}_0, \dots, \vec{v}_{n-1}$ such that $\vec{v}_0 = \alpha \vec{u}_0$ for some scalar α .**

(Hint: Use the Gram-Schmidt process on the list of $n + 1$ vectors obtained by starting with the given vector and appending the standard basis — i.e. the columns of the identity matrix. Follow the procedure as in lecture.)

- (c) For the main proof that every real symmetric matrix is diagonalized by a matrix of its orthonormal eigenvectors, we will proceed by formal induction. Recall that for a proof by induction, we have to start with a base case - this is also the base case in a recursive derivation.

Consider the trivial case of S having dimensions $[1 \times 1]$ ($n = 1$). **Does S have an eigenvector? Does S have an eigenbasis? Can this eigenbasis be made orthonormal? Is the matrix diagonal in this basis? Are the entries real?**

- (d) After the base case, we do an inductive stage of the main proof. The first step in the inductive stage is to write down the induction hypothesis. Assume that the property that every real symmetric matrix is diagonalized by a matrix of its orthonormal eigenvectors holds for all symmetric matrices with size $[(n - 1) \times (n - 1)]$. **Complete the proof template below by filling in the blanks.** (Hint: In general for proofs by induction, you want to start with the strongest version of what you want to prove. This gives you the most powerful inductive hypothesis.)

Goal: we want to show that any $[n \times n]$ real symmetric matrix S can be diagonalized by a matrix of its orthonormal real eigenvectors.

Base case: when $n = 1$, this holds for $[1 \times 1]$ matrix $S = [s]$ as you showed in Part (d).

Inductive hypothesis: Let Q be an $[(n - 1) \times (n - 1)]$ real symmetric matrix with eigenvectors $\vec{u}_0, \dots, \vec{u}_{n-2}$, then we can express Q as $U \Lambda_Q U^T$, where Λ_Q is a _____ matrix with _____ eigenvalues of Q along its _____, and U is an $[(n - 1) \times (n - 1)]$ matrix, where the columns are _____ of Q .

- (e) Now think about a symmetric matrix S with size $[n \times n]$. Consider a real eigenvalue λ_0 of S and the corresponding eigenvector $\vec{u}_0$ (a column vector with size n). **Use an appropriate orthonormal change of basis V to show that $S = V X V^T$, where X is of the form**

$$X = \begin{bmatrix} \lambda_0 & x_{1,2} & \cdots & x_{1,n} \\ 0 & x_{2,2} & \cdots & x_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & x_{n,2} & \cdots & x_{n,n} \end{bmatrix}.$$

That is, the first column of X is $\begin{bmatrix} \lambda_0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$.

(Hint: Follow the proof strategy from lecture. What should the first column of V be? How can you fill out the rest?)

- (f) Continue the previous part **to show that in fact S can be written as**

$$S = V \begin{bmatrix} \lambda_0 & \vec{0}^T \\ \vec{0} & Q \end{bmatrix} V^T$$

where Q is an $[(n-1) \times (n-1)]$ symmetric matrix and V is the matrix that you found above.

Hint: Define $\vec{v}_0 = \frac{\vec{u}_0}{\|\vec{u}_0\|}$, think of V as $V = \begin{bmatrix} \vec{v}_0 & R \end{bmatrix}$, where the dimension of R is $[n \times (n-1)]$. Also recall that S is a symmetric matrix.

- (g) According to our induction hypothesis, we can write Q as $U\Lambda U^T$ where U is an orthonormal $[(n-1) \times (n-1)]$ square matrix and Λ is a diagonal matrix with real entries along the diagonal and 0s everywhere else. **Substitute this in the previous part to show that indeed there must exist an orthonormal $[n \times n]$ square matrix W such that**

$$S = W \begin{bmatrix} \lambda_0 & \vec{0}^T \\ \vec{0} & \Lambda \end{bmatrix} W^T$$

(Hint: If R and U are both orthonormal matrices, is RU also orthonormal? What is $(RU)^T(RU)$?)

- (h) **Finally, prove that the eigenvalues λ of real, symmetric matrix S are real.** Hint: Suppose that S had a complex eigenvalue λ with eigenvector $\vec{v}$. Because S is a real matrix, what do you know about $S\bar{\vec{v}}$, where $S\bar{\vec{v}}$ is the complex conjugate of $\vec{v}$? What happens when you take a potentially complex number and multiply it by its own complex conjugate? Consequently, what do you know happens if you multiply a complex vector $\vec{v}$ by the conjugate of its transpose: i.e. consider $\bar{\vec{v}}^T \vec{v}$? Since S is symmetric, what do you know about $\bar{\vec{v}}^T S \vec{v}$?

By induction, we are now done since we have proved that having the desired property for $n-1$ implies that we have the property for n and we also have a valid base case at $n=1$.

According to the base case and inductive steps we just proved, the statement, “every real symmetric matrix is diagonalized by a matrix of its real orthonormal eigenvectors” is proved by induction.

4. Classification of Sinusoids

This HW problem can be viewed as a warm-up for the next topic in the course: which is going to be motivated by figuring out how to process signals recorded from the brain to decipher what a person wants to do in terms of a specific command to their robot arm. These kinds of problems are called “classification” problems. In this exercise, you will be using jupyter to classify sinusoids.

The iPython notebook `Sinusoidal_Projection_fa19_prob.ipynb` will guide you through the process of performing sinusoidal projections.

Suppose you already know the true potential frequencies f_i and potential phases ϕ_i of a set of sinusoidal signals

$$S := \{\sin(2\pi f_i k + \phi_i), i = 1, 2, \dots, n\}, \quad (1)$$

and you have some noisy samples of these true sinusoidal signals. You want to determine the true sinusoidal signal for each of these noisy samples—How would you approach the problem?

We will show in this problem that we can project noisy sinusoidal signals onto noiseless sinusoids to achieve good classification.

In the realistic world, one often doesn’t have the complete waveform of a continuous function, instead oftentimes one works with *samples* of the continuous function.

In our case, we generate noisy samples of the true sinusoidal signals in the following way. For each of the `num_sinusoids` true frequencies, each noisy sample y_i consists of N sample points sampled with a sampling rate of F_s sample rate, and corrupted by noise scaled by σ .

$$y_i(k) = \sin(2\pi f \cdot k / F_s) + \sigma \cdot \text{Noise}. \quad k = 1, 2, \dots, N.$$

In our example, we will work with `num_sinusoids = 3`.

A higher σ corresponds to more noise in our measurements.

Please complete the notebook by following the instructions given.

- (a) Run the first part of the jupyter notebook to generate our noisy data points. **Use $\sigma = 0.1, 1.0, 10.0, 100.0$ and comment on what you observe in the plots.**
- (b) **Complete part (b) of the notebook** to project noisy sinusoids onto potential true sinusoids. **Sketch the resulting 3D plot of projections qualitatively. Comment on what happens when you try the noise scalings $\sigma = 0.1, 1.0, 10.0, 100.0$.**
- (c) **Complete part (c) of the notebook** to classify the data points and calculate the number of misclassified points. **Report the number of misclassifications for $\sigma = 0.1, 1.0, 10.0, 100.0$. Explain what happens when there is a high level of noise.** Recall that our noisy process is random so that there can be cases where there are misclassifications even in low noise.
- (d) **For what qualitative regions of the noise level is it very beneficial for us to use projections?** For very low values of noise, do you have to do projections to successfully classify? What else could you have done? This question is asking you to reflect on what you have observed.

5. Using upper-triangularization to solve differential equations

You know that for any square matrix A with real eigenvalues, there exists a real matrix V with orthonormal columns and a real upper triangular matrix R so that $A = VRV^T$. In particular, to set notation explicitly:

$$V = [\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n]$$

$$R = \begin{bmatrix} \vec{r}_1^T \\ \vec{r}_2^T \\ \vdots \\ \vec{r}_n^T \end{bmatrix}$$

where the rows of the upper-triangular R look like

$$\begin{aligned} \vec{r}_1^T &= [\lambda_1, r_{1,2}, r_{1,3}, \dots, r_{1,n}] \\ \vec{r}_2^T &= [0, \lambda_2, r_{2,3}, r_{2,4}, \dots, r_{2,n}] \\ \vec{r}_i^T &= [\underbrace{0, \dots, 0}_{i-1 \text{ times}}, \lambda_i, r_{i,i+1}, r_{i,i+2}, \dots, r_{i,n}] \\ \vec{r}_n^T &= [\underbrace{0, \dots, 0}_{n-1 \text{ times}}, \lambda_n] \end{aligned}$$

where the λ_i are the eigenvalues of A .

Suppose our goal is to solve the n -dimensional system of differential equations written out in vector/matrix form as:

$$\begin{aligned} \frac{d}{dt} \vec{x}(t) &= A\vec{x}(t) + \vec{u}(t), \\ \vec{x}(0) &= \vec{x}_0, \end{aligned}$$

where $\vec{x}_0$ is a specified initial condition and $\vec{u}(t)$ is a given vector of functions of time.

Assume that the V and R have already been computed and are accessible to you using the notation above.

Assume that you have access to a function $ScalarSolve(\lambda, y_0, \tilde{u})$ that takes a real number λ , a real number y_0 , and a real-valued function of time $\tilde{u}$ as inputs and returns a real-valued function of time that is the solution to the scalar differential equation

$$\frac{d}{dt} y(t) = \lambda y(t) + \tilde{u}(t) \tag{2}$$

with initial condition $y(0) = y_0$.

Also assume that you can do regular arithmetic using real-valued functions and it will do the right thing. So if u is a real-valued function of time, and g is also a real-valued function of time, then $5u + 6g$ will be a real valued function of time that evaluates to $5u(t) + 6g(t)$ at time t .

Use V, R to construct a procedure for solving this differential equation

$$\begin{aligned} \frac{d}{dt} \vec{x}(t) &= A\vec{x}(t) + \vec{u}(t), \\ \vec{x}(0) &= \vec{x}_0, \end{aligned}$$

for $\vec{x}(t)$ by filling in the following template in the spots marked ♣, ◇, ♥, ♠.

(Note: It will be useful to upper triangularize A by change of basis to get a differential equation in terms of R instead of A .)

(Note: We use the notation $\vec{v}[i]$ to be the i th component of the vector $\vec{v}$)

(HINT: The process here should be similar to diagonalization with some modifications. Start from the last row of the system and work your way up to understand the algorithm.)

- 1: $\vec{x}_0 = V^T \vec{x}_0$ ▷ Change the initial condition to be in V -coordinates
- 2: $\vec{u} = V^T \vec{u}$ ▷ Change the external input functions to be in V -coordinates
- 3: **for** $i = n$ down to 1 **do** ▷ Iterate up from the bottom row
- 4: $\tilde{u}_i = \clubsuit + \sum_{j=i+1}^n \spadesuit$ ▷ Make the effective input for this level
- 5: $\tilde{x}_i = \text{ScalarSolve}(\diamond, \tilde{x}_0[i], \tilde{u}_i)$ ▷ Solve this level's scalar differential equation
- 6: **end for**
- 7: $\vec{x}(t) = \heartsuit \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \vdots \\ \tilde{x}_n \end{bmatrix} (t)$ ▷ Change back into original coordinates

- (a) Give the expression for $\heartsuit$ on line 7 of the algorithm above. (i.e. How do you get from $\tilde{x}(t)$ to $\vec{x}(t)$?)
- (b) Give the expression for $\diamond$ on line 5 of the algorithm above. (i.e. What are the λ arguments to ScalarSolve , equation (2), for the i^{th} iteration of the for-loop?)
(HINT: Convert the differential equation to be in terms of R instead of A . It may be helpful to start with $i = n$ and develop a general form for the i th row.)
- (c) Give the expression for $\clubsuit$ on line 4 of the algorithm above.
- (d) Give the expression for $\spadesuit$ on line 4 of the algorithm above.
- (e) (OPTIONAL) Let us complete the algorithm by investigating how $\text{ScalarSolve}(\lambda, y_0, \tilde{u})$ works.
Consider an input that is a weighted polynomial times and exponential.

$$\tilde{u}(t) = \alpha t^\beta e^{\gamma t}$$

Here, α is a real constant, β is a non-negative integer, and γ is a real exponent. In addition, we will assume that $\gamma = \lambda$ for simplicity. We encourage you to attempt solving this system if $\gamma \neq \lambda$ if you are curious.

What function should $\text{ScalarSolve}(\lambda, y_0, \tilde{u})$ return for the above $\tilde{u}$? Remember, we are only considering the case where $\gamma = \lambda$. Express the answer in terms of α, β, γ .

6. Homework Process and Study Group

Citing sources and collaborators are an important part of life, including being a student!

We also want to understand what resources you find helpful and how much time homework is taking, so we can change things in the future if possible.

(a) **What sources (if any) did you use as you worked through the homework?**

(b) **If you worked with someone on this homework, who did you work with?**

List names and student ID's. (In case of homework party, you can also just describe the group.)

Contributors:

- Siddharth Iyer.
- Yu-Yun Dai.
- Sanjit Batra.
- Anant Sahai.
- Sidney Buchbinder.
- Gaoyue Zhou.
- Kuan-Yun Lee.
- Daniel Abraham.

EECS 16B Designing Information Devices and Systems II
UC Berkeley

Homework 11

1. Reading Lecture Notes

Staying up to date with lectures is an important part of the learning process in this course. Here are links to the notes that you need to read for this week: [Note 12](#)

- (a) Consider two vectors $\vec{x} \in \mathbb{R}^m$ and $\vec{y} \in \mathbb{R}^n$, what is the dimension of the matrix $\vec{x}\vec{y}^\top$ and what is the rank of it?
- (b) Consider a matrix $A \in \mathbb{R}^{m \times n}$ and the rank of A is r . Suppose its SVD is $A = U\Sigma V^\top$ where $U \in \mathbb{R}^{m \times m}$, $\Sigma \in \mathbb{R}^{m \times n}$, and $V \in \mathbb{R}^{n \times n}$. Can you write A in terms of the singular values of A and outer products of the columns of U and V ?

2. Proofs

In this problem we will review some of the important proofs we saw in the lecture as practice. Let's define $S = A^\top A$ where A is an arbitrary $m \times n$ matrix. $V = [\vec{v}_1 \cdots \vec{v}_n]$ is the matrix of normalized eigenvectors of S with eigenvalues $\lambda_1, \dots, \lambda_n$.

- (a) Let $\vec{v}_1, \dots, \vec{v}_r$ be those normalized eigenvectors that correspond to non-zero eigenvalues (i.e. $\|\vec{v}_i\| = 1$ and $\lambda_i \neq 0$ for $i = 1, \dots, r$). Show that $\|A\vec{v}_i\|^2 = \lambda_i$.
- (b) Following the assumptions in part (a), show that $A\vec{v}_i$ is orthogonal to $A\vec{v}_j$.
- (c) Show that if $V \in \mathbb{R}^{n \times n}$ is an orthonormal square matrix, then $\|\vec{x}\|^2 = \|V\vec{x}\|^2$ for all $\vec{x} \in \mathbb{R}^n$. *Hint: Write the norm as an inner product instead of trying to do this elementwise.*

3. SVD

- (a) Consider the matrix

$$A = \begin{bmatrix} -1 & 1 & 5 \\ 3 & 1 & -1 \\ 2 & -1 & 4 \end{bmatrix}.$$

Observe that the columns of matrix A are mutually orthogonal with norms $\sqrt{14}$, $\sqrt{3}$, $\sqrt{42}$.

Verify numerically that columns $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 5 \\ -1 \\ 4 \end{bmatrix}$ are orthogonal to each other.

- (b) **Write $A = BD$, where B is an orthonormal matrix and D is a diagonal matrix.** What is B ? What is D ?
- (c) **Write out a singular value decomposition of $A = U\Sigma V^T$ using the previous part.** Note the ordering of the singular values in Σ should be from the largest to smallest.
(*HINT: From the previous part $A = BDI_{3 \times 3}$. Now, re-order to have eigenvalues in decreasing order.*)

4. The Moore-Penrose pseudoinverse for “wide” matrices

Say we have a set of linear equations given by $A\vec{x} = \vec{y}$. If A is invertible, we know that the solution for $\vec{x}$ is $\vec{x} = A^{-1}\vec{y}$. However, what if A is not a square matrix? In 16A, you saw how this problem could be approached for tall “standing up” matrices A where it really wasn’t possible to find a solution that exactly matches all the measurements, using linear least-squares. The linear least-squares solution gives us a reasonable answer that asks for the “best” match in terms of reducing the norm of the error vector.

This problem deals with the other case — when the matrix A is wide — with more columns than rows. In this case, there are generally going to be lots of possible solutions — so which should we choose? Why? We will walk you through the **Moore-Penrose pseudoinverse** that generalizes the idea of the matrix inverse and is derived from the singular value decomposition.

This approach to finding solutions complements the OMP approach that you learned in 16A and that we used earlier in 16B in the context of outlier removal during system identification.

(a) Say you have the matrix

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}.$$

To find the Moore-Penrose pseudoinverse we start by calculating the SVD of A . That is to say, calculate U, Σ, V such that

$$A = U\Sigma V^T$$

where U and V are orthonormal matrices.

Here we will give you that the decomposition of A is:

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

where:

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 2 & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{bmatrix}$$

$$V^T = \begin{bmatrix} 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

It is a good idea to be able to calculate the SVD yourself as you may be asked to solve similar questions on your own in the exam. For this subpart, verify that this decomposition of A is correct. You may use a computer to do the matrix multiplication if you want, but it is better to verify by hand.

(b) Let us now think about what the SVD does. Consider a rank m matrix $A \in \mathbb{R}^{m \times n}$, with $n > m$. Let $U \in \mathbb{R}^{m \times m}$, $\Sigma \in \mathbb{R}^{m \times n}$, $V \in \mathbb{R}^{n \times n}$. Let us look at matrix A acting on some vector $\vec{x}$ to give the result $\vec{y}$. We have

$$A\vec{x} = U\Sigma V^T \vec{x} = \vec{y}.$$

We can think of $V^\top \vec{x}$ as a rotation of the vector $\vec{x}$, then Σ as a scaling, and U as another rotation (multiplication by an orthonormal matrix does not change the norm of a vector, try to verify this for yourself). We will try to "reverse" these operations one at a time and then put them together to construct the Moore-Penrose pseudoinverse.

If U "rotates" the vector $(\Sigma V^\top) \vec{x}$, what operator can we derive that will undo the rotation?

- (c) **Derive a matrix $\tilde{\Sigma}$ that will "unscale", or undo the effect of Σ where it is possible to undo.** Recall that Σ has the same dimensions as A .

Hint: Consider

$$\tilde{\Sigma} = \begin{bmatrix} \frac{1}{\sigma_0} & 0 & \dots & 0 \\ 0 & \frac{1}{\sigma_1} & \dots & 0 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & \frac{1}{\sigma_{m-1}} \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix}.$$

Then compute $\tilde{\Sigma}\Sigma$.

- (d) **Derive an operator that would "unrotate" by $V^\top$.**
- (e) **Try to use this idea of "unrotating" and "unscaling" to write out an "inverse", denoted as $A^\dagger$.** That is to say,

$$\hat{\vec{x}} = A^\dagger \vec{y}$$

where $\hat{\vec{x}}$ is the recovered $\vec{x}$. The reason why the word inverse is in quotes (or why this is called a pseudo-inverse) is because we're ignoring the "divisions" by zero and $\hat{\vec{x}}$ isn't exactly equal to $\vec{x}$.

- (f) **Use $A^\dagger$ to solve for a vector $\vec{x}$ in the following system of equations.**

$$\begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix} \vec{x} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

- (g) Now we will see why this matrix, $A^\dagger = V\tilde{\Sigma}U^\top$, is a useful proxy for the matrix inverse in such circumstances. **Show that the solution given by the Moore-Penrose pseudoinverse satisfies the minimality property that if $\hat{\vec{x}} = A^\dagger \vec{y}$ is the pseudo-inverse solution to $A\vec{x} = \vec{y}$, then $\hat{\vec{x}}$ has no component in the nullspace of A .**

Hint: To show this recall that for any vector $\vec{x}$, the vector $V^{-1}\vec{x}$ represents the coordinates of $\vec{x}$ in the basis of the columns of V . Compute $V^{-1}\vec{x}$ and show that the last $(n - m)$ rows are all zero.

This minimality property is useful in many applications. You saw a control application in lecture. This is also used all the time in machine learning, where it is connected to the concept behind what is called ridge regression or weight shrinkage.

- (h) Consider a generic wide matrix A . We know that A can be written using $A = U\Sigma V^\top$ where U and V each are the appropriate size and have orthonormal columns, while Σ is the appropriate size and is a diagonal matrix — all off-diagonal entries are zero. Further assume that the rows of A are linearly independent. **Prove that $A^\dagger = A^\top(AA^\top)^{-1}$.**

(HINT: Just substitute in $U\Sigma V^\top$ for A in the expression above and simplify using the properties you know about U, Σ, V . Remember the transpose of a product of matrices is the product of their transposes in reverse order: $(CD)^\top = D^\top C^\top$.)

5. Frobenius Norm

In this problem we will investigate the basic properties of the Frobenius norm.

Similar to how the norm of vector $\vec{x} \in \mathbb{R}^N$ is defined as $\|x\| = \sqrt{\sum_{i=1}^N x_i^2}$, the Frobenius norm of a matrix $A \in \mathbb{R}^{N \times N}$ is defined as

$$\|A\|_F = \sqrt{\sum_{i=1}^N \sum_{j=1}^N |A_{ij}|^2}.$$

A_{ij} is the entry in the i^{th} row and the j^{th} column. This is basically the norm that comes from treating a matrix like a big vector filled with numbers.

(a) With the above definitions, **show that for a 2×2 matrix A :**

$$\|A\|_F = \sqrt{\text{Tr}\{A^\top A\}}.$$

Think about how this generalizes to $n \times n$ matrices. *Note:* The trace of a matrix is the sum of its diagonal entries. For example, let $A \in \mathbb{R}^{N \times N}$, then,

$$\text{Tr}\{A\} = \sum_{i=1}^N A_{ii}$$

(b) **Show that if U and V are square orthonormal matrices, then**

$$\|UA\|_F = \|AV\|_F = \|A\|_F.$$

(*HINT: Use the trace interpretation from part (a).*)

(c) **Use the SVD decomposition to show that $\|A\|_F = \sqrt{\sum_{i=1}^N \sigma_i^2}$, where $\sigma_1, \dots, \sigma_N$ are the singular values of A .**

(*HINT: The previous part might be quite useful.*)

6. SVD from the other side

In lecture, we thought about the SVD for a wide matrix M with n rows and $m > n$ columns by looking at the big $m \times m$ symmetric matrix $M^T M$ and its eigenbasis. This question is about seeing what happens when we look at the small $n \times n$ symmetric matrix $Q = M M^T$ and its orthonormal eigenbasis $U = [\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n]$ instead. Suppose we have sorted the eigenvalues so that the real eigenvalues $\tilde{\lambda}_i$ are sorted in descending order where $\lambda_i \vec{u}_i = Q \vec{u}_i$.

(a) **Show that $\tilde{\lambda}_i \geq 0$.**

(*HINT: You want to involve $\vec{u}_i^T \vec{u}_i$ somehow.*)

(b) Suppose that we define $\vec{w}_i = \frac{M^T \vec{u}_i}{\sqrt{\tilde{\lambda}_i}}$ for all i for which $\tilde{\lambda}_i > 0$. Suppose that there are ℓ such eigenvalues. **Show that $W = [\vec{w}_1, \vec{w}_2, \dots, \vec{w}_\ell]$ has orthonormal columns.**

7. Homework Process and Study Group

Citing sources and collaborators are an important part of life, including being a student!

We also want to understand what resources you find helpful and how much time homework is taking, so we can change things in the future if possible.

(a) **What sources (if any) did you use as you worked through the homework?**

(b) **If you worked with someone on this homework, who did you work with?**

List names and student ID's. (In case of homework party, you can also just describe the group.)

Contributors:

- Kourosh Hakhamaneshi.
- Elena Jia.
- Siddharth Iyer.
- Justin Yim.
- Anant Sahai.
- Gaoyue Zhou.
- Aditya Arun.

EECS 16B Designing Information Devices and Systems II

UC Berkeley

Homework 12

1. Reading Lecture Notes

Staying up to date with lectures is an important part of the learning process in this course. Here are links to the notes that you need to read for this week: [Note 13](#), [Note 14](#)

Consider a matrix $A \in \mathbb{R}^{m \times n}$ with $\text{rank}(A) = r$ and $r \leq m < n$ (wide matrix with full row rank). We will consider the representations of the full, compact, and outer product forms of the SVD of A :

$$\begin{aligned}
 A &= \begin{bmatrix} U_r & U_{m-r} \end{bmatrix} \begin{bmatrix} \Sigma_r & 0_{r \times (n-r)} \\ 0_{(m-r) \times r} & 0_{(m-r) \times (n-r)} \end{bmatrix} \begin{bmatrix} V_r^T \\ V_{n-r}^T \end{bmatrix} \\
 A &= U_r \Sigma_r V_r^T \\
 A &= \sum_{i=1}^r \vec{u}_i \sigma_i \vec{v}_i^T
 \end{aligned}$$

Where $\vec{u}_i$ and $\vec{v}_i$ are the i^{th} column vectors of U and V respectively. **Explain what spaces the columns of U_r, U_{m-r}, V_r , and V_{n-r} span. What is the rank of the matrix $\vec{u}_i \sigma_i \vec{v}_i^T$?**

2. SVD

(a) Given the matrix

$$A = \frac{1}{\sqrt{50}} \begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix} + \frac{3}{\sqrt{50}} \begin{bmatrix} -4 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix},$$

write out a singular value decomposition of matrix A in the form $U\Sigma V^T$. Note the ordering of the singular values in Σ should be from the largest to smallest.

HINT: You don't have to compute any eigenvalues for this. Some useful observations are that

$$\begin{bmatrix} 3, 4 \end{bmatrix} \begin{bmatrix} -4 \\ 3 \end{bmatrix} = 0, \quad \begin{bmatrix} 1, -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 0, \quad \left\| \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\| = \left\| \begin{bmatrix} -4 \\ 3 \end{bmatrix} \right\| = 5, \quad \left\| \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\| = \left\| \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\| = \sqrt{2}.$$

(b) Define the matrix

$$A = \begin{bmatrix} -1 & 4 \\ 1 & 4 \end{bmatrix}.$$

Find the SVD of A . Then, find the eigenvectors and eigenvalues of A . Is there a relationship between the eigenvalues or eigenvectors of A with the SVD of A ?

3. More SVD Proofs (PRACTICE)

Let $A \in \mathbb{R}^{m \times n}$ be an arbitrary matrix. Let the SVD of $A = U\Sigma V^T$. Let $r = \text{rank}(A)$. Let the columns of U be $\vec{u}_1, \dots, \vec{u}_m$, the columns of V be $\vec{v}_1, \dots, \vec{v}_n$, and the singular values of A be $\sigma_1, \dots, \sigma_r, 0, \dots, 0$ since the remaining $\min\{m, n\} - r$ singular values are 0.

- (a) **Show that** $A = \sum_{i=1}^r \sigma_i \vec{u}_i \vec{v}_i^T$.

HINT: It might be helpful to use the outer product version of matrix multiplication, i.e. $XY = \sum_{i=1}^k \vec{x}_i \vec{y}_i^T$ where $\vec{x}_i, \vec{y}_i$ are the columns of X and Y respectively.

- (b) **Prove that if $\vec{x}$ is a linear combination of $\vec{v}_{r+1}, \dots, \vec{v}_n$, then $A\vec{x} = 0$ so $\vec{x} \in \text{Null}(A)$.**

- (c) **Now we prove the opposite of the previous part, so if $\vec{x} \in \text{Null}(A)$, then $\vec{x}$ is a linear combination of the vectors $\vec{v}_{r+1}, \dots, \vec{v}_n$.**

The last two parts together prove that $\text{Null}(A) = \text{span}\{\vec{v}_{r+1}, \dots, \vec{v}_n\}$. Moreover, since the $\vec{v}_i$ are orthonormal (which implies linear independence), the vectors $\vec{v}_{r+1}, \dots, \vec{v}_n$ form an orthonormal basis for $\text{Null}(A)$.

- (d) **Show that $U_r = \begin{bmatrix} \vec{u}_1 & \dots & \vec{u}_r \end{bmatrix}$ is a basis for the column space of A .**

4. Generalizing Least Squares using the Pseudoinverse

This question will show you how to generalize the orthogonal projection from normal least squares to work for any set of vectors. On the way, we will show some powerful properties of the Moore-Penrose pseudoinverse.

- (a) Suppose we have a matrix $A \in \mathbb{R}^{m \times n}$. Now consider a vector $\vec{y} \in \mathbb{R}^m$. We are interested in projecting $\vec{y}$ onto the column space of A , i.e. finding an $\vec{x}$ such that $\|A\vec{x} - \vec{y}\|^2$ is minimized. For that $\vec{x}$, $A\vec{x}$ is known as the orthogonal projection of $\vec{y}$ onto the column space of A (this was actually how we derived $\vec{x}$ in 16A). **If $m \geq n$ and A has linearly independent columns, find $\vec{x}$ and then the projection $A\vec{x}$.**

HINT: Remember what least squares does.

- (b) The previous part only worked when restricting A to have linearly independent columns so that $A^T A$ is invertible. Thus it does not work for a wide A with more columns than rows. (Try to think why a wide matrix can't have linearly independent columns on your own.)

We want to remove these restrictions so now we consider an arbitrary A . In this case, A can have a nontrivial nullspace, and so there can be infinite such solutions for $\vec{x}$ that minimize the objective. Thus, we want to impose an extra condition that out of all the minimizing vectors $\vec{x}$, we want to find the one with the minimum squared norm $\|\vec{x}\|^2$. This is called the **minimum-norm least squares problem**:

$$\operatorname{argmin}_{\vec{x}} \|\vec{x}\|^2 \quad \text{s.t.} \quad \|A\vec{x} - \vec{y}\|^2 = \min_{\vec{s}} \|A\vec{s} - \vec{y}\|^2 \quad (1)$$

(Note that if A doesn't have a nullspace, then the least squares problem has a unique solution, so it is automatically the minimum norm solution.)

To solve this problem, we will use the SVD and the Moore-Penrose pseudoinverse. Let the SVD of $A = U\Sigma V^T$. We define the Moore-Penrose pseudoinverse $A^\dagger = V\Sigma^\dagger U^T$ from the last homework where

$$\Sigma^\dagger = \left[\begin{array}{c|c} \frac{1}{\sigma_1} & \\ & \ddots \\ & \frac{1}{\sigma_r} \\ \hline 0_{(n-r) \times r} & 0_{(n-r) \times (m-r)} \end{array} \right].$$

First show that $\|A\vec{x} - \vec{y}\|^2 = \|\Sigma V^T \vec{x} - U^T \vec{y}\|^2$.

HINT: Remember that orthonormal transformations don't affect norms of vectors (discussion 11A).

- (c) We now let $\vec{z} = V^T \vec{x}$ and $\vec{w} = U^T \vec{y}$. Since V^T is orthonormal, we know that $\|\vec{x}\|^2 = \|V^T \vec{x}\|^2 = \|\vec{z}\|^2$. Additionally since V^T is invertible, we can convert between $\vec{x}$ and $\vec{z}$, which means we can equivalently optimize over $\vec{z}$ instead of $\vec{x}$ in (1). Combining this with the result from part (b), our minimum-norm least squares problem simplifies to

$$\operatorname{argmin}_{\vec{z}} \|\vec{z}\|^2 \quad \text{s.t.} \quad \|\Sigma \vec{z} - \vec{w}\|^2 = \min_{\vec{s}} \|\Sigma \vec{s} - \vec{w}\|^2. \quad (2)$$

Using the fact that Σ has a simple form with singular values on its diagonal, **find the minimum-norm least squares solution for (2). Write your answer in terms of $\vec{w}$ and $\Sigma^\dagger$ (not Σ).**

HINT: Show that

$$\Sigma \vec{z} - \vec{w} = \begin{bmatrix} \sigma_1 z_1 - w_1 \\ \vdots \\ \sigma_r z_r - w_r \\ -w_{r+1} \\ \vdots \\ -w_m \end{bmatrix}$$

where $r = \text{rank}(A)$, and see which entries you can minimize by choosing $\vec{z}$.

- (d) Since we found the $\vec{z}$ that solved (2), we can convert this solution back to $\vec{x}$ and it should solve (1). **Do this and show that the minimum-norm least squares solution for (1) is $\vec{x} = A^\dagger \vec{y}$.**
- (e) Thus, we just proved that applying the pseudoinverse exactly gives the minimum-norm least squares solution which works for all A matrices!

Now, we can finally go back to part (a) and address the restrictions on A . Using your new-found tool, **determine the projection of $\vec{y}$ onto the column space of A , where now A can be any arbitrary matrix.**

- (f) Let $r = \text{rank}(A)$. We will define $U_r = [\vec{u}_1 \ \dots \ \vec{u}_r]$ which is the first r columns of U . **Show that the expression for the projection from part (e) can be simplified to $U_r U_r^T \vec{y}$ by plugging in the SVD and pseudoinverse.**

HINT: Write out both Σ and $\Sigma^\dagger$ and multiply them out.

This simplification of the projected vector is very interesting, and there actually is an alternative reasoning to arrive at it. We know that U_r is a basis for the column space of A , which was proved in lecture. Then projecting onto the column space of A should be the same as projecting onto U_r . But now, note that U_r has linearly independent columns, so we can just use part (a) replacing A with U_r . Thus, the projected vector is $U_r (U_r^T U_r)^{-1} U_r^T \vec{y} = U_r I^{-1} U_r^T \vec{y} = U_r U_r^T \vec{y}$, where $U_r^T U_r = I$ since U_r has orthonormal columns.

5. Brain-machine interface and neuron sorting

The iPython notebook `pca_brain_machine_interface.ipynb` will guide you through the process of analyzing brain machine interface data leveraging the SVD. This will help you to prepare for the project.

For SIXT33N (your robot car), you will need to use the SVD and PCA to classify your sound inputs so that your car does what you tell it to do.

Brain-Machine interfaces (BMIs) are a way for a brain to directly communicate to an external device. They're often used for research, mapping, assisting or repairing human cognitive or sensory-motor function.

Data was collected that shows waveform traces of (simulated) brain waves of a subject whose arm is pointing in certain directions over time. These waveform traces are gathered from electrodes inserted into the brain, but the electrodes are generally recording more than 1 neuron at the same time — the brain is crowded after all. To make predictions based on neuron firing rates, we need to be able to distinguish waveforms that come from different neurons.

Each neuron has its own waveform "signature" shape unique to that neuron. This is due to the physical characteristics of neurons, such as their physical shape and structure. It is impossible to know beforehand how many neurons an electrode measures or what each neuron's waveform looks like, so we must first separate the waveforms from the different neurons near the electrode.

The goal of this problem is to see how we can use the SVD (in its PCA manifestation) and clustering to decide which neuron fired. We will first look at a case where there are only two neurons, then a case where there are three neurons. The neurons have also been presorted using a professional software package, so we can check our model against presorted data.

Please complete the notebook by following the instructions given.

Task 1: Two Neuron Waveform Sorting

- The data you have are traces of length 32 timesteps. The data is arranged along the rows of the matrix, each row is one trace. Import the data sets and see the average waveform for each presorted neuron by running the corresponding cells in the `.ipynb`. **What is the shape of the training data matrix `three_neurons_training`?**
- What do the columns and rows of the training data matrix: `three_neurons_training` represent?**
We can approximate each waveform in a lower dimensional space using PCA. In your implementation of PCA you will decide how to choose these components.
- You will be using the SVD in your PCA function. **What matrices does the SVD return? What are the dimensions of these matrices for the SVD of `three_neurons_training`?**
- To represent each waveform in a lower dimensional space we want a basis for the time signals of the waveforms of the neurons. To construct this basis we start by taking the SVD (of `three_neurons_training`). **Which of the U and V matrices can we use to construct our desired basis?**
- Read through `PCA_train` and implement `PCA_project`.** We will use these functions throughout the rest of the notebook, so make sure you understand them.
- Call `PCA_train` on `two_neurons_training` and plot the 2 principal components.** Note that since the dataset is randomized, you might get different plots every time you run the second code cell of this notebook (the `_make_training_set` function). Note that these components have no real 'physical'

relationship to the actual shape of the neuron plots. They are a part of a basis, not the representation of exemplar traces.

- (g) Before we project onto our principal components, let us project our data onto 2 random 32 length vectors. **Run the corresponding part in the .ipynb file and comment on the separation of the 2 neuron's distinctive trace shapes in this projected basis.** Do you think that it would be easy to tell them apart just by looking at their projection into these two random dimensions?
- (h) **Project two_neurons_test onto the two principal components found using the SVD and produce a 2D scatter plot in the new basis.** We will also try projecting the presorted data containing 2 neurons so we can see how the model behaves on the 2 neurons. **Comment on the difference between the projections here and part (g) where you projected onto random directions.**
- (i) Now we will repeat this process for three principal components. **Do you see a significant improvement with 3 principal components?**

Task 2: Three Neuron Sorting

- (j) In the previous part we sorted 2 neurons using two or three principal components. In this part we are now dealing with 3 neurons. **Replicate what we did in the previous parts by finding and projecting our data on the first two principal components of this three neuron dataset in the .ipynb.**
- (k) Now **call PCA_train on three_neurons_training and plot the 3 principal components. Do the same classification process as the 2 neuron data, but now with the 3 neuron data. Compare your model's behavior with that of the presorted data.** The plot_3D function will be useful to view the results.
- (l) **How many principal components do you actually need to cluster the 3 neurons? (Hint: Think about whether there was an increase in separability between the last two parts)**

Task 3: Determining Neurons (OPTIONAL) Now that we know how to project our data onto a basis where it is clearly separable we can classify our points and determine which neuron fired. (This is what we need to know if we want to use the firings of different neurons to build a BMI.)

To classify our points we use the following algorithm:

- i. Find centroids: points that represent the average waveform of a particular neuron firing, in the basis of principal components.
 - ii. Classify each incoming point by assigning it the value of the centroid closest to it (i.e. declare that the neuron waveform that a point represents is the same as the neuron waveform of the centroid the point is closest to).
- (m) **OPTIONAL** When we are doing such problems in practice, we will not have a presorted data set. In such cases we can use algorithms like k -means clustering to determine our centroids (You should have seen this algorithm in CS61A during the Maps project). As a refresher, k -means will do the same thing as in (I)i. but with a couple modifications. A step is added at the beginning to randomly assign each data point to some centroid and a step is added to the end such that if the classifications remained the same as last time, exit. Else, go back to step 2.

Scipy happens to have a built in function to compute k -means given clustered data, which we have conveniently formatted for you.

Run the code that we have provided to do k -means for you and find the centroids. **Then use 'which_neuron' on the two_classified data set and count the number of times each neuron fired. Are these values the same as the previous part ?**

In later courses, you will build on the basic techniques taught here. But in reality, the ideas that you have learned in 16AB so far already form a conceptual core for machine learning. You can go very far with just the ideas you have already learned.

6. Movie Ratings and PCA

Recall from the lecture on PCA that we can think of movie ratings as a structured set of data. For every person i and movie j , we have that person's rating $R_{i,j}$ (thought of as a real number).

Suppose that there are m movies and n people. Let's think about arranging this data into a big $n \times m$ matrix R with people corresponding to rows and movies corresponding to columns. So the entry in the i -th row and j -th column should be $R_{i,j}$ above. Note that this is organized differently from how it was in lecture. Each row corresponds to a unique person and each column to a unique movie.

$$R = \begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1m} \\ R_{21} & R_{22} & \cdots & R_{2m} \\ \vdots & \vdots & \cdots & \vdots \\ R_{n1} & R_{n2} & \cdots & R_{nm} \end{bmatrix}$$

- (a) Suppose we believe that there is actually an underlying pattern to this rating data and that a separate study has revealed that every movie is characterized by a set of characteristics: say action and comedy. This means that every movie j has a pair of numbers $a[j], c[j]$ that define it. At the same time, every person i has a pair of sensitivities $s_a[i], s_c[i]$ that essentially define that person's preferences. A person i will rate the movie j as $R_{i,j} = s_a[i]a[j] + s_c[i]c[j]$.

If we arrange the sensitivities into a pair of n -dimensional vectors $\vec{s}_a, \vec{s}_c$, and the movie characteristics into a pair of m -dimensional vectors $\vec{a}, \vec{c}$, **use outer products to express the rating matrix R in terms of these vectors $\vec{s}_a, \vec{s}_c, \vec{a}, \vec{c}$.**

- (b) Now suppose that we want to discover the underlying nature of movies from the data R itself.

Suppose for this part, that you have four observed rating data vectors (corresponding to four different movies being rated by six individuals).

All of the movie data vectors just happened to be multiples of the following 6-dimensional vector

$$\vec{w} = \begin{bmatrix} 2 \\ -2 \\ 3 \\ -4 \\ 4 \\ 0 \end{bmatrix}. \text{ (For your convenience, note that } \|\vec{w}\| = 7\text{.)}$$

You arrange the movie data vectors as the columns of a matrix R given by:

$$R = \begin{bmatrix} | & | & | & | \\ -\vec{w} & -2\vec{w} & 2\vec{w} & 4\vec{w} \\ | & | & | & | \end{bmatrix} \quad (3)$$

You want to perform PCA (for movies) using the SVD of the matrix R to better understand the pattern in your data.

The first "principal component vector" is a *unit vector* that tells which direction we would want to project the columns of R onto to get the best rank-1 approximation for R .

Find this first principal component vector of the columns of R to explain the nature of your movie data points.

(HINT: You don't need to actually compute any SVDs in this simple case. Also, be sure to think about what size vector you want as the answer. Don't forget that you want a unit vector!)

- (c) Suppose that now, we have two more data points (corresponding to two more movies being rated by the same set of six people, i.e. we added two columns to our matrix) that are multiples of a different vector $\vec{p}$ where:

$$\vec{p} = \begin{bmatrix} 6 \\ 3 \\ -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}. \text{ (For your convenience, note that } \|\vec{p}\| = 7 \text{ and that } \vec{p}^\top \vec{w} = 0.)$$

We augment our ratings data matrix with these two new data points to get:

$$R = \begin{bmatrix} | & | & | & | & | & | \\ -\vec{w} & -2\vec{w} & 2\vec{w} & 4\vec{w} & -3\vec{p} & 3\vec{p} \\ | & | & | & | & | & | \end{bmatrix} \quad (4)$$

Find the first two principal components corresponding to the nonzero singular values of R . This is what we would use to best project the movie data points onto a two-dimensional subspace.

What is the first principal component vector? What is the second principal component vector? Justify your answer.

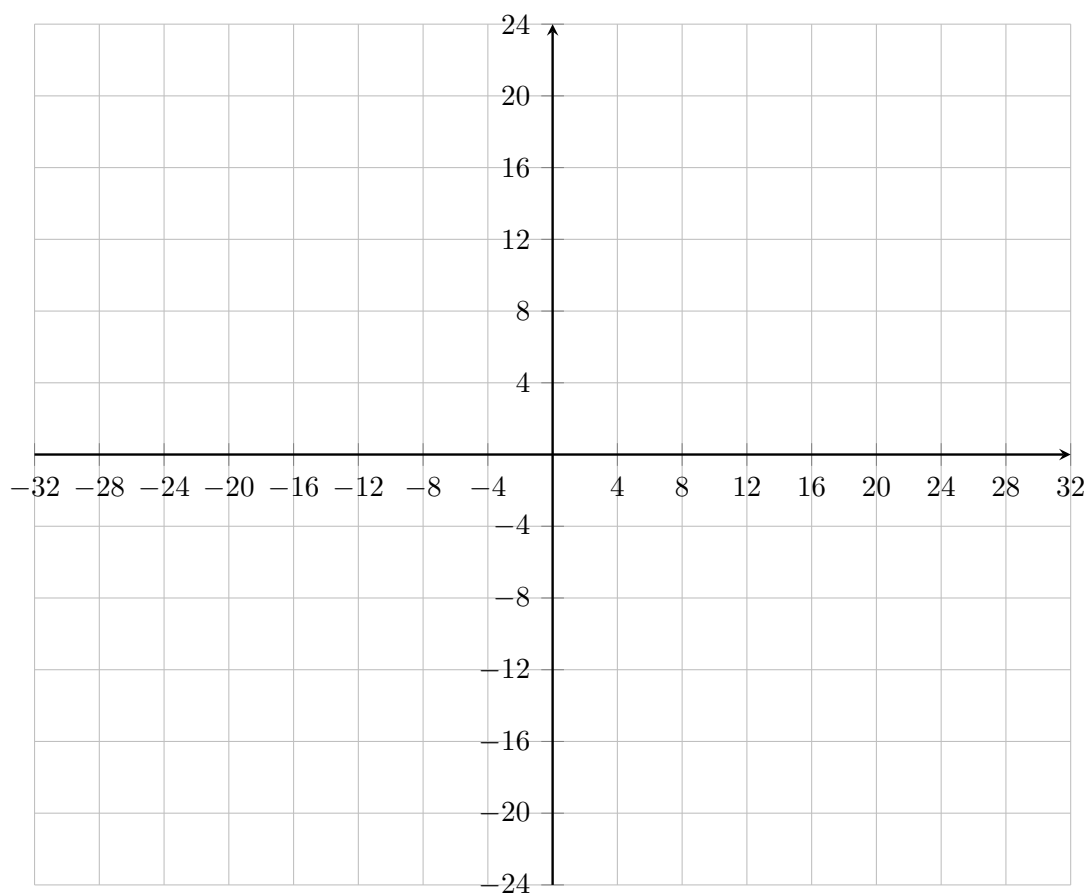
(Hint: Think about the inner product of $\vec{w}$ and $\vec{p}$ and what that implies for being able to appropriately decompose R . Again, very little computation is required here.)

- (d) In the previous part, you had

$$R = \begin{bmatrix} | & | & | & | & | & | \\ -\vec{w} & -2\vec{w} & 2\vec{w} & 4\vec{w} & -3\vec{p} & 3\vec{p} \\ | & | & | & | & | & | \end{bmatrix}$$

with $\|\vec{w}\| = 7$ and $\|\vec{p}\| = 7$, satisfying $\vec{p}^\top \vec{w} = 0$.

If we use $\vec{r}_i$ to denote the i -th column of R , **plot the movie data points $\vec{r}_i$ projected onto the first and second principal component vectors along the columns of R .** The coordinate along the first principal component should be represented by horizontal axis and the coordinate along the second principal component should be the vertical axis. **Label each point, and the axes. Remember that principal component vectors are normalized.**



7. Image Compression With SVD

Open the **image_compression.ipynb** Jupyter Notebook. Make sure to read through the theory section to understand how we will use what we know about PCA and SVD in order to compress images! Once you have read through the math and are comfortable with it, go ahead and start running the coding cells and answer the questions below.

- Run all cells up to and throughout Part 1. Thoroughly read through the ‘rank_k_approx’ function. **Since we are now only using the top k singular values, in terms of k, m , and n , how many real numbers do we require to describe the rank k approximation of A ?**
- Run the cells in Part 2 of the notebook. Notice there are two images: One compressed, one uncompressed. **Can you easily visually detect any difference between the two images even though we are using only about 40% of the data?**
- Run the cells that plot the singular values. **Given the plots above, why did the image look so good even with only about 40% of the data?**
- Play around with the slider for the kitten picture. **What is the lowest acceptable value of k that you can go without losing too much image quality? What were the memory savings?**
Note: Use your best judgment since this depends on eyesight, screen resolution, etc.
- Plot the singular values for the image of the US flag. **What do you notice about the singular values here in comparison to the kitten’s singular values? What does this mean for our low rank compression?**
- Play around with the interactive slider for the US flag. **What is the lowest acceptable value of k here? What is the memory saving?**

8. Homework Process and Study Group

Citing sources and collaborators are an important part of life, including being a student!

We also want to understand what resources you find helpful and how much time homework is taking, so we can change things in the future if possible.

- What sources (if any) did you use as you worked through the homework?**
- If you worked with someone on this homework, who did you work with?**
List names and student ID’s. (In case of homework party, you can also just describe the group.)

Contributors:

- Elena Jia.
- Neelesh Ramachandran.
- Aditya Arun.
- Daniel Abraham.
- Ashwin Vangipuram.
- Nikhil Shinde.
- Gaoyue Zhou.
- Nathan Lambert.
- Anant Sahai.

EECS 16B Designing Information Devices and Systems II
UC Berkeley

Homework 13

1. Reading Lecture Notes

Staying up to date with lectures is an important part of the learning process in this course. Here are links to the notes that you need to read for this week: [Note 15](#)

- (a) We know that a scalar function $f(x)$ can be linearly approximated around a particular point $x = x_*$ using Taylor's series expansion as follows:

$$f(x) \approx f(x_*) + \left. \frac{df}{dx} \right|_{x=x_*} \cdot (x - x_*)$$

What is the equivalent linear approximation of a multivariate scalar function $f(x, u)$ around a particular expansion point (x_*, u_*) ?

- (b) Now assume we have a vector valued function given by

$$\vec{f}(\vec{x}, \vec{u}) = \begin{bmatrix} f_1(\vec{x}, \vec{u}) \\ f_2(\vec{x}, \vec{u}) \\ \vdots \\ f_n(\vec{x}, \vec{u}) \end{bmatrix}$$

Let the state $\vec{x}$ be n dimensional, and control $\vec{u}$ be k dimensional. What is the linear approximation of the function $\vec{f}(\vec{x}, \vec{u})$ around a particular expansion point $(\vec{x}_*, \vec{u}_*)$?

2. Single-dimensional linearization

This is an exercise around linearization of a scalar system. The scalar nonlinear differential equation we have is

$$\frac{d}{dt}x(t) = \sin(x(t)) + u(t). \quad (1)$$

- (a) The first thing we want to do is find equilibria (DC operating points) that this system can support. Suppose we want to investigate potential expansion points (x^*, u^*) with $u^* = 0$. **Sketch $\sin(x^*)$ for $-4\pi \leq x^* \leq 4\pi$ and intersect it with the horizontal line at 0.** This will show us the equilibria points, where $\sin(x^*) + u^* = 0$.
- (b) **Show that all $x(t) = x_m^* = m\pi$ satisfy (1) together with $u^* = 0$, i.e. $u(t) = u^* = 0 \forall t$.**

Let us zoom in on two choices: $x_{-1}^* = -\pi$ and $x_0^* = 0$. Looking at the sketch we made, these seem like representative points.

- (c) Linearize the system (1) around the equilibrium $(x_0^*, u^*) = (0, 0)$. **What is the resulting linearized scalar differential equation for $x_\ell(t) = x(t) - x_0^* = x(t) - 0$, involving $u_\ell(t) = u(t) - u^* = u(t) - 0$?** Remember we are ignoring the higher order terms during linearization so we have to account for those using some $w(t)$ that can be thought of as noise.
- (d) For the linearized approximate system model that you found in the previous part, what happens if we try to discretize time to intervals of duration Δ ? Assume now we use a piecewise constant control input $u_\ell(t) = u_\ell[n]$ in the time interval $t \in [n\Delta, (n+1)\Delta)$, where Δ is small relative to the ranges of controls applied, and that we sample the state x every Δ (that is, at every $t = n\Delta$, where n is an integer) as well. **Write out the resulting scalar discrete-time control system model, i.e. what is $x_\ell[n+1]$ in terms of $x_\ell[n]$ and $u_\ell[n]$?** This model is an approximation of what will happen if we actually applied a piecewise constant control input to the original nonlinear differential equation at the operating point (x^*, u^*) . Here you can ignore the $w(t)$ term.
- (e) **Is the (approximate) discrete-time system you found in the previous part stable or unstable?**
- (f) Now linearize the system (1) around the equilibrium $(x_{-1}^*, u^*) = (-\pi, 0)$. **What is the resulting scalar differential equation for $x_\ell(t) = x(t) - (-\pi)$ involving $u_\ell(t) = u(t) - 0$?** Again, don't forget to include the $w(t)$ term to capture the approximation error.
- (g) For the linearized approximate system model that you found in the previous part, what happens if we try to discretize time to intervals of duration Δ ? Assume now we use a piecewise constant control input $u_\ell(t) = u_\ell[n]$ in the time interval $t \in [n\Delta, (n+1)\Delta)$, where Δ is small relative to the ranges of controls applied, and that we sample the state x every Δ (that is, at every $t = n\Delta$, where n is an integer) as well. **Write out the resulting scalar discrete-time control system model, i.e. what is $x_\ell[n+1]$ in terms of $x_\ell[n]$ and $u_\ell[n]$?** This model is an approximation of what will happen if we actually applied a piecewise constant control input to the original nonlinear differential equation at the operating point (x^*, u^*) . Here you can ignore the $w(t)$ term.
- (h) **Is the (approximate) discrete-time system you found in the previous part stable or unstable?**
- (i) Suppose for the two *linearized discrete-time systems* derived in parts (d) and (g), we chose to apply a feedback law

$$u_\ell[n] = -k(x_\ell[n] - x^*).$$

For what range of k values, would the resulting linearized discrete-time systems be stable? Your answer will depend on Δ .

(HINT: Your solution to part (d) should be

$$x_\ell[n+1] = e^\Delta x_\ell[n] + u_\ell[n](e^\Delta - 1)$$

and solution to part (g) should be

$$x_\ell[n+1] = e^{-\Delta} x_\ell[n] + u_\ell[n](1 - e^{-\Delta})$$

)

3. Linearizing for understanding amplification

Linearization isn't just something that is important for control, robotics, machine learning, and optimization — it is one of the standard tools used across different areas, including thinking about circuits.

The circuit below is a voltage amplifier, where the element inside the box is a bipolar junction transistor (BJT). You do not need to know what a BJT is to do this question.

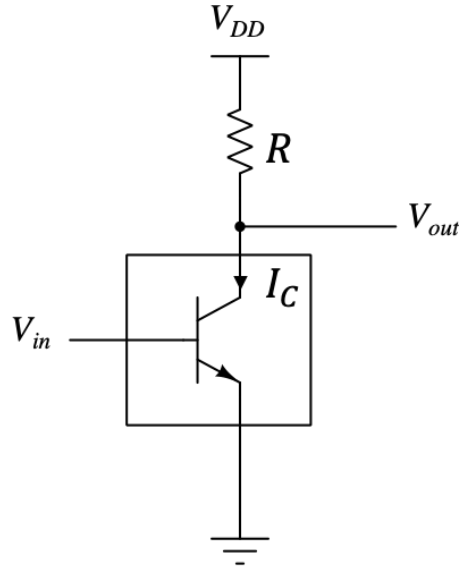


Figure 1: Voltage amplifier circuit using a BJT

The BJT in the circuit can be modeled quite accurately as a nonlinear, voltage-controlled current source, where the collector current I_C is given by:

$$I_C(V_{in}) = I_S \cdot e^{\frac{V_{in}}{V_{TH}}}, \quad (2)$$

where V_{TH} is the thermal voltage. We can assume $V_{TH} = 26\text{mV}$ at room temperature. I_S is a constant whose exact value we are not giving you because we want you to find ways of eliminating it in favor of other quantities whenever possible.

Let's consider the 2N3904 model of a BJT, where the above expression for $I_C(V_{in})$ holds as long as $0.2\text{V} < V_{out} < 40\text{V}$, and $0.1\text{mA} < I_C < 10\text{mA}$. (Note that the 2N3904 is a cheap transistor that people often use in personal projects. You can get them for 3 cents each if you buy in bulk.)

The goal of this circuit is to pick a particular point (V_{in}^*, V_{out}^*) so that any small variation δV_{in} in the input voltage V_{in} can be amplified to a relatively larger variation δV_{out} in the output voltage V_{out} . In other words, if $V_{in} = V_{in}^* + \delta V_{in}$ and $V_{out} = V_{out}^* + \delta V_{out}$, then we want the magnitude of the 'amplification gain' given by $\left| \frac{\delta V_{out}}{\delta V_{in}} \right|$ to be large. We're going to investigate this amplification using linearization.

- Write a symbolic expression for V_{out} as a function of I_C , V_{DD} and R in Fig 1.
- Now let's linearize I_C in the neighborhood of an input voltage V_{in}^* and a specific I_C^* . Assume that you have found a particular pair of input voltage V_{in}^* and current I_C^* that satisfy the current equation (2).

We can look at nearby input voltages and see how much the current changes. We can write the linearized expression for the collector current around this point as:

$$I_C(V_{in}) = I_C(V_{in}^*) + m(V_{in} - V_{in}^*) = I_C^* + m \delta V_{in} \quad (3)$$

where $\delta V_{in} = V_{in} - V_{in}^*$ is the change in input voltage.

What is m here as a function of I_C^* and V_{TH} ?

(If you take EE105, you will learn that this m is called the transconductance, which is usually written g_m , and is the single most important parameter in most analog circuit designs.)

(HINT: First just find m by taking the appropriate derivative and using the chain rule as needed. Then leverage the Taylor's series expansion of the exponential function to express it in terms of the desired quantities.)

- (c) We now have a linear relationship between small changes in current and voltage, $\delta I_C = m \delta V_{in}$ around a known solution (I_C^*, V_{in}^*) . This is called a “bias point” in circuits terminology. (This is also why related things in neural nets are called bias terms — their job is to get the nonlinearity to behave the way we want it to.)

As a reminder, **the goal of this problem is to pick a particular point (V_{in}^*, V_{out}^*) so that any small variation δV_{in} in the input voltage V_{in} can be amplified to a relatively larger variation δV_{out} in the output voltage V_{out} . In other words, if $V_{in} = V_{in}^* + \delta V_{in}$ and $V_{out} = V_{out}^* + \delta V_{out}$, then we want the magnitude of the ‘amplification gain’ given by $\left| \frac{\delta V_{out}}{\delta V_{in}} \right|$ to be large.**

Plug in your linearized equation for I_C in the answer from part (a). Define

$$V_{out}^* = V_{DD} - R I_C^*$$

so that it makes sense to view $V_{out} = V_{out}^* + \delta V_{out}$ when we have $V_{in} = V_{in}^* + \delta V_{in}$, and **find the approximate linear relationship between δV_{out} and δV_{in} .**

The ratio $\frac{\delta V_{out}}{\delta V_{in}}$ is called the small-signal voltage gain of this amplifier around this bias point.

- (d) Assuming that $V_{DD} = 10\text{V}$, $R = 1\text{k}\Omega$, and $I_C^* = 1\text{mA}$ when $V_{in}^* = 0.65\text{V}$, **verify that the magnitude of the small-signal voltage gain $\left| \frac{\delta V_{out}}{\delta V_{in}} \right|$ between the input and the output around this bias point is approximately 38.**

(HINT: Remember $V_{TH} = 26\text{mV}$)

- (e) If $I_C^* = 9\text{mA}$ when $V_{in}^* = 0.7\text{V}$ with all other parameters remaining fixed, **verify that the magnitude of the small-signal voltage gain $\left| \frac{\delta V_{out}}{\delta V_{in}} \right|$ between the input and the output around this bias point is approximately 350.**

- (f) If you wished to make an amplifier with as large of a small signal gain as possible, which operating (bias) point would you choose among $V_{in}^* = 0.65\text{V}$ (part d) and $V_{in}^* = 0.7\text{V}$ (part e)?

This shows you how by appropriately biasing (choosing an operating point), we can adjust what our gain is for small signals. Although here, we just wanted to show you this as a simple application of linearization, these ideas are developed a lot further in 105, 140, and other courses to create things like op-amps and other analog information-processing systems.

4. Inverse Kinematics

Inverse Kinematics is critical in robotics, control, and computer graphics applications.

We need to be able to go backward from what we want to have happen in the real (or virtual) world to how to set parameters.

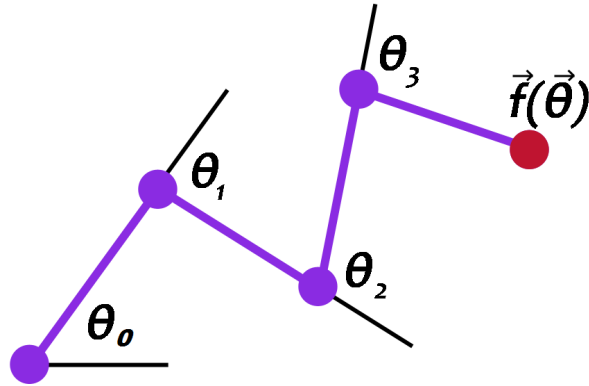


Figure 2: An example of an arm parameterized by θ_0 , θ_1 , θ_2 , and θ_3 with the end effector at point $\vec{f}(\vec{\theta})$.

Suppose you have a robotic arm composed of several rotating joints.

The lengths r_i of the arm are fixed, but you can control the arm by specifying the amount of rotation θ_i for each joint. If we have an arm with four joints, it can be parameterized by:

$$\vec{\theta} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}. \quad (4)$$

Suppose further that we have some target $\vec{t} \in \mathbb{R}^2$, which represents a point in the 2D space, and we would like for the end of the arm, called the end effector, to reach for the target. We have some function $\vec{f}(\vec{\theta})$ that rotates each joint of the arm according to the input and returns the position of the end effector. Figure 2 shows a visualization of an arm rotated by $\vec{\theta}$. To make the arm reach for the target $\vec{t}$, we want to find where the function $\vec{g}$ defined as

$$\vec{\theta} \in \mathbb{R}^4 \mapsto \vec{g}(\vec{\theta}) = \vec{f}(\vec{\theta}) - \vec{t} \quad (5)$$

is equal to $\vec{0}$. To accomplish this, we use the spirit of Newton's method for solving potentially nonlinear equations. (This is related to the spirit of the earlier problem on using iterative ways of solving least-squares problems.)

You might have seen Newton's method in your calculus course in the 1-D case. In this case, you have a real function g of a single parameter θ and we want to find a $\hat{\theta}$ so that $g(\hat{\theta}) = 0$. The method is the following. Step i of Newton's method does the following:

- (a) Linearize g around $\theta^{(i)}$, the current estimate of $\hat{\theta}$: $\hat{g}^{(i)}(\theta) = g(\theta^{(i)}) + g'(\theta^{(i)})(\theta - \theta^{(i)})$
- (b) $\theta^{(i+1)}$ solves $\hat{g}^{(i)}(\theta) = 0$. So we have

$$0 = g(\theta^{(i)}) + g'(\theta^{(i)})(\theta - \theta^{(i)}) \quad (6)$$

$$\theta = -\frac{g(\theta^{(i)})}{g'(\theta^{(i)})} + \theta^{(i)} \quad (7)$$

Therefore $\theta^{(i+1)} = -\frac{g(\theta^{(i)})}{g'(\theta^{(i)})} + \theta^{(i)}$. Note also that doing this is solving: $g'(\theta^{(i)})(\theta - \theta^{(i)}) = g(\theta^{(i)})$, which reduces to "inverting" the linear operator $z \mapsto g'(\theta^{(i)})z$.

We will iterate this until $g(\theta)$ is close enough to 0 for our application. In practice, instead of solving exactly for $\hat{g}^{(i)} = 0$, in the second step of iteration i , we may choose to move $\theta^{(i)}$ by a fixed step-size η in the direction that the first-order approximation to the function suggests, but not all the way. This is done because the derivative $g'(\theta)$ where $\hat{g}(\theta) = 0$ might be very different from $g'(\theta^{(i+1)})$ where $\theta^{(i+1)}$. After all, linearization is only valid in a local neighborhood. (This is the spirit of gradient descent as well.)

While you might have seen Newton's method as described above in your calculus courses, you might not have seen the vector-generalization of it. It follows exactly the same spirit. The first-order approximation to the vector valued function $\vec{g}(\vec{\theta})$ at $\vec{\theta}^{(i)}$ is now $\vec{g}(\vec{\theta}^{(i)}) + J_{\vec{g}}(\vec{\theta}^{(i)})(\vec{\theta} - \vec{\theta}^{(i)})$ where $J_{\vec{g}}(\vec{\theta})$ is the Jacobian matrix of the function $\vec{g}(\vec{\theta})$. For this problem, we will be using a robotic arm with 4 joints in a 2-dimensional space. Therefore, the Jacobian of $\vec{g}(\vec{\theta})$ will be a 2x4 matrix, and it is computed by calculating the partial derivatives of $\vec{g}(\vec{\theta})$:

$$J_{\vec{g}} = \begin{bmatrix} \frac{\partial g_x(\vec{\theta})}{\partial \theta_1} & \frac{\partial g_x(\vec{\theta})}{\partial \theta_2} & \frac{\partial g_x(\vec{\theta})}{\partial \theta_3} & \frac{\partial g_x(\vec{\theta})}{\partial \theta_4} \\ \frac{\partial g_y(\vec{\theta})}{\partial \theta_1} & \frac{\partial g_y(\vec{\theta})}{\partial \theta_2} & \frac{\partial g_y(\vec{\theta})}{\partial \theta_3} & \frac{\partial g_y(\vec{\theta})}{\partial \theta_4} \end{bmatrix}. \quad (8)$$

In this notation, we use $\vec{g}(\vec{\theta}) = [g_x(\vec{\theta}) \quad g_y(\vec{\theta})]^T$ where $g_x(\vec{\theta})$ is the x coordinate of the end effector and $g_y(\vec{\theta})$ is the y coordinate in our 2D space. There is nothing mysterious about this, if you think about it, this matrix of partial derivatives (a partial derivative is just a regular derivative with respect to a particular variable, treating all the other variables as constants) is the natural n-d candidate to replace g' .

The Newton algorithm in this case is an iterative method that gives us successively better estimates for our vector $\vec{\theta}$. If we start with some guess $\vec{\theta}^{(i)}$, then the next guess is given by

$$\vec{\theta}^{(i+1)} = \vec{\theta}^{(i)} - \eta J_{\vec{g}}^{\dagger}(\vec{\theta}^{(i)}) \vec{g}(\vec{\theta}^{(i)}) \quad (9)$$

where η is adjusted to determine how large of a step we make between $\vec{\theta}^{(i)}$ and $\vec{\theta}^{(i+1)}$. Notice that we need to invert the Jacobian matrix of first-partial-derivatives, and this matrix is not square. It is in fact a wide matrix. Fortunately, we know how to "invert" wide matrices, using the Moore Penrose pseudo-inverse that you saw in a previous homework. The minimality property of the Moore-Penrose pseudoinverse that we studied then is useful here because we would rather take small steps than big ones. And when tracking a moving reference, we'd like to have the joint angles change in a minimal way rather than in some very convoluted fashion.

The following problem will guide you step-by-step through the implementation of the pseudoinverse. The three steps of the pseudoinverse algorithm are:

- First, compute the compact-form SVD of the input matrix.
- Next, we compute Σ^{-1} by inverting each singular value σ_i .
- Finally, we compute the pseudoinverse by multiplying the matrices together in the right order.

The next three parts guide you through the code to be written for the pseudoinverse.

- (a) In the "pseudoinverse" function, **compute** the SVD of the input matrix A by using the appropriate NumPy function.

(HINT: It is useful to read the documentation for the numpy functions involved so that you call them with the right arguments. For example, for this problem you need to call `svd(A, full_matrices=False)`, What is the default? What do you want the function to return? What exactly does the SVD function return in numpy?)

- (b) To save memory space, the NumPy algorithm returns the matrix Σ as a one-dimensional array of the singular values.

Use this vector to **compute the diagonal entries of Σ^{-1}** . Be careful of numerical issues: first threshold the singular values, and only invert the singular values above a certain value ϵ , considering smaller ones are 0.

The reason is that you don't want to have very big entries in the pseudoinverse because that will defeat the point of you using a small step-size η to stay within the rough neighborhood that your linear approximation is valid. So you need to stop that from happening. That is what considering small singular values as being 0 does.

- (c) We now have all of the parts to compute the relevant pseudoinverse of A . **Add this computation to the function.**

(HINT: `np.diag` can be a very useful tool in converting a vector into a square diagonal matrix. Also remember that numpy knows how to multiply matrices and using `.T` compute transposes.)

- (d) There are three test cases that you can use to determine if your pseudoinverse function works correctly. In the first case, the arm is able to reach the target, and the end of the arm will be touching the target. In the second case, the arm should be pointing in a straight line towards the blue circle. The last case is the same as the second with the addition that a singular value will be very close to zero to test your pseudoinverse function's ability to handle small singular values. There is also an animated test case that will move the target in and out of the reach of the arm. Verify that the arm follows the target correctly and points towards the target when it is out of reach.

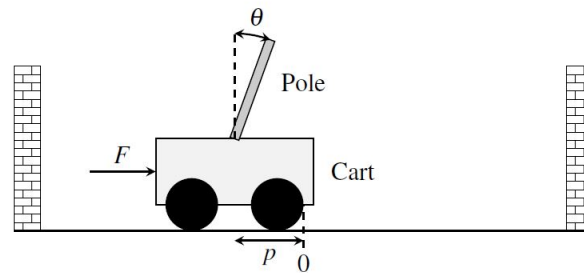
5. Segway Tours

A segway is a stand on two wheels, and can be thought of as an inverted pendulum. The segway works by applying a force (through the spinning wheels) to the base of the segway. This controls both the position on the segway and the angle of the stand. As the driver pushes on the stand, the segway tries to bring itself back to the upright position, and it can only do this by moving the base.

You may recall analyzing a problem related to segway in 16A homework. You were given a linear discrete time representation of the segway dynamics, and were asked to analyze if it's possible to make the segway reach some desired states. Now, we will see how to derive the linear discrete time system from the equations of motion and do some further detailed analysis.

Note that this problem is completely independent from the 16A version and does not require any prior knowledge or results of the previous problem.

Is it possible for the segway to be brought upright and to a stop from any initial configuration? There is only one input (force) used to control two outputs (position and angle). Let's model the segway as a cart-pole system and analyze.



A cart-pole system can be fully described by its position p , velocity $\frac{dp}{dt}$, angle θ , and angular velocity $\frac{d\theta}{dt}$. We can write this as the continuous time state vector $\vec{x}$ as follows:

$$\vec{x} = \begin{bmatrix} p \\ \frac{dp}{dt} \\ \theta \\ \frac{d\theta}{dt} \end{bmatrix}$$

The input to this system is a scalar quantity $u(t)$ at time t , which is the force F applied to the cart (or base of the segway). Let the co-efficient of friction be k .

The equations of motion for this system are as follows:

$$\begin{aligned} \frac{d^2 p}{dt^2} &= \frac{1}{\frac{M}{m} + \sin^2 \theta} \left(\frac{u}{m} + \left(\frac{d\theta}{dt} \right)^2 l \sin \theta - g \sin \theta \cos \theta - \frac{k}{m} \frac{dp}{dt} \right) \\ \frac{d^2 \theta}{dt^2} &= \frac{1}{l \left(\frac{M}{m} + \sin^2 \theta \right)} \left(-\frac{u}{m} \cos \theta - \left(\frac{d\theta}{dt} \right)^2 l \cos \theta \sin \theta + \frac{M+m}{m} g \sin \theta + \frac{k}{m} \frac{dp}{dt} \cos \theta \right) \end{aligned} \quad (10)$$

The derivation of these equations is a mechanics problem and not in 16B syllabus, but interested students can look up the details online.

- (a) First let us linearize the system of equations in (10) about the upright position at rest, i.e. $\theta_* = 0$ and $\frac{d\theta}{dt}_* = 0$. **Show that the linearized system of equations is given by the following state space form:**

$$\frac{d\vec{x}(t)}{dt} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{k}{M} & -\frac{m}{M}g & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{k}{Ml} & \frac{M+m}{Ml}g & 0 \end{bmatrix}}_A \vec{x}(t) + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{M} \\ 0 \\ -\frac{1}{Ml} \end{bmatrix}}_{\vec{b}} u(t) \quad (11)$$

(HINT: Since we are linearizing around $\theta_* = 0$ and $\frac{d\theta}{dt}_* = 0$, you can use the following approximations for small values of θ :

$$\begin{aligned} \sin \theta &\approx \theta \\ \sin^2 \theta &\approx 0 \\ \cos \theta &\approx 1 \\ \left(\frac{d\theta}{dt}\right)^2 &\approx 0. \end{aligned}$$

You do not have to do the full linearization using Taylor series, you can just substitute the values above. You will get the same answer as doing the linear Taylor series approximation.)

- (b) For all subsequent parts, assume that $m = 1$, $M = 10$, $g = 10$, $l = 1$ and $k = 0.1$. Let's consider the discrete time representation of the state space (11) at time $t = n\Delta$. For simplicity, assume $\Delta = 1$. The discrete time state $\vec{x}_d$ follows the following linear model:

$$\vec{x}_d[n+1] = A_d \vec{x}_d[n] + \vec{b}_d u_d[n] \quad (12)$$

where $A_d \in \mathbb{R}^{4 \times 4}$ and $\vec{b}_d \in \mathbb{R}^{4 \times 1}$. **Find A_d and $\vec{b}_d$.** Use the Jupyter notebook **segway.ipynb** for numerical calculations, and approximate the results to 2 or 3 significant figures.

(HINT: Recall that the continuous time scalar differential equation

$$\frac{dz(t)}{dt} = \lambda z(t) + cw(t)$$

can be represented in discrete time ($n\Delta = t$) as follows:

$$z_d[n+1] = \begin{cases} (e^{\lambda\Delta}) \cdot z_d[n] + \left(\frac{e^{\lambda\Delta}-1}{\lambda}\right) \cdot cw_d[n] & \text{if } \lambda \neq 0 \\ (1) \cdot z_d[n] + (\Delta) \cdot cw_d[n] & \text{if } \lambda = 0 \end{cases}$$

Use the eigendecomposition of $A = V\Lambda V^{-1}$ to do change of basis variables, and you should finally reach

$$\vec{x}_d[n+1] = \underbrace{V\Lambda_d V^{-1}}_{A_d} \vec{x}_d[n] + \underbrace{V M_d V^{-1} \vec{b}}_{\vec{b}_d} u_d[n]$$

What are the elements of Λ_d and M_d in terms of the elements of Λ ?)

- (c) **Show that the discrete time system in (12) is controllable by using the appropriate matrix in the Jupyter notebook.**

(HINT: Is the controllability matrix full rank? You have to use numerical values of A_d and $\vec{b}_d$ from the previous part. Use the Jupyter notebook for all numerical calculations.)

- (d) Since the discrete time system is controllable, it is possible to reach any final state $\vec{x}_{d,f}$ starting from initial state $\vec{x}_{d,i}$ using an appropriate sequence of inputs in exactly 4 steps, provided that the deviations are small enough so that the linearization approximation is valid. **Set up a set of linear equations to solve for the $u_d[0], u_d[1], u_d[2], u_d[3]$ given the initial and final states. Find the input sequence to reach the upright position $\vec{x}_{d,f} = \vec{x}_d[4] =$**

$$\vec{x}_{d,i} = \vec{x}_d[0] = \begin{bmatrix} -2 \\ 3.1 \\ 0.3 \\ -0.6 \end{bmatrix} \quad \text{starting from an initial state} \quad \vec{x}_{d,f} = \vec{x}_d[4] = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Explain qualitatively what you observe from the segway simulation.

(HINT: Use (12) and loop unrolling to express $\vec{x}_d[4]$ as a linear combination of $\vec{x}_d[0], u_d[3], u_d[2], u_d[1], u_d[0]$.)

- (e) Now suppose we try to use an initial state $\vec{x}_{d,i} = \vec{x}_d[0] = \begin{bmatrix} -2 \\ 3.1 \\ 3.3 \\ -0.6 \end{bmatrix}$ that does not satisfy the linearization

constraint, since $\theta_i = 3.3$ is very far from the linearization point $\theta_* = 0$. **Using the equations derived in the previous part, use the Jupyter notebook to determine the input sequence to reach the same final upright position. Explain qualitatively what you observe from the segway simulation.** Use the Jupyter notebook for all numerical calculations and simulation.

Compare the simulation results in parts (d) and (e). In both cases, the segway finally stabilizes to an upright position at rest. However, in part (d) the behavior of the segway looks more realistic whereas in part (e) it is doing some wild unexpected rotations.

This is because the linearization approximation is valid with the small initial values of θ and $\frac{d\theta}{dt}$ in part (d). So this discrete time linear model is a good representation of the original continuous time non-linear system. Hence the trajectory taken by the segway from the initial to the final position is similar to what we may expect from real life physics.

However in part (e), the linearization approximation is violated. The model still converges to the final upright position because (12) is controllable as we proved in part (c). However, since the approximation is not valid anymore, this discrete time linear model is **not** a good representation of the original continuous time non-linear system. Hence the trajectory is extremely weird with the segway undergoing a few full rotations, and does not match what we would expect from the real system.

We can still analyze the system in continuous time by directly solving the set of non-linear differential equations in (10) (out of 16B scope) or in discrete time using a discretized version of (10). Note that there are two independent distinctions we are making, i.e. continuous vs discrete, and linear vs non-linear. Part (e) failed because it's beyond the scope of the linear model, not because we are using a discrete time system. A non-linear discrete time analysis would also give the correct solution.

- (f) We have attached two videos: **cart_pole_non_linearized.mp4** and **cart_pole_linear.mp4**. These videos show the behavior of the continuous time segway system before (10) and after (11) we linearize

it, when no control input is applied. More specifically, we use a starting state of

$$\vec{x}(0) = \begin{bmatrix} 0 \\ 0 \\ 0.1\text{rad} \\ 0 \end{bmatrix}$$

and we set the input $u(t) = 0$ for the entire simulation. **Contrasting the non-linear and linearized continuous time simulations in the two videos, do you notice any differences in the trajectory when the angle θ gets large? Why is this the case?**

6. Homework Process and Study Group

Citing sources and collaborators are an important part of life, including being a student!

We also want to understand what resources you find helpful and how much time homework is taking, so we can change things in the future if possible.

(a) **What sources (if any) did you use as you worked through the homework?**

(b) **If you worked with someone on this homework, who did you work with?**

List names and student ID's. (In case of homework party, you can also just describe the group.)

Contributors:

- Ayan Biswas.
- Anant Sahai.
- Kris Pister.
- Alex Devonport.
- Regina Eckert.
- Stephen Bailey.
- Daniel Abraham.

EECS 16B Designing Information Devices and Systems II
UC Berkeley Homework 14

1. Reading Lecture Notes

Staying up to date with lectures is an important part of the learning process in this course. Here are links to the notes that you need to read for this week: [Note 16](#), [Note 17](#)

- (a) Write out an $N \times N$ orthonormal matrix U whose columns represent the DFT basis.

2. Linearization to help classification: discovering logistic regression and how to solve it

You can, in spirit, reduce the problem of linear multi-class classification to that of binary classification by picking vectors that correspond to each of the categories “X” as compared with all the other examples categorized into a hybrid synthetic category of “not-X”. This will give rise to vectors corresponding to each category with the winner selected by seeing which one wins. However, we will focus here on the binary problem since that is the conceptual heart of this approach.

As was discussed in lecture, the naive straightforward way of picking the decision boundary (by looking at the mean example of each category and drawing the perpendicular bisector) is not always the best. The included Jupyter Notebook includes synthetic examples that illustrate the different things that can happen so that you can better appreciate the pathway that leads us to discover logistic regression as a natural approach to solve this problem based on the conceptual building blocks that you have already seen.

It is no exaggeration to say that logistic regression is the default starting method for doing classification in machine learning contexts, the same way that straightforward linear regression is the default starting method for doing regression. A lot of other approaches can be viewed as being built on top of logistic regression. Consequently, getting to logistic regression is a nice ending-point for this part of the 16AB story as pertains to classification.

Let’s start by giving things some names. Consider trying to classify a set of measurements $\vec{x}_i$ with given labels ℓ_i . For the binary case of interest here, we will think of the labels as being “+” and “-”. For expository convenience, and because we don’t want to have to carry it around separately, we will fold our threshold implicitly into the weights by augmenting our given measurements with the constant “1” in the first position of each $\vec{x}_i$. Now, the classification rule becomes simple. We want to learn a vector of weights $\vec{w}$ so that we can deem any point with $\vec{x}_i^\top \vec{w} > 0$ as being a member of the “+” category and anything with $\vec{x}_i^\top \vec{w} < 0$ as being a member of the “-” category.

The way that we will do this, is to do a minimization in the spirit of least squares. Except, instead of necessarily using some sort of squared loss function, we will just consider a generic cost function that can depend on the label and the prediction for the point. For the i -th data point in our training data, we will incur a cost $c(\vec{x}_i^\top \vec{w}, \ell_i)$ for a total cost that we want to minimize as:

$$\arg \min_{\vec{w}} c_{total}(\vec{w}) = \sum_{i=1}^m c(\vec{x}_i^\top \vec{w}, \ell_i) \quad (1)$$

Because this can be a nonlinear function, our goal is to solve this iteratively as a sequence of least-squares problems that we know how to solve.

Consider the following algorithm:

- 1: $\vec{w} = \vec{0}$ ▷ Initialize the weights to $\vec{0}$
- 2: **while** Not done **do** ▷ Iterate towards solution
- 3: Compute $\vec{w}^\top \vec{x}_i$ ▷ Generate current estimated labels
- 4: Compute $\frac{d}{d\vec{w}} c(\vec{w}^\top \vec{x}_i, \ell_i)$ ▷ Generate derivatives with respect to $\vec{w}$ of the cost for update step
- 5: Compute $\frac{d^2}{d\vec{w}^2} c(\vec{w}^\top \vec{x}_i, \ell_i)$ ▷ Generate second derivatives of the cost for update step
- 6: $\vec{\delta w} = \text{LeastSquares}(\cdot, \cdot)$ ▷ We will derive what to call least squares on
- 7: $\vec{w} = \vec{w} + \vec{\delta w}$ ▷ Update parameters
- 8: **end while**
- 9: **Return** $\vec{w}$

The key step above is figuring out with what arguments to call LeastSquares while only having the labels ℓ_i and the points $\vec{x}_i$.

When the function $\vec{f}(\vec{x}, \vec{y}) : \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^m$ takes in vectors and outputs a vector, the relevant derivatives for linearization are also represented by matrices:

$$D_{\vec{x}} \vec{f} = \begin{bmatrix} \frac{\partial f_1}{\partial x[1]} & \cdots & \frac{\partial f_1}{\partial x[n]} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x[1]} & \cdots & \frac{\partial f_m}{\partial x[n]} \end{bmatrix}.$$

$$D_{\vec{y}} \vec{f} = \begin{bmatrix} \frac{\partial f_1}{\partial y[1]} & \cdots & \frac{\partial f_1}{\partial y[k]} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial y[1]} & \cdots & \frac{\partial f_m}{\partial y[k]} \end{bmatrix}.$$

where

$$\vec{x} = \begin{bmatrix} x[1] \\ \vdots \\ x[n] \end{bmatrix} \quad \vec{y} = \begin{bmatrix} y[1] \\ \vdots \\ y[n] \end{bmatrix}.$$

Then, the linearization (first-order expansion) becomes

$$\vec{f}(\vec{x}, \vec{y}) \approx \vec{f}(\vec{x}_0, \vec{y}_0) + D_{\vec{x}} \vec{f} \cdot (\vec{x} - \vec{x}_0) + D_{\vec{y}} \vec{f} \cdot (\vec{y} - \vec{y}_0). \quad (2)$$

(a) Now, suppose we wanted to approximate the cost for each data point

$$c_i(\vec{w}) = c(\vec{x}_i^\top \vec{w}, \ell_i) \quad (3)$$

where

$$\vec{w} = \begin{bmatrix} w[1] \\ \vdots \\ w[n] \end{bmatrix}$$

in the neighborhood of a weight vector $\vec{w}_*$. Our goal is to write out the first-order expression for approximating the cost function $c_i(\vec{w}_* + \delta \vec{w})$. This should be something in vector/matrix form like you have seen for the approximation of nonlinear systems by linear systems. We don't want to take any second derivatives just yet — only first derivatives. We have outlined a skeleton for the derivation with some parts missing. Follow the guidelines in each sub-section.

i) Comparing to eq. (2), we know that $c_i(\vec{w}_* + \delta \vec{w}) \approx c_i(\vec{w}_*) + \frac{d}{d\vec{w}} c_i(\vec{w}_*) \delta \vec{w}$. **Write out the vector form of $\frac{d}{d\vec{w}} c_i(\vec{w}_*)$.**

ii) **Write out the partial derivatives of $c_i(\vec{w})$ with respect to $w[g]$, the g^{th} component of $\vec{w}$.** (HINT: Use the linearity of derivatives and sums to compute the partial derivatives with respect to each of the $w[g]$ terms. Don't forget the chain rule and the fact that $\vec{x}_i^\top \vec{w} = \sum_{j=1}^n x_i[j] w[j] = x_i[g] w[g] + \sum_{j \neq g} x_i[j] w[j]$.)

iii) With what you had above, **can you fill in the missing part to express the row vector $\frac{d}{d\vec{w}} c_i(\vec{w})$?**

$$\frac{d}{d\vec{w}} c_i(\vec{w}) = c'(\vec{x}_i^\top \vec{w}, \ell_i) \underline{\hspace{2cm}}$$

- (b) Now, we want a better approximation that includes second derivatives. For a general function, we would look for

$$f(\vec{x}_0 + \delta\vec{x}) \approx f(\vec{x}_0) + f'(\vec{x}_0)\delta\vec{x} + \frac{1}{2}\delta\vec{x}^\top f''(\vec{x}_0)\delta\vec{x} \quad (4)$$

where $f'(\vec{x}_0)$ is an appropriate row vector and, as you've seen in the note, $f''(\vec{x}_0)$ is called Hessian that represents the second derivatives.

- i) Comparing to eq. (4), we know that

$$c_i(\vec{w}_* + \delta\vec{w}) \approx c_i(\vec{w}_*) + \frac{d}{d\vec{w}}c_i(\vec{w}_*)\delta\vec{w} + \frac{1}{2}\delta\vec{w}^\top \frac{d^2}{d\vec{w}^2}c_i(\vec{w}_*)\delta\vec{w}$$

Write out the matrix form of $\frac{d^2}{d\vec{w}^2}c_i(\vec{w}_*)$.

- ii) **Take the second derivatives of the cost $c_i(\vec{w})$, i.e. solve for $\frac{\partial^2 c_i(\vec{w})}{\partial w[g]\partial w[h]}$.**

(HINT: You should use the answer to part (a) and just take another derivative. Once again, use the linearity of derivatives and sums to compute the partial derivatives with respect to each of the $w[h]$ terms. This will give you $\frac{\partial^2}{\partial w[g]\partial w[h]}$. Don't forget the chain rule and again use the fact that $\vec{x}_i^\top \vec{w} = \sum_{j=1}^n x_i[j]w[j] = x_i[h]w[h] + \sum_{j \neq h} x_i[j]w[j]$.)

- iii) The expression in part (ii) is for the $[g, h]$ -th component of the second derivative. $\frac{1}{2}$ times this times $\delta w[g]$ times $\delta w[h]$ would give us that component's contribution to the second-derivative term in the approximation, and we have to sum this up over all g and h to get the total contribution of the second-derivative term in the approximation. Now, we want to group terms to restructure this into matrix-vector form by utilizing the outer-product form of matrix multiplication. **What should the space in the following expression be filled with?**

$$\frac{d^2}{d\vec{w}^2}c_i(\vec{w}) = c''(\vec{x}_i^\top \vec{w}, \ell_i) \underline{\hspace{2cm}}$$

- (c) Now we have successfully expressed the second order approximation of $c_i(\vec{w}_* + \delta\vec{w})$. Since we eventually want to minimize the total cost $c_{total}(\vec{w}) = \sum_{i=1}^m c_i(\vec{w})$, **can you write out the second order approximation of $c_{total}(\vec{w}_* + \delta\vec{w})$ using results from (a) and (b)?**
- (d) Now in this part, we want to re-write $c_{total}(\vec{w}_* + \delta\vec{w})$ in form of $C + \sum_{i=1}^m (\vec{q}_i^\top \delta\vec{w} - b_i)^2$.
- i) Let's first rewrite a general second order polynomial $f(x) = ax^2 + bx + c$ in the form of $f(x) = r + (px + q)^2$. **Find p, q, r in terms of a, b, c .** This procedure is called "completing the square". Then, **use this to argue that**

$$\arg \min_x ax^2 + bx + c = \arg \min_x (px + q)^2$$

- ii) Now rewrite $c_{total}(\vec{w}_* + \delta\vec{w})$ in the form of $C + \sum_{i=1}^m (\vec{q}_i^\top \delta\vec{w} - b_i)^2$. **What are C, q_i , and b_i ?**

- (e) Consider a least squares problem:

$$\vec{x}^\star = \arg \min_{\vec{x}} \left\| A\vec{x} - \vec{b} \right\|^2$$

Show that:

$$\left\| A\vec{x} - \vec{b} \right\|^2 = \sum_{i=1}^n (\vec{a}_i^\top \vec{x} - b_i)^2$$

where

$$A = \begin{bmatrix} - & \vec{a}_1^\top & - \\ - & \vec{a}_2^\top & - \\ & \vdots & \\ - & \vec{a}_n^\top & - \end{bmatrix}.$$

Use this to interpret your expression from Part (d) as a standard least squares problem. What are the rows of A ?

- (f) Consider the following cost functions:

squared error: $c_{sq}^+(p) = (p - 1)^2$, $c_{sq}^-(p) = (p + 1)^2$;

exponential: $c_{exp}^+(p) = e^{-p}$, $c_{exp}^-(p) = e^p$;

and logistic: $c_{logistic}^+(p) = \ln(1 + e^{-p})$, $c_{logistic}^-(p) = \ln(1 + e^p)$.

Compute the first and second derivatives of the above expressions with respect to p .

- (g) Run the Jupyter Notebook and answer the following questions.

i) **In Example 2, why does mean classification fail?**

ii) **In Example 3, for what data distributions does ordinary least squares fail?**

iii) Run the code cells in Example 4. By performing updates to $\vec{w}$ according to what you derived in previous parts of the question, **how many iterations does it take for exponential and logistic regression to converge?**

Congratulations! You now know the basic optimization-theoretic perspective on logistic regression. After you understand the probability material in 70 and 126, you can understand the probabilistic perspective on it as well. After you understand the optimization material in 127, you will understand even more about the optimization-theoretic perspective on the problem including why this approach actually converges.

3. Extending Orthonormality to Complex Vectors

So far in the course, we have only dealt with real vectors. However, it is often useful to also think about complex vectors, as you'll soon see with the DFT. In this problem, we will extend several important properties of orthonormal matrices to the complex case.

The main difference is that the normal Euclidean inner product is no longer a valid metric, and we must define a new complex inner product as

$$\langle \vec{u}, \vec{v} \rangle = \overline{\vec{v}^T} \vec{u} = \vec{v}^* \vec{u} = \sum_{i=1}^k u_i \overline{v_i},$$

where the $*$ operation will complex conjugate and transpose its argument (order doesn't matter), and is aptly called the *conjugate transpose*. Note that for real numbers, the complex inner product simplifies to the real inner product. In all the theorems you've seen in this class, you can replace every inner product with the complex inner product to show an analogous result for complex vectors: least squares becomes $\hat{x} = (A^* A)^{-1} A^* \vec{b}$, upper triangularization becomes $A = UTU^*$, Spectral Theorem becomes $A = U\Lambda U^*$, SVD becomes $A = U\Sigma V^*$.

- (a) To get some practice computing complex inner products, what is $\left\langle \begin{bmatrix} 1+j \\ 2 \end{bmatrix}, \begin{bmatrix} -3-j \\ 2+j \end{bmatrix} \right\rangle$ and $\left\langle \begin{bmatrix} -3-j \\ 2+j \end{bmatrix}, \begin{bmatrix} 1+j \\ 2 \end{bmatrix} \right\rangle$? Does the order of the vectors in the complex inner product matter i.e. is it commutative?

- (b) Let $U = \begin{bmatrix} \vec{u}_1 & \cdots & \vec{u}_n \end{bmatrix}$ be an n by n complex matrix, where its columns $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$ form an orthonormal basis for $\mathbb{C}^n$, i.e.

$$\vec{u}_i^* \vec{u}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Such a complex matrix is called *unitary* in math literature, to distinguish from real orthonormal matrices. **Show that $U^{-1} = U^*$, where U^* is the conjugate transpose of U .**

- (c) **Show that U preserves complex inner products, i.e. if $\vec{v}, \vec{w}$ are vectors of length n , then**

$$\langle \vec{v}, \vec{w} \rangle = \langle U\vec{v}, U\vec{w} \rangle.$$

HINT: Note that $(AB)^ = B^* A^*$. This is since $(AB)^* = \overline{(AB)^T} = \overline{B^T A^T} = \overline{B^T}^T \overline{A^T}^T = B^* A^*$*

- (d) **Show that if $\vec{u}_1, \dots, \vec{u}_n$ are the columns of a unitary matrix U , they must be linearly independent.**

(Hint: Suppose $\vec{w} = \sum_{i=1}^n \alpha_i \vec{u}_i$, then first show that $\alpha_i = \langle \vec{w}, \vec{u}_i \rangle$. From here ask yourself whether a nonzero linear combination of the $\{\vec{u}_i\}$ could ever be identically zero.)

This basic fact shows how orthogonality is a very nice special case of linear independence.

- (e) Now let V be another $n \times n$ matrix, where the columns of the matrix form an orthonormal basis for $\mathbb{C}^n$, i.e. V is unitary. **Show that the columns of the product UV also form an orthonormal basis for $\mathbb{C}^n$.**
- (f) We can also extend the idea of symmetric matrices to complex vectors, though we again will need to replace the transpose with the conjugate transpose. If $M = M^*$, then M is called a *Hermitian* matrix, and the Spectral Theorem will say it can be diagonalized by a unitary U , i.e. $M = U\Lambda U^*$, where Λ is a diagonal matrix with the eigenvalues along the diagonal. **Show that M has real eigenvalues.**

HINT: Use the fact that $M^ = M$.*

4. Roots of Unity

An N th root of unity is a complex number ω satisfying the equation $\omega^N = 1$ (or equivalently $\omega^N - 1 = 0$). In this problem we explore some properties of the roots of unity, as they end up being essential for the DFT.

- (a) **Show that the polynomial $z^N - 1$ factors as**

$$z^N - 1 = (z - 1) \left(\sum_{k=0}^{N-1} z^k \right).$$

- (b) **Show that any complex number of the form $\omega_N^k = e^{j\frac{2\pi}{N}k}$ for $k \in \mathbb{Z}$ is an N -th root of unity.** From here on, let $\omega_N = e^{j\frac{2\pi}{N}}$.
- (c) For a given integer $N \geq 2$, using the previous parts, **give the complex roots of the polynomial $1 + z + z^2 + \dots + z^{N-1}$.**
- (d) **What are the fourth roots of unity? Draw the fourth roots of unity in the complex plane. Where do they lie in relation to the unit circle?**
- (e) **What are the fifth roots of unity? Draw the fifth roots of unity in the complex plane.**
- (f) **For $N = 5$, $\omega_5 = e^{j\frac{2\pi}{5}}$, simplify ω_5^{42} such that the exponent is less than 5 and greater than 0.**
- (g) Let's generalize what you saw in the previous part. **Prove that $\omega_N^{k+N} = \omega_N^k$ for all integers k , both positive and negative.** This shows that the roots of unity have a periodic structure.
- (h) **What is the complex conjugate of ω_5 in terms of the 5th roots of unity? What is the complex conjugate of ω_5^{42} in terms of the 5th roots of unity? What is the complex conjugate of ω_5^4 in terms of the 5th roots of unity?**
- (i) **Compute $\sum_{m=0}^{N-1} \omega_N^{km}$ where ω_N is an N th root of unity.** Does the answer make sense in terms of the plot you drew?

5. Discrete Fourier Transform (DFT)

In order to get practice with calculating the Discrete Fourier Transform (DFT), this problem will have you calculate the DFT for a few variations on a cosine signal.

Consider a sampled signal that is a function of discrete time $x[t]$. We can represent it as a vector of discrete samples over time $\vec{x}$, of length N .

$$\vec{x} = \begin{bmatrix} x[0] & \dots & x[N-1] \end{bmatrix}^T \quad (5)$$

We can represent the DFT basis with the matrix U , and it is given by

$$U = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & e^{j\frac{2\pi}{N}} & e^{j\frac{2\pi(2)}{N}} & \dots & e^{j\frac{2\pi(N-1)}{N}} \\ 1 & e^{j\frac{2\pi(2)}{N}} & e^{j\frac{2\pi(4)}{N}} & \dots & e^{j\frac{2\pi 2(N-1)}{N}} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & e^{j\frac{2\pi(N-1)}{N}} & e^{j\frac{2\pi 2(N-1)}{N}} & \dots & e^{j\frac{2\pi(N-1)(N-1)}{N}} \end{bmatrix}.$$

In other words, for the ij^{th} entry of U , $U_{ij} = \frac{1}{\sqrt{N}} e^{j\frac{2\pi ij}{N}}$. From this, we can see that the DFT basis matrix is symmetric, so $U = U^T$. Another very important property of the DFT basis is that U is orthonormal, so $U^*U = I$. We want to find the coordinates of $\vec{x}$ in the DFT basis, and we know these coordinates are given by

$$\vec{X} = U^{-1}\vec{x}.$$

We call the components of $\vec{X}$ the *DFT coefficients* of the time-domain signal $\vec{x}$. We can think of the components of $\vec{X}$ as weights that represent $\vec{x}$ in the DFT basis. As we will explore in the problem, each coefficient can be thought of as a measurement for which frequency is present in the signal.

You can use Numpy or other calculation tools to evaluate cosines or do matrix multiplication, but you will not get credit if you directly calculate the DFT using a function in Numpy. You must show your work to get credit.

(a) **What is U^{-1} ?**

(b) Let the columns of $U = [\vec{u}_0 \ \vec{u}_1 \ \dots \ \vec{u}_{N-1}]$. **Prove that $\overline{\vec{u}_k} = \vec{u}_{N-k}$ for $k = 1, 2, \dots, N-1$.**

(c) **Decompose $\cos\left(\frac{2\pi}{7}n\right)$ into a sum of complex exponentials.**

(d) If $x_1[n] = \cos\left(\frac{2\pi n}{7}\right)$, **write $x_1[n]$ as a linear combination of $y_+[n] = e^{j\frac{2\pi n}{7}}$ and $y_-[n] = e^{-j\frac{2\pi n}{7}}$.**

(e) Now think of $\vec{x}_1$ as a length $N = 7$ vector which is $x_1[n]$ sampled at $n = 0, 1, \dots, 6$. We do the same to similarly get $\vec{y}_+, \vec{y}_-$. **Write $\vec{y}_+$ and $\vec{y}_-$ in terms of the columns of $U = [\vec{u}_0 \ \vec{u}_1 \ \dots \ \vec{u}_6]$.**

HINT: Use part (b).

(f) Using the last 3 parts, **compute the DFT coefficients $\vec{X}_1$ for signal $\vec{x}_1$.**

(g) **Plot the time domain representation of $x_1[n]$. Plot the magnitude, $|X_1[n]|$, and plot the phase, $\angle X_1[n]$, for the DFT representation $\vec{X}_1$.** You should notice that the DFT coefficients correspond to which frequencies were present in the original signal, which is why the DFT coefficients are called the *frequency domain* representation.

- (h) We define $x_2[n] = \cos\left(\frac{4\pi}{7}n\right)$ for $N = 7$ samples $n \in \{0, 1, \dots, 6\}$. **Compute the DFT coefficients $\vec{X}_2$ for signal $\vec{x}_2$.**
- (i) **Plot the time domain representation of $x_2[n]$. Plot the magnitude, $|X_2[n]|$, and plot the phase, $\angle X_2[n]$, for the DFT representation $\vec{X}_2$.**
- (j) To generalize this result, say we have some $p \in \{1, 2, 3\}$ which scales the frequency of our signal $\vec{x}_p$, which we define as $x_p[n] = \cos\left(\frac{2\pi}{7}pn\right)$ for $N = 7$ samples $n \in \{0, 1, \dots, 6\}$. **Compute the DFT coefficients $\vec{X}_p$ for signal $\vec{x}_p$ in terms of this scalar p .**
- (k) Let's see what happens when we have an *even* number of samples. We define $\vec{s} = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}^T$, which has $N = 6$ samples. **Compute the DFT coefficients $\vec{S}$ for signal $\vec{s}$.**
Hint: Write $\vec{s}$ as $a + b \cos\left(\frac{2\pi}{6}pn\right)$ for some constants a, b, p , and use the fact that this signal has period 2.

6. Survey

Please fill out the survey [here](#). As long as you submit it, you will get credit. Thank you!

7. Homework Process and Study Group

Citing sources and collaborators are an important part of life, including being a student!

We also want to understand what resources you find helpful and how much time homework is taking, so we can change things in the future if possible.

(a) **What sources (if any) did you use as you worked through the homework?**

(b) **If you worked with someone on this homework, who did you work with?**

List names and student ID's. (In case of homework party, you can also just describe the group.)

Contributors:

- Kourosh Hakhamaneshi.
- Kuan-Yun Lee.
- Nathan Lambert.
- Sidney Buchbinder.
- Gaoyue Zhou.
- Anant Sahai.
- Ashwin Vangipuram.
- John Maidens.
- Geoffrey Négier.
- Yen-Sheng Ho.
- Harrison Wang.
- Regina Eckert.